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वार्षिक प्रतिवेदन



भाकृअनुप - नार्म

ICAR - NAARM





भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी
राजेन्द्रनगर, हैदराबाद-500030, तेलंगाणा, भारत
ICAR-National Academy of Agricultural Research Management
(ISO 9001:2015 Certified)
Rajendranagar, Hyderabad - 500 030, Telangana, India
<https://www.naarm.org.in>



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Abbreviations

ABF	Agri-Biotech Foundation
ABM	Agribusiness Management
AERA	Agricultural Economics Research Association
AI	Artificial Intelligence
AICTE	All India Council for Technical Education
a-IDEA	Association for Innovation Development of Entrepreneurship in Agriculture
ANGRAU	Acharya N. G. Ranga Agricultural University
AP	Asia Pacific
APMC	Agricultural Produce Market Committee
ASCI	Administrative Staff College of India
ASRB	Agricultural Scientists Recruitment Board
ASTI	Agricultural Science and Technology Indicators
ATARI	Agricultural Technology Application Research Institutes
ATMA	Agriculture Technology Management Agency
AUs	Agricultural Universities
BSMA	Broad Subject Matter Area
BIRAC	Biotechnology Industry Research Assistance Council
CAB	College of Agricultural Banking
CAI	Centre for Agri-Innovation
CASFOS	Central Academy of State Forest Services
CII	Caspian Impact Assessment
CCARI	Central Coastal Agricultural Research Institute
CGIAR	Consultative Group for International Agricultural Research
CIFA	Central Institute of Freshwater Aquaculture
CIFE	Central Institute of Fisheries Education
CIMMYT	International Maize and Wheat Improvement Center
CIWA	Central Institute for Women in Agriculture
COLLAGe	Centre for Open and Lifelong Learning in Agriculture
CRIDA	Central Research Institute for Dryland Agriculture
CRISP	Centre for Remote Imaging, Sensing and Processing
CTCRI	Central Tuber Crops Research Institute
CVI	Cumulative Vulnerability Index
DBT	Department of Biotechnology
DFI	Doubling Farmers Income
DKMA	Directorate of Knowledge Management in Agriculture
DST	Department of Science and Technology
DWRP	Developing Winning Research Proposal

EDP	Executive Development Program
e-NAM	National Agriculture Market
FAI	Fertiliser Association of India
FET	Field Experience Training
FLD	Front Line Demonstrations
FFP	Farmer FIRST Programme
FIRST	Farm, Innovations, Resources, Science and Technology
FOCARS	Foundation Course on Agricultural Research Management
FOCFAU	Foundation Course for Faculty of Agricultural Universities
FOSS	Free and Open Source Software
GHMC	Greater Hyderabad Municipal Corporation
GIS	Geographic Information System
GoI	Government of India
GPS	Global Positioning System
GRA	Global Research Alliance
GST	Goods and Services Tax
GTWG	Global Technology Watch Group
IARI	Indian Agricultural Research Institute
IASRI	Indian Agricultural Statistics Research Institute
ICAR	Indian Council of Agricultural Research
ICBS	International Conference on Biocontrol for Sustainable Insect Pest Management
ICRISAT	International Crops Research Institute for the Semi-Arid Tropics
ICT	Information and Communication Technology
IFAF	Indian Fisheries and Aquaculture Forum
IFPRI	International Food Policy Research Institute
IGKV	Indira Gandhi <i>Krishi Vishwavidyalaya</i>
IGNFA	Indira Gandhi National Forest Academy
IIHR	Indian Institute of Horticultural Research
IIM	Indian Institute of Management
IIMR	Indian Institute of Millets Research
IIRR	Indian Institute of Rice Research
IISASTR	International Institute of SAS Training and Research
IISWC	Indian Institute of Soil and Water Conservation
IIWM	Indian Institute of Water Management
ILWIS	Integrated Land and Water Information System
IPE	Institute of Public Enterprise
IPNI	International Plant Names Index
IoT	Internet of Thing (IoT)
IRC	Institute Research Council
ISB	Indian School of Business
ISHS	International Society of Horticulture Sciences
ITEC	Indian Technical and Economic Cooperation
ISSN	International Journal of Agriculture Sciences

ITK	Indigenous Technical Knowledge
JAM	Jan Dhan account, Aadhar and Mobile
JNKVV	Jawaharlal Nehru Krishi Vishwa Vidyalaya
KVAFSU	Karnataka Veterinary, Animal and Fisheries Sciences University
KVKs	Krishi Vigyan Kendras
MAHYCO	Maharashtra Hybrid Seeds Company Private Limited
MANAGE	National Institute of Agricultural Extension Management
MANUU	Moulana Azaad National Urdu University
MBA	Master of Business Administration
MCCIA	Maharashtra Chamber of Commerce, Industry and Agriculture
MDP	Management Development Programme
MoEFCC	Ministry of Environment, Forest and Climate Change
MOOC	Massive Open Online Courses
MoU	Memorandum of Understanding
MPCC	Monthly Per Capita Consumption
MPI	Ministry of Primary Industries
MWDS	Mean Weighted Discrepancy Scores
NAARM	National Academy of Agricultural Research Management
NABARD	National Bank for Agriculture and Rural Development
NAAS	National Academy of Agricultural Sciences
NAIF	National Agriculture Innovation Fund
NAPCC	National Action Plan for Climate Change
NARES	National Agricultural Research and Education System
NARS	National Agricultural Research Systems
NASC	National Agricultural Science Complex
NASSCOM	The National Association of Software and Services Companies
NBA	National Board of Accreditation
NBP	Need-based programmes
NBSSLUP	National Bureau of Soil Survey and Land Use Planning
NCDEX	National Commodity & Derivatives Exchange Limited
NDRI	National Dairy Research Institute
NDVI	Normalized Difference Vegetation Index
NFCL	Nagarjuna Fertilizers and Chemicals Limited
NFDB	National Fisheries Development Board
NGOs	Non-Government Organizations
NIAP	National Institute of Agricultural Economics and Policy Research
NIFTEM	National Institute of Food Technology Entrepreneurship and Management
NIPHM	National Institute of Plant Health Management
NIRD & PR	National Institute of Rural Development and Panchayat Raj
NITHM	National Institute of Tourism and Hospital Management
NSTEDB	National Science & Technology Entrepreneurship Development
NSTMIS	National Science & Technology Management Information System
OER	Open Educational Resources

PCs	Producer Companies
PGDMA	Post Graduate Diploma in Management- Agriculture
PGDTMA	Post Graduate Diploma in Technology Management in Agriculture
PJTSAU	Professor Jayashankar Telangana State Agricultural University
PME	Priority setting, Monitoring and Evaluation
PRSI	Public Relations Society of India
PVNRTVU	P.V. Narasimha Rao Telangana Veterinary University
QGIS	Quantum Geographic Information System
QRT	Quinquennial Review Team
RGMVP	Rajiv Gandhi Mahila Vikas Paryojana
RKVY	Rashtriya Krishi Vikas Yojana
RVSKVK	Rajamata Vijayaraja Scindia Krishi Vishwavidyalaya
SAIDP	State Agriculture Infrastructure Development Plans
SAMETI	State Agricultural Management and Extension Training Institute
SAS	Statistical Analysis System
SAU's	State Agricultural Universities
SDG	Sustainable Development Goals
SEVI	Socio-Economic Vulnerability Index
SFAC	Small Farmers' Agribusiness Consortium
SHUATS	Sam Higginbottom University of Agriculture Technology and Sciences
SLA	Students' Learning Approaches
SMM	Smart Micro-Mandi
SMD	Subject Matter Division
SPIC-MACAY	Society for the Promotion of Indian Classical Music and Culture Amongst Youth
SSRN	Social Science Research Network
STEM	Society for Technology Management
TAAS	Trust for Advancement of Agricultural Sciences
TAR	Technology Assessment and Refinement
TANUVAS	Tamil Nadu Veterinary and Animal Sciences University
TBI	Technology Business Incubator
TELAge	Technology Enhanced Learning in Agricultural Education
TIFAC	Technology Information, Processing and Assessment Council
TISS	Tata Institute of Social Sciences
TNAU	Tamil Nadu Agricultural University
TNJFU	Tamil Nadu Dr.Jayalalitha's Fisheries University
TPACK	Technology Pedagogy and Content Knowledge Model
UAHS	University of Agricultural & Horticultural Sciences
UNFCCC	United Nations Framework Convention on Climate Change
UoH	University of Hyderabad
VNMAU	Vasantrao Naik Marathwada Krishi Vidyapeeth
WDP	Wheat Development Programme

निदेशक की कलम से



डा. सीएच. श्रीनिवास राव
निदेशक

विश्वविद्यालयों (AUs) के संकाय सदस्यों का सतत क्षमता निर्माण करने और उनके ज्ञान को अद्यतन बनाने का प्रयास किया जाता है। इस पृष्ठभूमि के साथ, मुझे कृषि एवं सम्बद्ध विज्ञान के क्षेत्र में क्षमता निर्माण, अनुसंधान शिक्षा एवं उद्यमशीलता विकास के क्षेत्रों में वर्ष 2018-19 के दौरान अकादमी की प्रगति और उपलब्धियों को यहां प्रस्तुत करने पर प्रसन्नता का अनुभव हो रहा है। अकादमी द्वारा मौजूदा प्रशिक्षण कार्यक्रमों में सुधार करके और जरूरत आधारित कार्यक्रमों को शामिल करके तथा साथ ही अकादमी के रणनीतिपरक विषयी क्षेत्रों में क्षमता निर्माण में शामिल विविध पोर्टफोलियो का विकास करने हेतु कार्यक्रमों में विशेषज्ञों को आमंत्रित करके स्वयं में सुधार करने का लगातार प्रयास किया जाता है। वर्ष 2018-19 के दौरान, देशभर से लगभग 3341 प्रतिभागियों को अकादमी के विभिन्न कार्यक्रमों के माध्यम से प्रशिक्षण प्रदान किया गया।

वर्ष 2018-19 के दौरान, भाकृअनुप - राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद द्वारा कृषि अनुसंधान प्रणाली (ARS) प्रोबेशनरों (108वां एवं 109वां बैच) के लिए दो बुनियादी पाठ्यक्रम आयोजित किए गए जिनमें कुल 167 नव नियुक्त एआरएस वैज्ञानिक प्रोबेशनरों को बुनियादी प्रशिक्षण प्रदान किया गया। साथ ही अकादमी द्वारा वर्ष 2018-19 के दौरान, कृषि विश्वविद्यालयों के संकाय उन्नयन के लिए एक माह का सेतु कार्यक्रम भी आयोजित किया गया। इन प्रमुख क्षेत्रों के अलावा, अकादमी द्वारा एमडीपी, ईडीपी, पुनश्चर्या पाठ्यक्रम, जरूरत आधारित कार्यक्रम एवं ऑफ-कैम्पस प्रशिक्षण कार्यक्रम भी आयोजित किए गए।

अकादमी के MOOCs कार्यक्रम द्वारा ऐसे पेशेवरों को महत्वपूर्ण लिंक प्रदान किया जाता है जो कि कई कारणों से वास्तविक समय वाले प्रशिक्षण कार्यक्रमों में भाग नहीं ले पाते। मुझे यह प्रस्तुत करते हुए गर्व का अनुभव हो रहा है कि मैसिव ओपन ऑन-लाइन कोर्स (MOOCs) द्वारा न केवल विभिन्न पाठ्यक्रमों के लिए पंजीकरण की संख्या के संबंध में वरन् इन पाठ्यक्रमों की

पूर्णता प्रतिशत के संबंध में भी स्वयं को लगातार बेहतर बनाया जा रहा है।

क्षमता निर्माण पर हमारे फोकस के अलावा, अकादमी द्वारा अनुसंधान कार्यक्रम भी चलाये जाते हैं जिनके परिणामों का जानकारी के प्रवाह को शामिल करके क्षमता निर्माण कार्यक्रमों में सुधार लाने पर सीधा प्रभाव पड़ता है। इसके साथ ही अकादमी द्वारा विभिन्न सार्वजनिक एवं निजी एजेन्सियों से अनुबंधीय तथा परामर्शी परियोजनाओं के माध्यम से बड़ी मात्रा में राजस्व का सृजन करने में भी सफलता मिली है जिससे न केवल अनुसंधान प्रसार करने में मदद मिलती है वरन् इससे अपने हितधारकों की दृष्टि में अकादमी के संकाय की विश्वसनीयता भी बढ़ती है। अनुबंधीय अनुसंधान के माध्यम से लगातार उच्च गुणवत्ता वाला अनुसंधान आउटपुट उत्पन्न किया जा रहा है जो कि उच्च प्रभाव वाली पत्रिकाओं में प्रकाशित हुआ है।

अकादमी का अग्रणी शैक्षणिक कार्यक्रम दो वर्ष के लिए पूर्ण समय हेतु कृषि के प्रबंधन में स्नातकोत्तर डिप्लोमा (PGDMA) है जो कि देशभर के छात्रों को बड़ी संख्या में आकर्षित करता है। भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद इस प्रबंधन पाठ्यक्रम में उच्च गुणवत्ता वाली शिक्षा प्रदान करने में अत्यंत सफल रहा है जिसके परिणामस्वरूप प्रतिष्ठित कृषि व्यवसाय कम्पनियों में छात्रों को 100 प्रतिशत रोजगार हासिल हुआ है। प्रौद्योगिकी प्रबंधन में स्नातकोत्तर डिप्लोमा को उत्साहवर्धक प्रतिक्रिया मिल रही है। इसके अलावा, दूरवर्ती शिक्षा मोड में शिक्षा प्रौद्योगिकी प्रबंधन में स्नातकोत्तर डिप्लोमा (PGDETM) पाठ्यक्रम को इस वर्ष से प्रारंभ किया गया जिसे उच्च स्तरीय स्वीकार्यता प्राप्त हो रही है।

अकादमी में उत्कृष्टता केन्द्र यथा कृषि इनोवेशन के लिए केन्द्र (CAI) तथा कृषि में ओपन एवं लाइफ लॉग लर्निंग के लिए केन्द्र (COLLAge) की सुविधा विद्यमान है। रिपोर्टाधीन वर्ष में दोनों केन्द्रों द्वारा उल्लेखनीय प्रगति की गई है। कृषि इनोवेशन केन्द्र (CAI) के अंतर्गत प्रौद्योगिकी व्यवसाय इनक्यूबेटर केन्द्र यथा a-IDEA। के तहत एग्री-उडान को चलाया जा रहा है जिसमें बड़ी संख्या में एग्री स्टार्ट-अप और कृषि उद्यमियों को अपनी ओर आकर्षित किया गया है। इस केन्द्र को सरकार के विज्ञान एवं प्रौद्योगिकी विभागों यथा विज्ञान व प्रौद्योगिकी विभाग (DST), बीआईआरएसी (BIRAC), जैव प्रौद्योगिकी विभाग (DBT), भारत सरकार से वित्तीय सहायता प्राप्त हो रही है और इसे एग्री बायोटेक फाउण्डेशन (ABF), राजेन्द्रनगर, हैदराबाद के साथ भागीदारी मोड में चलाया जाता है। COLLAge के अंतर्गत, भारतीय कृषि अनुसंधान परिषद की उत्कृष्ट परियोजना के आला क्षेत्र के तहत विकसित की गई सुविधा में सुधार किया गया है जिसमें रिपोर्टाधीन वर्ष में

ग्रीन मैट प्रौद्योगिकी को स्थापित किया गया है और साथ ही दूरवर्ती तथा ऑन-लाइन कार्यक्रमों में उपयोग किए गए डिजिटल माइयूल्स का विकास करने तथा विच्युल कक्षाओं, MOOCs आदि के आयोजन के लिए संवेदनशील उपकरण से सुसज्जित किया गया है।

राष्ट्रीय ग्रामीण विकास संस्थान (NIRD) तथा राष्ट्रीय पादप स्वास्थ्य प्रबंधन संस्थान (NIPHM) सहित राष्ट्रीय प्रशिक्षण संस्थानों के साथ भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद द्वारा समझौता ज्ञापन (MoU) किए गए हैं ताकि अपने कार्यक्रमों के लिए गुणवत्ता प्रशिक्षण माइयूल्स का सृजन किया जा सके। COLLAGe में स्थित मॉडर्न प्रयोगशाला सुविधा द्वारा अकादमी के लिए सतत आधार पर राजस्व का सृजन किया जा रहा है। स्वच्छ भारत, डिजिटल इंडिया, मेरा गांव – मेरा गौरव, फार्मर फर्स्ट परियोजना आदि जैसे राष्ट्रीय महत्व के विभिन्न कार्यक्रमों में अकादमी की भागीदारी सार्वजनिक एवं निजी संस्थानों के साथ निरन्तर बनी रही।

मुझे यह बताते हुए हर्ष हो रहा है कि रिपोर्टाधीन अवधि के दौरान अकादमी को अनेक पुरस्कार प्रदान किए गए जिनमें विज्ञान एवं प्रौद्योगिकी विभाग से “उभरते प्रौद्योगिकी व्यवसाय इनक्यूबेटर के लिए राष्ट्रीय पुरस्कार 2017 (National Award for Emerging Technology Incubator 2017)” भी शामिल है। डॉ. Ch. श्रीनिवास राव, निदेशक, भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद एवं अध्यक्ष, a-IDEA ने यह पुरस्कार दिनांक 11 मई, 2018 को विज्ञान भवन, नई दिल्ली में आयोजित समारोह में डॉ. हर्षवर्धन, माननीय केन्द्रीय विज्ञान व प्रौद्योगिकी, पृथ्वी विज्ञान, पर्यावरण, वन एवं जलवायु परिवर्तन मंत्री, भारत सरकार के कर-कमलों से प्राप्त किया। भारत के माननीय राष्ट्रपति श्री राम नाथ कोविंद ने भी इस कार्यक्रम की शोभा बढ़ाई।

पद्मश्री डॉ. मंगला राय की अध्यक्षता में भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM) के पंचवर्षीय समीक्षा दल (QRT) ने अपनी समीक्षा में अकादमी की प्रगति पर अपनी खुशी प्रकट की और अकादमी के विभिन्न कार्यकलापों को मजबूती प्रदान करने हेतु अपनी अनेक सिफारिशें दीं। यह अकादमी, निरन्तर मार्गदर्शन और प्रोत्साहन प्रदान करने के लिए डॉ. त्रिलोचन महापात्र, सचिव, डेयर एवं महानिदेशक, भाकृअनुप एवं श्री सुशील कुमार, अपर सचिव, डेयर एवं सचिव, भाकृअनुप के प्रति हार्दिक आभारी है। इसके साथ ही अकादमी की गतिविधियों और अकादमी के संकाय एवं स्टाफ द्वारा की गई प्रगति को आगे बढ़ाने में अपना हार्दिक सहयोग देने के लिए डॉ. एन.एस. राठौर, उप महानिदेशक (शिक्षा) के प्रति हम अपना आभार प्रकट करते हैं। भारतीय कृषि अनुसंधान परिषद के लगातार एवं बिना किसी शर्त वाले सहयोग के बिना अकादमी की यह प्रगति संभव नहीं हो पाती।

मैं, अकादमी को मजबूती प्रदान करने में भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद के संयुक्त निदेशक प्रभारी, समस्त संभागाध्यक्षों, विभिन्न अनुभागों के अध्यक्ष, संयुक्त निदेशक (प्रशासन) मुख्य वित्त व लेखा अधिकारी और समस्त स्टाफ द्वारा किए गए योगदान के प्रति आभार व्यक्त करता हूं। इस वार्षिक प्रतिवेदन का संकलन व सम्पादन करने में किए गए अथक प्रयासों के लिए मैं, सम्पादन मण्डल के सदस्यों डॉ. एम. कृष्णन, डॉ. आई सेकर, डॉ. बी.एस. सोन्ताकी, डॉ. रणजीत कुमार, डॉ. आर.वी.एस. राव, डॉ. डी. थम्मी राजू, डॉ. मंजू गेरार्ड, डॉ. पी.डी. श्रीकांत, श्री शेखर रेड्डी, श्री पी. नामदेव, श्रीमती जी. अनीजा, श्री एम. रवि एवं श्रीमती वी. शैलजा की सराहना करता हूं।

भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद के मुखिया के रूप में मैं, वर्ष 2018-19 को सफल बनाने में की गई कड़ी मेहनत और दृढ़ता के लिए समस्त नार्म परिवार की सराहना करता हूं।

दिनांक : 01 जुलाई, 2019

स्थान : हैदराबाद

(Ch. श्रीनिवास राव)

निदेशक

From the Director's Desk



Dr Ch. Srinivasa Rao

Director

ICAR-NAARM is the nodal training and capacity building institute of the ICAR and strives to continuously update and build the capacity of agricultural scientists and university faculty of National Agricultural Research

Education and Extension System (NAREES) in India. With this backdrop, I am happy to record the progress and achievements of the Academy during the year 2018-19 in areas of capacity building, research education, and entrepreneurship development in the field of agriculture and allied sciences. The Academy consistently tries to improve upon itself by refining the existing training programmes and adding need-based programmes and also by inviting experts to the programmes to develop a diversified portfolio of engagements in capacity building in strategic thematic areas of the Academy. During the year 2018-19 about 3,341 participants across the country have been imparted training through various programmes of the Academy.

ICAR-NAARM conducted two foundation courses for ARS Probationers (108th and 109th batches) during the year 2018-19 through which 167 newly recruited ARS Scientist Probationers have been imparted foundation training. The Academy also conducted one-month abridged programmes for faculty improvement of agricultural universities during the year 2018-19. In addition to these core programmes, the Academy also conducted MDP, EDP, Refresher Courses, need-based programmes and off-campus training programmes.

The MOOCs program of the academy provides the vital link to those professionals who are constrained

owing to several factors from participating in real time training programmes. I am proud to place on record that the Massive Open Online Courses (MOOCs) is continuously bettering itself not only in terms of number of registrations for various courses but also the percentage of completion of these courses.

In addition to our focus on capacity building, the academy also has research programmes, the output of which has direct relevance on improving the capacity building programmes by adding to the flow of knowledge. The Academy is also successful in garnering large funds through contractual and consultancy projects from various public and private agencies which not only adds to research extension but also to the confidence of the faculty in the eyes of its stakeholders. Contractual research being continuous in nature generates high quality research output which gets published in high impact journals

The Flagship education programme of the Academy is the two-year full time Post-Graduate Diploma in Management of Agriculture (PGDMA) which attracts large number of students from all over the country. ICAR-NAARM has been very successful in imparting high quality education in this management course which resulted in 100% placement of the students in reputed agri-business companies. The one-year Post-Graduate Diploma in Technology Management (PGDTMA) is receiving overwhelming response. In addition, the Post-Graduate Diploma in Educational Technology Management (PGDETM) course on distance education mode has been launched from this year onwards, which has also received high level of acceptance.

The Centres of Excellence at the Academy viz. Centre for Agri-innovation (CAI) and Centre for Open and Lifelong Learning in Agriculture (COLLAge). Both the centres have made commendable progress during

the year. The Technology Business Incubator Centre viz. a-IDEA under CAI is running the Agri-UDAAN which has attracted large number of agri-startups and agri-preneur enthusiasts. This centre is receiving financial support from government S&T departments like DST, BIRAC, DBT, GoI, and runs in partnership with Agri-Biotech Foundation (ABF), Rajendranagar, Hyderabad. Under COLLAGe, the facility developed under niche-area of excellence project of ICAR has improved its facilities by instilling Green Mat Technology during the year and has housed with sophisticated equipment for developing digital modules used in distance and online programmes and conduct of virtual classes, MOOCs, etc.

National training institutions including the National Institute of Rural Development (NIRD) and National Institute of Plant Health Management (NIPHM) have entered into an MOU with ICAR-NAARM to produce quality training modules for their programmes. The modern lab facility at COLLAGe has been consistently generating revenue for the Academy. The academy has continued to partner with public and private institutions in various programmes of national importance for example, Swatchh Bharath, Digital India, Mera Gaon Mera Gaurav, Farmers' First project etc.

It is my pleasure to place on record that the Academy has received several awards like "National Award for Emerging Technology Business Incubator 2017" from Department of Science & Technology, Dr Ch. Srinivasa Rao, Director, NAARM & President, a-IDEA received the award from Dr Harsh Vardhan, Hon'ble Union Minister for Science & Technology, Earth Sciences, Environment, Forests and Climate Change at a function held on Technology Day at Vigyan Bhawan,

New Delhi on May 11, 2018. The event was graced by Hon'ble President of India, Shri Ram Nath Kovind.

The Quinquennial Review Team (QRT) of ICAR-NAARM headed by Padma Shri Dr Mangla Rai expressed his happiness on the progress of the academy during its review and made several recommendations for strengthening the various arms of the Academy. The Academy is grateful to Dr Trilochan Mohapatra, Secretary-DARE and Director General, ICAR and Shri Sushil Kumar, Addl. Secretary-DARE & Secretary-ICAR for their continuous guidance and encouragement. The Academy is also grateful to Dr N.S. Rathore, DDG (Edn.) for providing whole-hearted support in pursuing the activities of the academy and furthering the progress made by the faculty and staff of the Academy. All this could not have been possible but for the unstinted and unconditional support of ICAR.

I acknowledge the contributions of Joint Director In-charge, all Heads of Divisions, various section In charges, Joint Director (Admn.) & Registrar, Chief Finance & Accounts Officer and all staff of ICAR-NAARM in strengthening the Academy. I sincerely appreciate the efforts put in by Drs. M. Krishnan, I. Sekar, B.S. Sontakki, Ranjit Kumar, R.V.S.Rao, D. Thammi Raju, Manju Gerard, P.D. Sreekanth, Mr. Shekhar Reddy, Mr P. Namdev, Mrs.G. Aneja, M. Ravi and Mrs. V. Shailaja, for compiling and editing this report.

As Head of NAARM family, I place on record the hard work and sincere efforts of all in making the year 2018-19 successful and rewarding.



(Ch. Srinivasa Rao)
DIRECTOR

Date: 1st July, 2019

Place: Hyderabad

कार्यकारी सारांश

भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), भारतीय कृषि अनुसंधान परिषद (ICAR) का एक नोडल क्षमता निर्माण संगठन है। अकादमी द्वारा भारत में सम्पूर्ण राष्ट्रीय कृषि अनुसंधान, शिक्षा एवं प्रसार प्रणाली (NARES) के पेशवरों की क्षमता को बढ़ाने में केन्द्रीय भूमिका निभाई जाती है।

वर्ष 2018-19 में, अकादमी द्वारा अनुसंधान योगदान, नीतिगत सहयोग और शिक्षा के साथ साथ राष्ट्रीय कृषि अनुसंधान, शिक्षा एवं प्रसार प्रणाली (NARES) के भीतर एवं बाहर विभिन्न हितधारकों के बीच शिक्षा अनुसंधान के विकास कार्य में उल्लेखनीय योगदान दिया गया। इसके अलावा, कृषि सेक्टर में उद्यमशीलता विकास के क्षेत्र में अकादमी द्वारा किए गए प्रयासों को सहर्ष स्वीकार किया गया जिसकी कि प्रत्येक जन तथा सभी ने सराहना की। अकादमी में विभिन्न कृषि विश्वविद्यालयों के कृषि पेशवरों और कृषि विज्ञान केन्द्रों (KVK) के समन्वयकों की क्षमता को बढ़ाने में योगदान किया जाता है। चूंकि, क्षमता निर्माण करना अकादमी का प्रमुख क्षेत्र है, इसलिए वर्ष 2018-19 के दौरान अनेक क्षमता निर्माण कार्यक्रम आयोजित किए गए। इसे जरूरत आधारित कार्यक्रमों, ऑफ कैम्पस कार्यक्रमों व अन्य में वर्गीकृत किया जा सकता है।

भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद द्वारा देश के विभिन्न भागों में 48 जरूरत आधारित कार्यक्रम तथा 17 ऑफ-कैम्पस कार्यक्रम आयोजित किए गए। आयोजित किए गए कार्यक्रमों में कुल 1779 प्रशिक्षुओं को लाभ पहुंचाया गया। प्रशिक्षण कार्यक्रमों में मैसिव ओपन ऑन-लाइन कोर्स (MOOC) भी शामिल है जिसमें कुल 1192 प्रतिभागियों को शामिल किया गया।

प्रत्येक वर्ष की भांति, वर्ष 2018-19 में भी एआरएस प्रोबेशनर्स के लिए कृषि अनुसंधान सेवा का बुनियादी पाठ्यक्रम (FOCARS) आयोजित किया गया जो कि जुलाई, 2018 तथा जनवरी, 2019 में प्रारंभ हुआ। इनमें कुल 167 नव नियुक्त एआरएस वैज्ञानिक प्रोबेशनर्स को प्रशिक्षित किया गया। इसी प्रकार, फोकार्स की तर्ज पर वर्ष के दौरान कृषि विश्वविद्यालयों से 48 प्रतिभागियों के लिए बुनियादी पाठ्यक्रम (FOCEAU) आयोजित किए गए जो कि एक माह की अवधि वाले थे।

इस वर्ष भारतीय कृषि अनुसंधान परिषद के नव नियुक्त अनुसंधान प्रबंधन पद संवर्ग के लिए कार्यकारी विकास कार्यक्रम में प्रतिभागी शामिल हुए तथा भारतीय कृषि अनुसंधान परिषद संस्थानों के प्री- RMP संवर्ग के लिए प्रबंधन विकास कार्यक्रम (MDP) में कुल 47 प्रतिभागी थे। भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद द्वारा राष्ट्रीय कृषि अनुसंधान शिक्षा प्रणाली (NARES) के भीतर विभिन्न संवर्गों को प्रशिक्षण की जिम्मेदारियों को पूरा किया जा रहा है।

इस वर्ष अकादमी द्वारा जरूरत आधारित कार्यक्रमों का आयोजन करके भारतीय कृषि अनुसंधान परिषद के विभिन्न संस्थानों के कुल 134 तकनीकी अधिकारियों एवं प्रशासनिक पेशवरों को प्रशिक्षण प्रदान किया गया। कुशल सहायी स्टाफ की क्षमता में बढ़ोतरी करने के पूर्ववर्ती वर्ष के प्रशिक्षण कार्यक्रमों के साथ निरंतरता में इस वर्ष एक प्रशिक्षण कार्यक्रम भी आयोजित किया गया जिसमें कुल 27 सहायी स्टाफ ने भाग लिया।

इसके साथ ही अकादमी द्वारा विभिन्न विश्वविद्यालयों यथा प्रोफेसर जय शंकर तेलंगाना राज्य कृषि विश्वविद्यालय (PJTSAU), आचार्य एन.जी. रंगा कृषि विश्वविद्यालय (ANGRAU), जवाहर लाल नेहरू कृषि विश्वविद्यालय (JNKVV), टीएनजेएफयू, आरवीएसकेवीवी, कृषि विज्ञान विश्वविद्यालय (UAS) तथा कृषि एवं बागवानी विज्ञान विश्वविद्यालय (UAHS) आदि की प्रशिक्षण जरूरतों की भी पूर्ति की जाती है। इस प्रक्रिया में, अकादमी द्वारा वर्ष 2018-19 में विभिन्न प्रायोजित कार्यक्रमों के माध्यम से लगभग रुपये 88.61 लाख का राजस्व सृजित किया।

रिपोर्टाधीन अवधि के दौरान, भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद द्वारा दो प्रमुख शैक्षणिक कार्यक्रमों का आयोजन किया गया। पहला कार्यक्रम कृषि में स्नातकोत्तर डिप्लोमा (PGDMA) है। यह दो वर्षीय पाठ्यक्रम पूरी तरह से आवासीय है जिसमें 357 आवेदन प्राप्त हुए और इनमें से प्रवेश के लिए 27 का चयन किया गया। पूर्ववर्ती बैच को विभिन्न कम्पनियों में रोजगार हासिल हुआ और इसमें अधिकतम वेतन पैकेज प्रति वर्ष रुपये 9.5 लाख हासिल किया गया।

कृषि में प्रौद्योगिकी प्रबंधन में एक वर्षीय स्नातकोत्तर डिप्लोमा (PGDTMA) में पिछले वर्ष कुल 65 छात्रों ने प्रवेश लिया, अब इस वर्ष उसे पूरा कर लिया गया है। वर्ष 2018-19 के दौरान, नए छात्रों को कृषि में प्रौद्योगिकी प्रबंधन में एक वर्षीय स्नातकोत्तर डिप्लोमा (PGDTMA) में प्रवेश दिया गया। इसके अलावा, अकादमी द्वारा कुल 48 छात्रों के इनटेक के साथ कृषि में शिक्षा प्रौद्योगिकी प्रबंधन में स्नातकोत्तर डिप्लोमा (PGDETMA) को प्रारंभ किया गया। इन कार्यक्रमों से अकादमी को वर्ष 2018-19 में रुपये 7.85 लाख का राजस्व सृजित करने में मदद मिली।

अनुसंधान के मोर्चे पर, भाकृअनुप – राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद में कुल 38 अनुसंधान परियोजनाएं विद्यमान थीं जिनमें 19 घरेलू परियोजनाएं तथा 19 बाह्य वित्त पोषित अनुबंधीय अनुसंधान परियोजनाएं एवं परामर्शी परियोजनाएं शामिल हैं।

प्रायोजित परियोजनाओं में विभिन्न संगठनों यथा भारतीय कृषि अनुसंधान परिषद (ICAR) स्कीम, एनएसएमआईएस, विज्ञान एवं प्रौद्योगिकी विभाग (DST), आईटीसी प्रा. लि., जैव प्रौद्योगिकी विभाग (DBT) विश्व बैंक, मायको मॉनसेन्टो बायोटेक (इंडिया) प्रा. लि., नीति आयोग, आईएफपीआरआई, टोबेको बोर्ड, आरजीएमवीपी – रायबरेली, प्रोफेसर जय शंकर तेलंगाना राज्य कृषि विश्वविद्यालय (PJTSAU), मैनेज तथा एनआईपीएचएम से जुड़ी परियोजनाएं शामिल हैं। अनुसंधान परियोजनाओं को छः विषयी क्षेत्रों में वर्गीकृत किया गया है जैसे कि सम्यक प्रगति के लिए कृषि व्यवसाय प्रबंधन; समृद्ध शिक्षण गुणवत्ता एवं वातावरण के लिए शिक्षा प्रणालियां प्रबंधन; नेतृत्वशीलता तथा शासन प्रणाली के लिए क्षमताओं में सुधार करना; बाजार चालित वातावरण में प्रसार प्रणालियां प्रबंधन; नवोन्मेष एवं शासन प्रणाली को बढ़ावा देने के लिए सूचना एवं संचार प्रबंधन; तथा नवोन्मेष एवं टिकाऊ विकास के लिए विज्ञान एवं प्रौद्योगिकी को गतिशील बनाना।

वर्ष 2018-19 में भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (ICAR - NAARM), हैदराबाद के संकाय सदस्यों द्वारा विभिन्न प्रारूपों में कुल 151 प्रकाशनों में अपना योगदान दिया गया। कुल 57 सहकर्मी समीक्षित अनुसंधान

Executive Summary

ICAR-National Academy of Agricultural Research Management is the nodal organization of ICAR for capacity building and engaged in enhancing the capacity of professionals in the entire National Agricultural Research Education and Extension Systems (NARES) in India.

During 2018-19, the contribution of ICAR-NAARM in agricultural education research, policy and education has been immense. ICAR-NAARM has made significant contributions in the field of entrepreneurship development in agriculture sector, which was lauded by all. The Academy also contributed towards the capacity of agriculture professionals of different Agricultural Universities and programme coordinators of Krishi Vigyan Kendras (KVKs). Since the main mandate of the Academy is capacity building, several need-based programmes, off-campus programmes in addition to the flagship training programmes were organized during 2018-19.

In total, 48 need-based programmes and 17 off-campus programmes were organized by the Academy in various parts of the country benefitting 1,779 trainees who attended the programmes. Training programmes also include Massive Open Online Course (MOOC) in which 1192 participants were involved.

The Foundation Course of Agriculture Research Service (FOCARS) has been the flagship programme of the Academy for ARS probationers. There were two batches (108 & 109th FOCARS) during 2018-19, one commenced in July, 2018 and another in January, 2019 in which a total of 167 newly recruited ARS Scientist Probationers were trained. In similar fashion, ICAR-NAARM has taken a pro-active role in training newly recruited Assistant Professors of State Agricultural Universities. The Foundation Course for the Agricultural Universities (FOCAU) was also

organized during the year for one month duration, which benefitted 48 participants of agricultural universities. The Academy also cater to the training needs of various universities like PJTSAU, ANGRAU, JNKVV, TNJFU, RVSKVV, UAS, UAHS etc. In this process, the Academy generated approx. Rs.88.61 lakhs in 2018-19 through various sponsored programmes.

The Executive Development Programme for newly recruited research managers of ICAR, the Management Development Programme (MDP), pre-RMP programme were organized fulfilling NAARM's responsibilities of training different cadres at different hierarchical levels within the NARES.

The Academy also trained 134 technical officers and administrative professionals of various ICAR institutes by conducting need based programmes during the year. In continuation with previous year's training programmes to enhance the capacity of skilled support staff, a training programme was organized in which 27 supporting staff attended.

The Post-Graduate Diploma in Management-Agriculture (PGDMA) is a two-year fully residential programme in which 27 students were selected for admission during the academic year 2018-20. The previous batch were able to get placements in various companies and the highest compensation package of Rs.9.5 lakhs per year was obtained. The one-year Post-Graduate Diploma in Technology Management in Agriculture (PGDTMA) admitted 34 students during 2018-19. In addition to this, the Academy initiated a new programme on Post Graduate Diploma in Educational Technology Management (PGDETM) with an intake of 48 students.

On the research front, the Academy had 38 research projects, which comprised of 19 In-house projects, and 19 externally funded contract research and consultancy

projects. The sponsored projects are from various organizations including ICAR Schemes, NSTMIS, DST, DBT World Bank, NITI Aayog, IFPRI, RGMVP-Rae Bareilly, PJTSAU, MANAGE, NIPHM, ITC Pvt. Ltd., Mahyco Monsanto Biotech (India) Pvt. Ltd., Tobacco Board. The research projects span over six thematic areas viz. Agricultural Business Management for Inclusive growth, Education Systems Management for Enriched Learning Quality and Environment, Enhancing Capacity for Leadership Governance, Extension Systems Management in Market driven Environment, Information and Communication Management for Promoting Innovation and Governance and Mobilizing Science & Technology for Innovation in Sustainable Development.

The faculty members of the Academy have brought out 151 publications in 2018-19 in various formats. NAARM faculty published fifty-seven peer review research papers, which is more than 37% of the total publications. The other publications include books, book chapters, popular articles, training manuals, seminar proceedings, research reports etc.

ICAR-NAARM is one of the consortia partner of Farmers FIRST Initiative launched by ICAR. Academy was involved in the capacity building activities and organized several workshops under Farmers FIRST programme. NAARM faculty constantly upgrade their knowledge and skills through attending various capacity building programmes across the country. In 2018-19, NAARM faculty have attended 10 different capacity building programmes.

The Centre for Agri-Innovation (CAI) is located in the campus provides hand-holding support to the agri-business incubators through a-IDEA, the technology business incubator centre of the Academy. So far, 15 incubatees have been inducted and four have already graduated. Different events were organized in various places which include AGRI-UDAAN, two sensitization workshops, 6 Startup Samvaad, 2 startups 360, one EDP Pitch Perfect, one workshop on data platform.

Among others, the Technology Enhanced Learning is state-of-the-art Agricultural Education Lab (TELAge), which was established in the Academy to develop digital courses, offer distance and online programmes, conduct virtual classes and support in-house programmes. This facility has advanced technologically as well as in terms of infrastructure. The facility is now being used to offer MOOCs consultancy support to other agencies in developing and hosting digital content. Other neighboring institutions like MANAGE, PJTSAU and NIPHM have been using this facility for generating their digital output. In this process, this facility has generated revenue of Rs. 21.6 lakhs during 2018-19. This facility is now renamed as Centre for Life Long Learning in Agricultural Education (COLLAge).

Number of dignitaries graced ICAR-NAARM during 2018-19. Some of the lead luminaries who visited the Academy include Shri Gajendra Singh Shekhawat, Union Minister of State for Agriculture & Farmers' Welfare, participated in the interactive Session held at the Academy on March 09, 2019. During the session, Shri Shekhawat emphasized that the government's priority is for making the agriculture profitable by reducing the input cost and providing the remunerative prices for enhancing the farmers' income. The Minister was of the opinion that by listening to the stakeholders could help in enhancing the government's functioning and decision making process. Shri Shekhawat had an interactive meeting with the farmers, farmer associations, scientists of all the ICAR Institutes located at Hyderabad, officials, students, start-ups and State department of Agri-Allied Sector and Universities, etc. The Minister also released a NAARM Hindi publication titled "AgRIM" during the occasion.

Dr. Ashok Dalwai, IAS, Chairman Doubling Farmers Income visited the Academy to inaugurate the 108th batch FOCARS programme. The academy and the probationers had the benefit of listening to his erudite lecture on the lives and successes of great Indian and international agricultural scientists. He motivated the young probationers to stay focused on work and laurels will follow on its own.

The Second 7th Research Advisory Committee, Quinquennial Review Team, Institute Management Committee & Institute Research Committee meeting was organized in the year 2018-19. During the year ICAR-NAARM has had collaborations with various organizations like Maharashtra Hybrid Seeds Company Private Limited (MAHYCO), Jalna, Maharashtra to promote the startups programmes/events organized by MAHYCO for the larger benefit of Indian agriculture. A memorandum of Understanding (MoU) between a-IDEA, NAARM, Hyderabad and Millet Processing & Incubation Centre, Professor Jayashankar Telangana State Agricultural University (PJTSAU), Hyderabad was signed to synergize the institutional capabilities and expertise to create complementing entrepreneurship ecosystem for millet based startups. The MoU was executed on Aug 16, 2018. MoU

a-IDEA, Technology Business Incubator of NAARM won “National Award for Emerging Technology Business Incubator 2017” from Department of Science & Technology, Govt. of India. Dr. Ch. Srinivasa Rao,

Director, NAARM & President, a-IDEA along with Mr. Vijay Nadiminti, COO, aIDEA received the award from Dr. Harsh Vardhan, Hon’ble Union Minister for Science & Technology, Earth Sciences, Environment, Forests and Climate Change at a function held on Technology Day at Vigyan Bhawan, New Delhi on May 11, 2018. The event was graced by Shri Ram Nath Kovind, Hon’ble President of India.

The Academy had participated in International Agricultural and Horti-Expo cum show on Government Achievements and Schemes during July 27-29, 2018 organized by NNS Media Group, New Delhi at Pragati Maidan. The stall was visited by different kinds of stakeholders which included farmers, students, teachers, researchers, entrepreneurs, ministers, government employees and general public. NAARM stall was adjudged ‘Best Stall’ during this 2-day International Expo based on the multidimensional showcase of NAARM’s Education, Extension and Training activities through audio visual as well as verbal presentation.



Chapter - 1
ICAR - NAARM:
An Overview

ICAR - NAARM: An Overview



Vision

A global knowledge institution enabling National Agricultural Research and Education System (NARES) adapt to change through continuous innovation.



Mission

To enhance leadership, governance and innovation capacities of National Agricultural Research and Education System (NARES) through capacity strengthening, education, research, consultancy and policy support.

The National Academy of Agricultural Research Management (NAARM) was established by the Indian Council of Agricultural Research in 1976 at Hyderabad. The major mandate of the Academy is to build capacity in agricultural research, education and extension education systems and provide policy advocacy for the National Agricultural Research and Education System (NARES). To fulfil these mandates, the Academy organizes various capacity building programmes for researchers, academicians, extension personnel, scholars and other stakeholders of NARES. The Academy strives to enhance individual and institutional capacity for innovation in NARES. Considering the strategic importance of agricultural research in food security and economic growth of the country, leadership, governance and innovation are emerging as prerequisite for the transformation of NARES into a more pluralistic innovation system. The Academy strives to generate a sense of fraternity and inculcate a scientific work culture amongst the agriculture scientists and faculty across the country. It plays a catalyzing role to usher planned transformation of the NARES in meeting its current challenges.

The Academy has leveraged its potential of excellent human resource management in the field of agricultural research into agribusiness management, technology management and education technology management to develop and nurture students. The Academy offer higher education programmes viz. Post Graduate

Mandate

- Enhance individual and institutional competencies in managing innovation through capacity development, research and policy advocacy.
- Serve as a think tank for National Agricultural Research and Education System and facilitate strategic management of human capital.

Diploma in Management - Agriculture (PGDMA) having AICTE approval and accreditation from National Board of Accreditation (NBA), Post Graduate Diploma in Technology Management in Agriculture (PGDTMA) and Post Graduate Diploma in Education Technology and Management (PGD-ETM) in distance mode along with University of Hyderabad (UoH). The Technology Business Incubator of the Academy-a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture) has earned laurels for its innovative approach and received national award for nurturing the Agri Startups. The Academy aims to emerge as an institution par excellence to facilitate and support a culture of dynamic management in agricultural research and education. The Academy is also ISO 9001: 2015 certified for Human Resource Development in Agricultural Research and Education System.



ISO 9001:2015 Certificate for Academy

1.1 Strategic framework

In tune with its Vision, Mission and Mandate, the Academy has developed a six-tier strategic framework during 12th Plan to steer different activities to achieve the set goals and targets. Those are

1. Agribusiness Management for Inclusive Growth

- Enhance capacities for food and agribusiness management education and research in NARES

- Enhance capacities for entrepreneurship development in NARES and strategic management of agribusinesses

2. Education System Management for Enriched Learning Quality & Environment

- Enhance capacities for agricultural education policy, planning and evaluation in institutions of NARES
- Enhance capacities for faculty excellence and technology-enhanced learning for increasing learning effectiveness and continuous learning by deploying emerging media to create new and vibrant education environments

3. Enhancing Capacities for Leadership and Governance

- Institutionalize a framework for leadership development at all levels of NARES - early, mid-career, senior professional and managerial across all functions (research, education, extension)
- Institutionalize good governance in NARES by building capacities for effective management of institutions, performance assessment, accountability and developing strong organizational value systems and work culture that promote innovation

4. Extension Systems Management in a Market driven Environment

- Enhance operational, adaptive and generational capacity of front-line agricultural extension systems to address emerging challenges
- Enhance capacities for use of ICTs to provide customized knowledge, skills and solutions to farmers, farmer groups, rural communities and for social networking

5. Information and Communication Management for Promoting Innovation and Governance

- Enhance capacities for integrating ICTs and Internet of Things (IoTs) in institutional management and governance

- Enhance capacities for managing data, information and knowledge in agriculture, scientific communication and institutional governance

6. Mobilizing Science and Technology for Innovation and Sustainable Development

- Enhance capacities for research policy, strategy, priority setting, planning, management, monitoring and evaluation for a more pluralistic NARES, in the emerging contexts of food security, climate change and globalization
- Enhance capacities for technology foresight and strategic management of intellectual property and commercialization of technologies.

1.2 Organization and Management

The academic, research and capacity building programmes of the Academy along the above strategic framework is carried out through six divisions

Agribusiness Management (ABM)

Education Systems Management (ESM)

Extension Systems Management (XSM)

Human Resources Management (HRM)

Information and Communication Management (ICM)

Research Systems Management (RSM)

1.2.1 Research Advisory Committee of Academy

The Indian Council of Agricultural Research (ICAR) constituted the Research Advisory Committee (RAC) for three years from November 24, 2016 to November 23, 2019 as given in Table 1.1

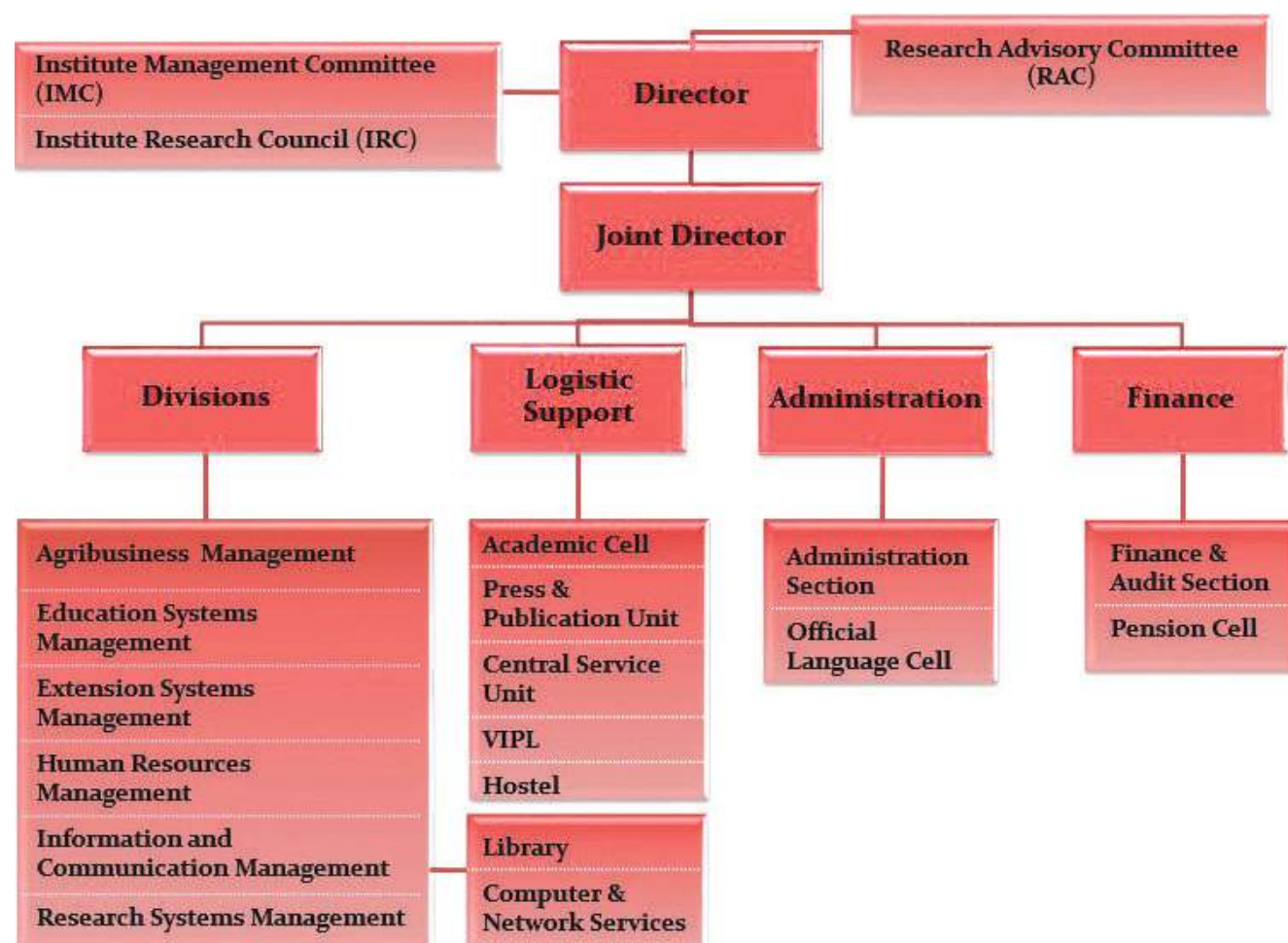


Fig. 1.1: Organogram of the Academy

Table 1.1: Research Advisory Committee of the Academy till November 23, 2019

Padma Shri Prof. S.K. Sopory Ex-Vice Chancellor, JNU, New Delhi	Chairman
Dr. K. C. Bansal Ex- Director ICAR-NBPGR and OSD (NAEP) Agril. Edu. Div., ICAR, New Delhi	Member
Dr. P.K. Joshi Director for South Asia International Food Policy Research Institute (IFPRI), New Delhi	Member
Dr. C. Shambu Prasad Professor Institute of Rural Management Anand, Gujarat	Member
Dr. Ishwar Murthy Professor Decision Science & Information Systems, Bengaluru	Member
Dr. R.K. Samanta Ex-Director ICAR-NAARM, Hyderabad	Member
Dr. Ch. Srinivasa Rao Director, ICAR-NAARM, Hyderabad	Member
Dr. N.S. Rathore DDG (Agric. Education) ICAR, New Delhi	Member
Shri Venkateshwarlu Vasanthapu Telangana	Member
Shri M. Dharma Rao Ex-MLA, Telangana	Member
Dr. S.K. Soam Head, ICM Division, ICAR-NAARM, Hyderabad	Member Secretary



**Members of Research Advisory Committee (RAC)
with Heads of Divisions of NAARM**

1.2.2 Institute Management Committee (IMC)

The following officials were nominated as the members

of the Institute Management Committee of NAARM for a period of three years with effect from September 01, 2016.

Table 1.2: Institute Management Committee of NAARM

Dr Ch. Srinivasa Rao Director, ICAR-NAARM, Hyderabad	Chairman
Addl. Director of Agriculture Department of Agriculture Govt. of Telangana, Hyderabad	Member
The Director of Agriculture Dept. of Agriculture Govt. of Andhra Pradesh Secretariat, Hyderabad	Member
Director (Research) Prof. Jayashankar Telangana State Agricultural University, Rajendranagar, Hyderabad	Member
Sh. Venkateshwarlu Vasanthapu Rajendranagar, Hyderabad, Telangana	Member
Sh. M. Dharma Rao Ex-MLA, Hanuma Konda, Warangal District, Telangana State	Member
Additional Secretary - DARE & Financial Advisor, ICAR or nominee ICAR Krishi Bhawan, New Delhi	Member
Dr A. Vishnu Vardhan Reddy Director Indian Institute of Oilseeds Research Rajendranagar, Hyderabad	Member
Joint Director (Admn.) & Registrar NAARM, Hyderabad	Member Secretary

1.3 Linkages/Networking/Collaborations

NAARM as an Academy has a strong network with 108 ICAR research institutes and 75 Agricultural Universities (AUs), and plays crucial role through capacity building of these institutions, agribusiness industry, scientists, and academia. Its network spans across various other National and International institutions.

1.3.1 National Institutes/Organizations

1. All ICAR institutes and Agricultural Universities.



2. Various departments of the Government of India like NITI (National Institution for Transforming India) Aayog, Department of Science and Technology (DST), Department of Biotechnology (DBT), National Science & Technology Entrepreneurship Development Board (NSTEDB).
 3. University of Hyderabad, Osmania University, Jawaharlal Nehru Technological University (JNTU), International Institute of Information Technology (IIIT), Hyderabad, Council of Scientific & Industrial Research (CSIR), Indian Council of Medical Research (ICMR) and various institutions located at Hyderabad.
 4. MoEF & CC, NBA, PPV&FRA, NFDB and State Biodiversity Boards.
 5. Management institutions located at Hyderabad viz. the Administrative Staff College of India (ASCI), Indian School of Business (ISB), National Institute of Agricultural Extension Management (MANAGE), National Institute of Rural Development and Panchayat Raj (NIRD & PR), and Institute of Public Enterprise (IPE); and Indian Institute of Management (IIMs) across the country.
 6. State Departments of Agriculture of various States (Telangana, Andhra Pradesh).
 7. Private sector organizations (Agri-Input Companies, Service Companies, etc.).
 8. Non-Governmental Organizations (NGOs).
 9. Other organizations viz. National Innovation Foundation (NIF), Federation of Indian Chamber of Commerce and Industry (FICCI), Centre for Innovation Incubation and Entrepreneurship (CIIE).
 10. Krishi Vigyan Kendras (KVKs).
- ### 1.3.2 International Organizations
1. CGIAR Institutions (CIMMYT, ICRISAT, IFPRI, etc.)
 2. World Bank, FAO and other United Nations Organizations.
 3. Department of International Development (DFID), UK.
 4. SAARC Agricultural Centre (SAC).
 5. NARS of South Asian Countries such as Bangladesh Agricultural Research Council (BARC), Nepal Agricultural Research Council (NARC), Pakistan Agricultural Research Council (PARC) and other NARS institutions in Afghanistan, Sri Lanka and other ASEAN countries.
 6. NARS in African and South East Asian countries such as Kenya, Mali, Tanzania, Nigeria, Malawi, Liberia and Philippines.
 7. Leading Land Grant and State Universities of USA and other Universities in Europe, Australia and other developing and developed countries.



Chapter - 2
Thematic Areas:
Research, Capacity
Building & Education

Thematic Areas: Research, Capacity Building & Education

Keeping in view the targets envisaged in the 2050 Vision documents of the Academy and ICAR and the premises of keeping innovation at the center of the debate and discussion, the Academy recognizes the need for developing knowledge-driven agriculture with new research governance models. Recognizing the enabling nature of role of emerging science and technologies across diverse sectors, it voices for their integration into agricultural value chain. Further, the evolving changes in agri-education sector shifting focus from ‘teaching to learning outcome’ signifies the requirement to develop newer institutional mechanisms and capacities of the system. The Academy also identified the crucial need for a continuous stakeholder engagement to usher effective and efficient performance in the NARES with more transparency and accountability systems in place. Therefore, the entire activities of research, capacity building and education have been aligned along with six-tier strategy approach. Through a structured deployment of this strategy, it is envisaged that the Academy can play an effective role in the planned transformation process of the NARES.

The subsequent sections present the research projects, capacity building programmes and the conferences/workshops conducted by the Academy under the six strategic areas which are already mentioned in the previous chapter.

2.1. Agribusiness Management for Inclusive Growth

With the shift in agricultural development strategies from a pure production-oriented approach to a broader systems perspective, the importance of the agribusiness sector has grown significantly. Seen as an engine for growth, agribusiness and its related industries are receiving increased attention in policies and strategies that aim to promote investments in agro-

enterprises and agro-based value chain development. This has prompted a need for deeper understanding of agribusinesses, agro-industries and agri-food value chains. The theme focuses on agribusiness management for inclusive growth through research, education and consultation activities for developing future leader in the field of agribusiness. It broadly aims to:

- Enhance capacities for food and agribusiness management education and research in NARES
- Enhance capacities for entrepreneurship development in NARES and strategic management of agribusinesses

Rapid urbanization, rising disposable income of households, expanding and diversifying food demand and globalization of agricultural economy have given impetus to research and capacity building in the area of Agribusiness Management. In order to meet the broader objectives of the theme, the Academy focused on the following thematic areas for research, capacity building and education:

2.1.1. Agribusiness environment: With major focus on value chain analysis for agri-commodities and services; consumers demands; price behavior and forecast analysis.

2.1.2. Public policy and impact: With focus on market reforms like e-NAM, GST, etc.; financial inclusion and its spillover; spread and convergence of crop insurance.

2.1.3. Markets and institutional innovation: Market integration of commodities; producer companies and FPOs; inputs and service delivery systems; diversification and intensification of agriculture for high value commodities; business analytics.

2.1.4. Nurturing future leaders in agribusiness: Through imparting quality education in agribusiness management.



2.1.1 Agribusiness Environment

The agribusiness sector, which comprises the business activities performed from farm to fork, is a major generator of employment and income worldwide. In contrast to the dynamism evident in other sectors, investment in Indian agriculture and agribusiness-public and private, domestic and foreign-has been low until recently, and has shown little growth. Private agribusiness investment depends on state and central regulatory policies and business environment including marketing, interstate movement, storage and taxation policies. Prospects for agricultural investment hinge largely on the environment for private investment by farmers and agribusinesses. Key factors likely to shape the climate for private investment by farmers and agribusinesses include the extent to which domestic demand for agricultural products make investment potentially profitable; a range of government regulatory, credit, tax, and other policies that affect incentives for domestic and foreign investment in Indian agriculture; and infrastructure constraints.

Mainstreaming the Small Ruminant Farmers into Efficient Value Chains in India

Small ruminant production systems are the only means of livelihood for landless and resource poor farmers in many parts of the country. They are climate resilient animals, suitable for arid and semi-arid regions. The sector is comparatively a neglected one in terms of required support from public agencies. It needs focused approach with higher investment in community approach and establishing meat processing plant and developing adequate market linkages. Goat rearing has tremendous potential in increasing the income of poor households and thereby helping farmers towards achieving doubling farmer's income. Lack of pastures and fodder shrubs and absence of farmers' group / organizations in linking them to markets, besides availing inputs and veterinary services are the major constraints.

It is in this context, the present study has been taken up with the following objectives, (i) to enumerate the socio-economic profile of the small ruminant farmers

and carry out livelihood analysis; (ii) to study their production and marketing practices with respect to small ruminants; (iii) to identify their constraints in small ruminant value chains in the study area; and (iv) to suggest strategies and policies for small ruminant farmers to increase their income through efficient value chains.

During the period under report, the study areas were finalized. In case of sheep, Telangana was chosen, and the districts selected for the primary survey were Mahabubnagar and Nalgonda, based on the population status as per 19th Livestock Census, 2012. About 100 sheep farmers from each district would be surveyed with the help of structured interview schedule. Besides, a sample size of five value chain actors in each district would also be interviewed to know their activities, the purchase and sale prices at each level in the market channel. The survey form and check-list have been prepared, and would be put to pre-testing in the coming months, before actual survey.

Adoption of Insect Resistance Management Practices in Bt- Cotton Cultivation in India

India tops in area under the cotton crop (12.2 million hectares) as well as its production (6.7 million tonnes or 37 million bales of 170 kg each) in the world. Besides, India exported 1.25 million tonnes of cotton in 2017-18. However, various reports since 2014 have highlighted the outbreak of pink bollworm, termed as PBW (*Pectinophora gossypiella*) on Bt cotton crops in states namely, Gujarat, Madhya Pradesh, Maharashtra, Karnataka and Andhra Pradesh. The current study attempted to find out the adoption level of different insect resistance management (IRM) practices to control pest infestation and their effect on cotton production and farm income of the cotton growers. The survey of 611 cotton growers was taken up in three major cotton growing states of India namely, Gujarat, Maharashtra and Karnataka. Total 11 farm practices were considered under Insect Resistance Management (IRM) practices. Farms adopting four or less number of practices, five to eight practices and above eight practices were categorized as 'Low', 'Medium' and 'High' adopter, respectively. It was found that the

important IRM practices like growing refuge crops and use of pheromone traps were hardly adopted by the farmers. There is a heavy reliance on cotton stubble for cooking purpose and hence they were reluctant to destroy it after the harvest. Instead, they cut the stubbles and store them either on the boundary of their field or near their home, mostly on the thatched roof

of animal shelter. Some of the farmers, mostly with larger crop area, leave the stubbles in the field itself, till the next season starts. According to the farmers' perception pink bollworm is the most predominant pests with high infestation level (around 80 per cent) across the states in the last two years.



Fig. 2.1: Farmers awareness and adoption of various IRM practices (% of total)

Table 2.1: Cost and return of cotton in the study area (₹/hectare)

Particulars	Gujarat			Maharashtra			Karnataka		
	Low	Med	High	Low	Med	High	Low	Med	High
IRM level	Low	Med	High	Low	Med	High	Low	Med	High
Total CoC	35,198	39,176	42,744	33,644	37,427	40,308	40,457	45,511	46,284
Yield (q/ha)	16.3	20.4	24.7	11.3	13.7	16.4	21.3	22.8	23.5
Avg. Price (₹/q)	4,318	4,486	4,473	3,988	4,011	4,026	3,384	4,022	3,985
Gross Revenue	70,168	91,520	1,10,598	45,162	54,957	65,921	71,905	91,793	93,542
Net Revenue	34,970	52,344	67,854	11,518	17,529	25,612	31,448	46,283	47,258

* CoC-Cost of Cultivation

Average seed cotton yield was estimated to be 24.75, 16.5 and 23.5 q/ha in the states of Gujarat, Maharashtra and Karnataka, respectively. Farmers belonging to 'High' IRM adoption level are able to get a higher yield

in all the selected states. It was found that higher the adoption of IRM practices, the more returns one gets across the states. The survey of ginning mills revealed that even though it is maintained in a good and clean



condition, there is possibility of maintaining continuity in the life cycle of pests especially pink bollworm due to storage and handling of infested seed cotton and not maintaining pheromone traps. The study recommends i) Implementation of Refuge-in-Bag system to be ensured immediately, ii) Developing short video clips in vernacular languages and distributing to the farmers through the seed dealers, iii) Simplification of IRM practices for the farmers so that farmers do not need to follow more than 3 activities under IRM, iv) Free distribution of pheromone trap through seed and fertilizer dealers in the cotton growing area, and v) Convincing the farmers for not going for long duration crop or stop picking after January-end.

2.1.2 Public Policy and Impact

Farmer Producer Companies in India: A Study on their Management Practices and Business Potential

In India, about 85 per cent of the farmers are small having less than 2 hectares of land. These farmers have low participation in wholesale markets or commodity markets and they sell their produce through village merchants who charges a high margin. Further, they have very poor bargain power both while purchasing the input as well as while disposing the output. Their risk bearing ability is also very low. This call for an intervention, which can increase their income and help them realize better value for their goods. One such intervention is Farmer Producer Company (FPC), which has been projected as an institutional arrangement to solve the problems of the small producers. Small Farmers' Agribusiness Consortium (SFAC), National Bank for Agriculture and Rural Development (NABARD) and RKVY-RAFTAR scheme have been actively pushing this intervention in all areas spread across the country. However, there have been questions raised on the management and sustainability of FPCs which need to be answered through a detailed scientific study.

The specific objectives of the study are (i) To study the growth and performance (financial and membership) of selected FPCs, (ii) To examine ongoing business and identify the business opportunities of selected

FPCs, (iii) To estimate the gains for the members after joining FPC and the externalities in the rural economy, and (iv) To understand the critical success factors and suggest possible changes in strategies for promoting the producer companies.

Through extensive review of various literatures, the critical success factors were identified for further testing in the field. They are broadly classified as market-related factors, participant/stakeholder related, strategy related, budget related, production related and R&D. Some examples of specified factors within the broad classification are built-in trust of the company among its producer members, providing crop and animal advisories, appointment of professional graduates, distribution of inputs at wholesale rates, reaping benefits from various schemes of Government and transparent system.

Collaboration with SAU has been initiated without financial obligation from different regions. TNAU Coimbatore for southern regions, JNKVV Jabalpur for central India, VNMKV, Parbhani for western region were approached for collaboration. These centres have already initiated project on FPCs in their respective areas.

Financial Inclusion of Smallholders and Landless Households in Madhya Pradesh

The study was carried out with the objectives of assessing the extent and level of financial inclusion for smallholders and landless households, estimating the plausible income gains to them with better financial inclusion, finding the determinants of financial inclusion in the study area, and suggesting strategies for enhancing the financial inclusion of smallholders and landless households. Two districts of Madhya Pradesh were selected for the study. The data for Jabalpur were collected during previous year, whereas for Morena which falls under low financial inclusion were collected during the current year and were analyzed. Data from 116 respondents have been collected from Morena district from cluster of 2 to 3 villages which were 10km, 15 km and 25 km from the district headquarter.



Fig. 2.2: Districts of Madhya Pradesh selected for the study

The salient findings of the study are given below:

- a) As per the secondary sources, one district falls under low financial inclusion category (Morena) whereas the other, under high financial inclusion category (Jabalpur). But, the study shows that the landless and smallholders in both the districts are having same characteristics and fall under low financial category.

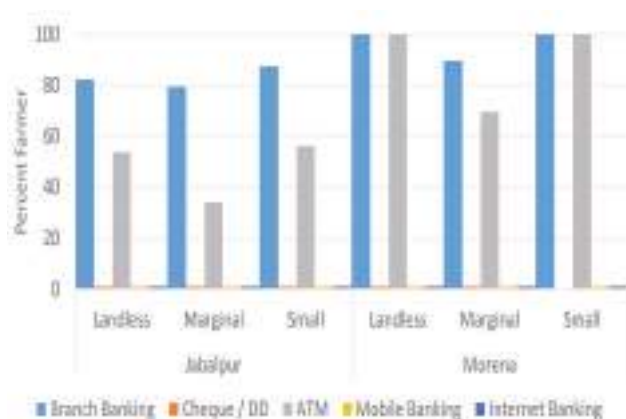


Fig. 2.3: Land holding wise use of bank services

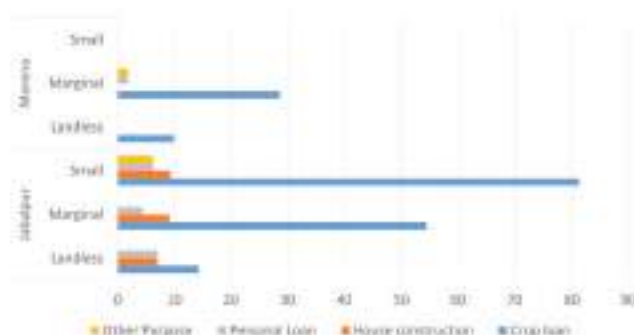


Fig. 2.4: Land holding wise type of loan availed

- b) Number of bank branches in Morena is very less (87) than Jabalpur (304). This could be one of the reasons for many people without having any account in the bank.
- c) Public sector banks such as State Bank of India (SBI) collectively lead in market penetration. Less than 15% population have account in Post Office, Co-operative and Regional rural banks.
- d) Population which availed loan is very less in both the districts and among them crop loan is maximum. This is because of *Kisan Credit Card* (KCC). No one was found to have availed loans like education loan, livestock purchase loan in both the districts.
- e) Despite having less branches in Morena, deposits in the form of Fixed Deposit (FD) and Recurring Deposit (RD) are higher (>90%) among all categories of farmers.

Adoption of Natural Farming and its Effect on Crop Yield and Farmers' Livelihood in India

The NAARM-CRIDA collaborative study on 'Adoption of Natural Farming and its Effect on Crop Yield and Farmers' Livelihood in India' has been sponsored by NITI Aayog to understand the perception of the farmers about Natural Farming, its adoption pattern and its impact on farm income. It is also envisaged to examine its sustainability and scalability among the farmers. Natural farming

(also called as Zero Budget Natural Farming or ZBNF) is considered to be significantly different from organic farming, however the farm produce remains organic in both the cases. The Natural Farming (NF) is agro-ecological based farming, under which rearing of indigenous cow is the pivotal factor. It is recommended that a farmer having one indigenous cow can cultivate five acre lands without purchasing any chemical fertilizers or pesticides from the market. The practices of Natural Farming include the addition of microbial cultures to enhance decomposition and nutrient recycling; use of local seeds; integration of crops, trees and livestock (mainly cows of native breeds); effective spacing of crops, contouring and bunds to conserve water; intensive mulching; extensive intercropping and crop rotations. There are four essential components of Natural Farming: a) *Jeevamritha*, a fermented microbial culture consisting of fresh cow dung, cow urine, jiggery, pulses flour and handful of bund soil, all mixed in water. The mixture is ready as nutrients after 48 hours to spray twice a month in the crop.

The 200 litres of *Jeevamritha* is contemplated to be sufficient for one acre of land; b) *Beejamrutha* for seed treatment prepared by cow dung and urine; c) *Acchadana - Mulching* for maintaining soil moisture; and d) *Whapsa* - irrigating the field only at noon, in alternate furrows.

Under the study, farmers survey was conducted in three states: Andhra Pradesh, Karnataka and Maharashtra in those districts having higher proportion of farmers adopting Natural Farming. Information about crop choice and cultivation practices were collected from NF-adopters and non-adopters in the same village to estimate the difference between the cost and profitability. Soil and plants samples were collected from the adopter and non-adopter farmers' fields to examine the microbial activity and soil nutrient status, and nutrients uptake by the plants.

Developing Business Proposal for Farmer Producer Companies

A training programme on 'Developing Business Proposal for Farmer Producer Companies' was



First meeting of the project team of "Natural Farming" conducted at ICAR-NAARM

organized in the Academy during August 06-10, 2018. The training was attended by 27 participants representing ICAR institutes, FPOs and agricultural universities from 12 states. The training programme was aimed to build an integrated view and enhance the knowledge base to cater to the agri-business management challenges in the current scenario. The sessions included business proposal development, quantitative methods for analyzing the financial viability of businesses, developing marketing plan, project risk management and winning pitch presentation. The participants also visited Suraksha Producer Company Ltd, Siddipet and interacted with the executives and farmer members.



Interaction of trainees during the program

Impact Assessment of Agricultural Research and Technologies

A Training Programme on, “Impact Assessment of Agricultural Research and Technologies” was held at ICAR - National Academy for Agricultural Research Management (NAARM), Hyderabad during December 04-07, 2018. The programme was conducted to orient the participants to the concepts and description of impact assessment of agricultural research; to acquaint the participants with techniques/tools and methodological approaches of impact assessment of agricultural research; and to discuss empirical studies and to synthesize experiences in application of Impact Assessment methods in the domain of agricultural research and technologies. A total of 16 participants, comprising 8 from ICAR institutions, 7 from SAUs and one from NITI Aayog participated in the training

programme. The programme consisted of a blend of lectures, case analysis, discussions and hands-on exercises.



Impact Assessment of Agricultural Research and Technologies programme

Entrepreneurship Development Programme for Student Entrepreneurs

a-IDEA, the Technology Business Incubator of ICAR-NAARM with support of Caspian Impact Investment (CII) took this initiative to catch at early stage the college students who have some interests in entrepreneurship. Both the organization i.e. a-IDEA and CII have been conducting similar programmes since last year and successfully organized Startup Samvaads at 3 locations, viz. Guwahati, Assam; Raipur, Chhattisgarh; and US Nagar, Uttarakhand. The Entrepreneurship Development Programme (EDP) for the Students was aimed to shape the students’ business ideas having a potential to scale-up in agribusiness sector. Besides, it was also envisaged to scout for those students who have presented his/her ideas at different forums. Caspian Impact Investment (CII) has kindly agreed to support the initiative. The 5 days programme was a mix of sessions by the domain experts, business game, activity, field visits to CCMB and T-Hub. Three of our successful startups also interacted with them. The programme covered all aspects of the agribusiness and agri-preneurship, like: Overview of agri-business eco-system in India, Product Development and Customer validation, Marketing of products, Team building, Essentials of communication, Business Law, etc.

Entrepreneurship Skill Development Programme for Agri-Graduates of ANGRAU

ICAR-NAARM has imparted a first-of-its-kind training programme to orient young minds towards entrepreneurship through a 21-days intensive residential training programme on “Entrepreneurship Skill Development Programme for Agri-graduates of ANGRAU” at NAARM. This programme was specifically designed for agri-graduates who are willing to take up entrepreneurship in the field of agribusiness. Forty-five B.Sc. Agriculture / B. Tech Agricultural Engineering students from five colleges of Acharya N.G. Ranga Agricultural University, namely, Agricultural College, Mahanandi; Agricultural College, Naira; Agricultural College, Bapatla; Dr NTR College of Agricultural Engineering, Bapatla; S.V. Agricultural College, Tirupati had taken part in the training programme. These participants were selected based on their inclination towards entrepreneurship and their past academic performances. This programme was intended to sharpen the entrepreneurial skills which is useful for achieving success in one’s life. The objectives were:

- To promote and nurture entrepreneurship abilities in agricultural students, and
- To equip them with management tools needed for making successful entrepreneurs

This programme has a right blend of teaching methods including lectures, hands-on practices, exposure visits to institutions and industry, experiential sharing sessions by entrepreneurs and group assignments. These budding entrepreneurs had visited two markets—one dealing mostly with vegetables and flowers and the other dealing with grains. They interacted with various players in the market and followed the farmers to the villages and studied their cultivation and marketing pattern. They also took part actively in the club activities of PGDMA where quiz, Just-a-Minute, sports and other similar activities were regularly conducted in the evenings.

a-IDEA, a facility at the Academy which is incubating several entrepreneurs working on agribusiness, conducted interactive sessions and hands-on sessions on business idea generation, business plan preparation and Pricing and business negotiation. Video conferencing facilities were also used in the programme to bring face-to-face interaction with entrepreneurs based in faraway locations.



Students with NAARM Faculty

Founder of Sahaja Aharam, Founder of Amay Biotech, HR of Coromandel, CEO of NeML, Senior Advisor of Caspian Capital, VP of e-Fresh, Co-founder, Kalgudi, Director of Ingrain Tech. and Former MD & CEO of Advanta Ltd., have shared their experiences and motivated these young minds to succeed in their endeavors.

Socio-Economic Impact Assessment of Research Programmes

This training was envisaged to impart knowledge related to impact assessment and hands-on practice in special statistical tools used in impact evaluation. The objectives were

- To equip the researchers with the conceptual framework of impact assessment.
- To acquaint them with the basics of statistics and have hands-on exercises in statistical software and
- To have interactive module of development of framework, identification of criteria and indicators and identifying the right tool of impact assessment.



The participants were from ICAR research institutes, KVK and faculty who are executing research projects including Farmer FIRST. The statistical software “Stata” was introduced to the participants and all practical exercises were demonstrated using the software. The topics covered were Overview of Impact Assessment (IA), Impact assessment framework for capacity building programme, conducting baseline study, case study and focus group discussion; scale construction and testing; criteria and indicators for IA of research projects; tools of IA, etc. The participants were provided with training manual, soft copies of course materials including presentations, codes and datasets for practicing. Greater emphasis was given to develop framework for IA for different types of projects. The participants were provided various topics related to IA which they prepared well and presented the same in group. The innovative approach in the programme was to ask the participants to bring data and cases from their projects and during hands-on sessions, they made use of their data to perform statistical analysis and interpret the data.

2.1.3 Markets and Institutional Innovation

Structural Changes in Agricultural Marketing in India: Effect on Inclusiveness, Efficiency & Profitability

According to Small Farmers Agribusiness Consortium (SFAC), the Electronic National Agricultural Market (e-NAM) implementing agency under Ministry of Agriculture & Farmers' Welfare, 585 APMC mandis across 16 States and 2 Union Territories have been integrated on e-NAM platform as on 30th March, 2019. The e-NAM platform trades in more than 114 commodities. The present study has been undertaken to examine the readiness of the APMC Mandis for implementation of e-NAM, awareness of different stakeholders about e-NAM, level of farmers' participation and estimate the benefits from this structural change in selected market. Current year, survey has been conducted in one mandi each in Madhya Pradesh, Maharashtra and Telangana state. The transaction data from e-NAM platform was also

collected for these states. The e-NAM transaction data revealed that several perishable fruits and vegetables are also being traded on e-NAM in these states. In many cases, traded prices are biased towards higher volume of the commodities by the farmers. Similarly, peak price time is away from the harvest time and due to lack of holding capacity of smallholders, the benefits of rise in price during off-season are cornered by the traders. The study suggested a new concept of Smart Micro-Mandi (SMM), which envisages combination of e-NAM platform, strength of JAM (Jan Dhan account, Aadhar & Mobile) and application of modern technologies like sophisticated camera & sensors, artificial intelligence (AI) and Internet of Thing (IoT). The proposed Smart Micro-Mandi (SMM) includes five strategies:

- a) Milkization of agricultural commodities:** after assaying & grading, pooling together similar produce (like liquid milk market) of all the farmers in the catchment area;
- b) Dematerialization of agricultural commodities:** converting the agricultural produce into small units like demat and crediting units into the respective farmer's virtual account;
- c) Instant partial payment based on warehouse receipt:** seamless integration of farmers' bank account with financial institution to make instant partial advance payment to the farmers against the produce deposited in the warehouse, and;
- d) End-to-end digitalization:** use of optical scanner & sensor, IoT, cloud computing, artificial intelligence, online bidding, etc. to capture all the information on real-time basis to all the stakeholders.
- e) Proximity to the farmers:** SMM is proposed to be set up in the radius of 5-6 km, thus catering to an area of about 100 sq. km. It would attract all the farmers to participate in the formal market in expectation of better price discovery. Having the catchment area of 100 sq. km. and excluding the districts with very low population density (<100 persons/sq. km.), there is need of 28,200 Smart Micro-Mandis in India.

The conceptual framework of the SMM would be like:



Fig. 2.5: Architecture and workflow of Smart Micro-Mandi

The SMM would function in nine stages (1) All the uniquely registered farmers in catchment area with 2-3 major crops would arrive at SMM with their produce. 2) The produce passes through quality check. Initially with maximum 3-4 quality standards (moisture, grain size, broken grain & admixture), the produce is categorised in Grade-I, II or III. 3) The produce of all the farmers are pooled together across these three grades (*milkization*). 4) Then, the entire lot of each farmer is standardized into small units, say each kg of grain is equivalent to one Unit. Accordingly, farmer's virtual demat account is credited with number of units (*dematerialization*). 5) The deposits of the produce would trigger the financial institutions to credit the partial payment to farmer's account, say the value of 1/3rd of the produce deposited in warehouse by each farmer. This would be helpful for the farmer to meet the immediate cash need. 6) Individual farmer may also stagger the sale as per his/her requirement and market condition. Using the advanced computing methods and sophisticated algorithms, the detailed information about the quantity of the produce under different grades would be adjusted and reflected on the dashboard. 7) The bidders from any part of the country may bid for the particular grade of the produce. The information of the highest bidder in each SMM would be sent to the warehouse of the respective SMM. 8) With the help of third-party logistics already registered with each SMM, the quantity of the produce would

be dispatched to the winner-bidder, after deposit of the amount. 9) The final payment to the farmers would be made according to the proportionate share of each farmer in each day's sale proceeds after adjusting the initial advance payment made. The details of the proposed Smart Micro-Mandi (SMM) can be accessed at https://naarm.org.in/wp-content/uploads/2018/09/E_Policy_Brief_SMM04082018.pdf.

2.1.4 Nurturing Future Leaders in Agribusiness

Rising Startups of Agri-entrepreneurs: A success story

In this study, success stories of ten Agri Startups of agricultural graduates were recorded and compiled. Out of 10 agripreneurs, two were women. The startups are distributed in the sectors of food processing, agri inputs, dairy & veterinary, supply chain & collateral management.



Fig. 2.6: Agri Startups by Agricultural graduates in different sectors

These startups have come up with innovative ideas such as strip based checking of milk adulteration, efficient soil & plant management, novel pet products, etc., and are making their presence felt in market in a gradual manner. All the startups acknowledge the role of Technology Business Incubators for leveraging the support specially in mentoring and fund generation. Their expectation from ICAR/SAUs is for providing technical support, popularizing entrepreneurship and in branding of the products.



Fig. 2.7: Flow chart depicting the type of sector of Agri Startups and their impact

Post Graduate Diploma in Management-Agriculture (PGDMA)

The Academy started 2-year fulltime AICTE-approved agribusiness management education programme PGDMA in the year 2009. The aim was to help the students build solid academic background, strong characters and leadership skills to meet the demand for employers ranging from the agricultural business, financial institutes to the public services. The 34 students of 9th batch of PGDMA (2017-19) successfully completed their final project and were awarded with provisional course completion certificate. During the academic year (2018-19), 24 companies of which 16 visited for the first time covering various sectors of agribusiness. The academy proved as one of the premier B-schools of the country for providing agribusiness management

professionals with a record of 100% placement for PGDMA students. The placement process witnessed participation of companies cutting across the major sectors of agribusiness: Input (Seeds, Fertilizers and Pesticides), Microfinance, Commodities Exchange, Food Retail, Crop Advisory, Market Research and Consultancy Services.



Fig. 2.9: Prestigious recruiters for 9th Batch of PGDMA

For 10th batch of PGDMA, 28 students took admission, out of which 12 are girls and 16 are boys. These students are from Agricultural Universities with different backgrounds of agriculture, horticulture, food technologies, agricultural engineering, agri business management etc. and all students got the summer placement.

Admission process of 11th batch of PGDMA also started in December 2018 by inviting applications from the graduating students from agricultural universities.



Fig. 2.8: Profile of the 10th batch of PGDMA



Fig. 2.10: The organizations which offered summer placements



Ms. Sheesham Rana, PGDMA 2015-17 and Ms. Naushaba Khatoun, PGDMA 2016-18 receiving Gold Medals

ICAR-NAARM Celebrated 3rd Graduation Ceremony of PGDMA

The Academy organized 3rd Graduation Ceremony on October 6, 2018 for awarding the diploma for the students of Post Graduate Diploma in Management-Agriculture (PGDMA). During the ceremony, 45 students of PGDMA received the Diploma certificate in person and 10 students were awarded certificate in absentia. Two students from each batch of PGDMA were also awarded Gold and Silver medals for the best academic performance. Sh. V. Vijay Kumar, MD & CEO, National Commodity & Derivative Exchange (NCDEX) was the Chief Guest. He urged all the graduating students to give their best for the organizational and societal goal and create credible impact. The Guest of Honour Sh. K.S. Raju, Chairman, Nagarjuna Group advised the young students to be prepared to accept different kinds of challenges, for which they need to think out of the box. Dr Ch. Srinivasa Rao, Director of the Academy informed that the Academy is transforming towards a leadership role in policy guidance in all spheres of agricultural research management. Agribusiness being the pivot element in accelerating the income of the farmers, all the young students should serve the country with zeal and compassion.

SANKALP' 18: 3rd Annual B-Fest

SANKALP, an annual business fest with an aim to provide a platform to young management graduates to develop organizing skill, to foster inter-learning and have an opportunity to network with some of the Agri-Startups and industry representatives. The Sankalp'18



received 2281 registrations from 300+B-schools including IIMs and IITs. Total nine competitive events were organized viz. Advitiya-Business Plan competition; Adhishtatha-The Best Manager; Analytica-about data analysis and interpretation; Manthan-Real-time Business Solutions; Samadhan-Case Study Competition; Darpan-Quizzing event; Ad-Mania-Ad Making Event; Flashgun-Capture the moment; and Marketing Catapult-Prototype Marketing. Apart from these events, a Workshop series called "DANCE"- Decoding Agri-Startup: Nurturing & Creating Entrepreneurs was held in which five Agri-Start Ups shared their journey experiences with the participants. After initial screening, 150 students in total were invited for the fest. All the events had attractive prizes for the winners ranging from ₹5K - 20K each. The Sankalp'18 received sponsorship from several agribusiness companies.



Winners of the event receiving the prize from Director, NAARM

Reunirse'18 - PDGMA Alumni Meet

PGDMA Alumni Meet Reunirse'18 was organized

at the Academy. More than 100 Alumni attended the meet and shared their experiences. Dr Ranjit Kumar, President of the Alumni Association welcomed all the alumni and congratulated those who received award and recognition in their sectors in the last one year. The alumni presented the sectoral views to sensitize the in-campus students to make them aware about the changing dynamics of various sectors. During the meeting, new website for alumni (<https://alumni.naarm.org.in/>) was launched by Dr Ch Srinivasa Rao, Chief Patron of Alumni and Director, NAARM and Dr S.K. Soam, Patron & Joint Director. New executive committee was also elected in the General Body Meeting. For details about PGDMA, visit <https://naarm.org.in/pgdma/>



Reunirse'18 Alumni Meet

2.2 Education System Management for Enriched Learning Quality & Environment

This theme is one of the core themes of the Academy, which essentially aims to:

1. Enhance capacities for agricultural education policy, planning and evaluation in institutions of NARES
2. Enhance capacities for faculty excellence and technology-enhanced learning for increasing learning effectiveness and continuous learning by deploying emerging media to create new and vibrant education environments.

In order to advance the objectives of this broad theme, the Academy focused on the following areas for its research, academic and capacity building activities:

2.2.1 Lifelong Learning for Professional Development

2.2.2 Education beyond Higher Educational Institutions

2.2.3 Technology Enhanced Learning

2.2.4 Policy Advocacy on Education Systems Management

2.2.1 Lifelong Learning for Professional Development

Professional development and competency enhancement is a continuous phenomenon for any organization for achieving the par excellence in the respective domains. The major thrust of the academy is providing the opportunities for lifelong learning for National Agricultural Research and Education System. In this endeavor, multipronged strategies are adopted. Processes and methodologies are envisioned to enhance the quality of agricultural education in the country. A unique initiative in this direction is competency enhancement of faculty through Foundation Courses and need based programmes. The outcomes, feedback from participants backed up by research studies formed the basis for designing the courses for lifelong learning.

Assessment of Technological Interventions in Teaching-Learning Process

The study was undertaken to assess the existing practices being followed by select SAUs where expand cells were established and the perception of teachers and students in use of educational technology for their teaching and learning environment. The main objectives of the study is (i) to study faculty perceptions on technology experiences and expectations through ET, (ii) to study the student's overall attributes towards technology adoption in learning process and (iii) to identify the constraints and to propose suitable strategies for effective functioning of ET cells. The study conducted in TNAU, Coimbatore reported utilization pattern of ET resources by the students. Prominent among them are use of e-learning platform (45%), access to online journals (20%), multimedia software (17%), computer lab with all gadgets (13%) and statistical software (5%). The perception of students on the benefits of technology-enhanced

courses include peer communication (87.7%), access to grades (84.8%), more convenient to use (83.3%), increases students' awareness on ET (79%), and access to course material (76.8%). The challenges in technology adoption for teaching learning process include faster obsolescence of technology (76%), lack of manpower support in instructional lab (72%), less time to develop lessons (69%), students scheduling for use of technology (68%) and constant change in the teaching style due to new technology development (56%). The administrators suggested more budget allocation for equipment and other facilities besides regular faculty capacity building programme on ET for quality education in agriculture.

Foundation Course for Faculty of Agricultural Universities (FOCAU)

The 6th Foundation Training for 49 faculty members & scientists of Acharya N.G. Ranga Agricultural University (ANGRAU) was organized at the Academy.



Sri V.V. Lakshminarayana, Former Addl. DGP, Govt. of Maharashtra releasing the souvenir

The course was primarily intended at intensive capacity building of the faculty with focus on hands-on learning with conceptual and practical orientation. The programme stressed on areas like National Agricultural Research, Education and Extension Systems; Teaching - Learning Management; Research Project Management; IP Informatics; Human Resource Management; Extension and Training Management; Administration and Finance Management; Personality including Health Management, and Communication Skills. There were 63 interactive sessions, 10 hands-on sessions, eight participant activities along with four institutional visits. The innovative efforts introduced

in the programme were: webinars, skype interactions, video conferencing with the national and international agriculture experts. Sri V.V. Lakshminarayana, Former Addl. DGP, Govt. of Maharashtra was the chief guest for the valedictory ceremony.

Training Programme on Agricultural Research and Education Management

Induction Course for newly Recruited Assistant Professors of PJTSAU was conducted in three phases with total 183 participants with 88 male and 95 female participants. The participants were trained in research, education and teaching methodologies, given exposure in Technology Assessment and Refinement (TAR) and Front Line Demonstrations (FLDs) and of Govt. of India efforts in Agri-entrepreneurship programmes for industrial skill development.



Trainee participants with NAARM Faculty

Digital Teaching Evaluation in Agricultural Education using TPACK Model

The study assesses the extent of overall teaching dimensions in the faculty in respect of proficiency in teaching content, pedagogy and technology using an established TPACK (Technology Pedagogy and Content Knowledge) Model. The specific objectives of the project : (i) to assess the digital teaching aptitude among teachers using established ET model(s) and (ii) to identify the issues and possible solutions to address the digital teaching for implementation in AUs. The existing instrument for using TPACK is studied and suitably modified to suit the educational scenario in agricultural universities.



Students' Learning Approaches (SLAs) for Enhanced Learning Outcomes in Agriculture Education: A Critical Analysis

Quality of education vis-a-vis learning has become a major issue in the Indian Higher Education. Several studies emphasized on the deterioration of quality of education vis a vis learning. An applied research project was taken up with the objectives (i) to analyze Students' Learning Approaches (SLAs) among agriculture and allied sciences for enhanced learning outcomes; (ii) to analyze the changes in students' learning approaches over time; (iii) to investigate the factors affecting students' learning approaches for effective learning outcomes; and to develop strategies for enhancing the quality of learning outcomes in agriculture education. An instrument ASSIST (developed by Marton and Säljö (1976) and revised by Entwistle (1997)) is modified through pre-testing. Further, a schedule has been developed to for identifying the factors affecting learning.

AgAcademy: A National Virtual Academy for Indian Agriculture

AgAcademy is a cognitive edtech platform with a vision to create a National Agricultural Platform that enables educators to create content and engage with learners. Basically it enables educators to develop courses with a Mobile App, and can be shared with different stakeholders. AgAcademy vision is to get the best minds from the country to share their knowledge in an easy-to-understand form. The study aims to develop an i) Artificial Intelligence driven framework for scaling up of electronic learning platform for Indian Agriculture; ii) the robust cloud based platform for hosting the courses developed by end users; iii) proper strategic, marketing and communication methodologies for popularizing the developed platform; iv) learning path for individual and customized learning of each and every end user; v) financially sustainable platform and vi) assess adaptation and impact of the platform among stakeholders. The technologies required for meeting the objectives were identified.

Competency Analysis among Faculty of Agricultural Universities in Technology Enhanced Learning

The study is undertaken with the objectives to study

the institutional practices in support of Technology Enhanced Learning (TEL), to understand faculty competency about their technology experiences and expectations for faculty development, to identify the constraints and training needs of faculty in adoption of technology enhanced learning and evolve suitable interventions. The expected outcomes include changed pedagogy and technology integration skills among faculty and approaches evolved for enhancing the quality education at institutional level.

2.2.2 Education beyond Higher Education Institutions

The endeavor to enhance the quality of agricultural education and providing access and equity to all stakeholders, the Academy took the path of reaching out the unreached with the help of technologies; thus contributing to redefining the educational process.

Massive Open Online Course on Dynamics of Teaching and Learning

The Academy organised a Massive Open Online Course (MOOC) on **Dynamics of Teaching-Learning** from November 1-30, 2018, 4th in the series. The course attracted registrations from 1194 participants. Majority of the participants were from agriculture and veterinary sciences apart from other non-agriculture disciplines such as engineering, management, and arts and science. The main objective of the course is to make the participants understand various dimensions of teaching management and develop effective teaching styles for quality education. During the course, 15 video modules with an aggregate duration of 4 hours 12 minutes were provided along with 111 pages of reading material and 216 power point slides of various topics. The participants reported 15% gain in knowledge, as evident from pre-knowledge test scores and final knowledge test scores. The average score in online pre-knowledge test (administered before the start) was 50.0% which was enhanced to 65.0% by the end of course (after one month). About 482 participants (40.36%) became eligible for either successful completion / participation certificates. This MOOC has generated Rs. 3, 65,000 income from certification charges.

Digital Teaching Techniques

Training programme on Digital Teaching Techniques to enhance pedagogical methods with digital technology and to sensitize on digital learning management was conducted in the Academy and off campus at University of Agricultural Sciences, Bengaluru and generated revenue of Rs. 5.46 lakhs.



Off Campus training programme organized by ICAR-NARM at UAS, Bengaluru

2.2.3 Technology Enhanced Learning

Enriched learning quality in Agricultural Education necessitates integration of technology in teaching-learning process. Several technological advancements are bringing significant changes in the society and education sector is no exception. In this direction, the Academy focused on building capacities of NARES on technological integration with sound research base. The Academy's mission on technology enhanced learning stems from earlier experience on eLearning.



Release of Digital course content developed under TELAgE Project by Dr Trilochan Mohapatra, Secretary DARE & DG, ICAR

Technology Enhanced Learning in Agricultural Education

The Technology Enhanced Learning processes could consolidate the earlier outcomes in developing digital content and evolving a self-supportive model. TEL also helped realize EFC objective of establishing Centre for Lifelong Learning in Agricultural Education (COLLAge). Revenue Generation Model, Mobile based learning and digital course development in interactive mode were the major outcomes, besides offering MOOC for the fourth time with over 40% course completion rate. Technologically, the Chroma-key application was extensively used for developing digital content. The facility has resulted in unique digital content like short films for the first time.

Major Activities of project include:

- New academic program of the Academy - one year PG Diploma in Educational Technology Management (PGD-ETM) in distance mode is offered in one of its kind online mode with the support of TELAgE Resource centre support. The content is produced in TELAgE lab while the same is launched through eLearning portal which was established under TELAgE. Eighty modules have been prepared in digital mode for online access.

- Digital content development in consultancy mode was offered for NIPHM, MANAGE, PJTSAU for developing content on Plant Biosecurity, Agri-warehousing and library information retrieval respectively, with a total revenue generation of Rs. 21.6 lakhs.
- Digital courses in interactive mode which has video & text material with offline access. Courses on “Pests of crops and their Management” and “Veterinary Physiology” are completed.
- MOOC on Dynamics of Teaching & Learning was offered with course completion rate of 40% and revenue generation of 3.6 lakhs.

Digital certificate with QR code generation is done for MOOCs. The MOOC offered during last four years has demonstrated that technology can ease the processes of learning and teaching including certification. A comparative analysis of the 4 MOOCs is shown in Table.

Table 2.2: Comparative analysis of 4 MOOCs

S. No.	Activity	MOOC-I	MOOC-II	MOOC-III	MOOC-IV
1	Course title	Competency Enhancement Through Micro Teaching Methodology	Competency Enhancement for Effective Teaching	Teaching Management	Dynamics of Teaching-Learning
2	Users	318	1006	982	1192
3	Completion	64	352	421	267
4	Participation	198	181	95	215
5	Certification	98	529	516	482
6	Key features	➤ Online mode ➤ Evaluation instruments Online discussion and online assignment ➤ Access and Certification: free	➤ Online mode ➤ Evaluation instruments ➤ Online discussion, assignment and quiz ➤ Free Access & Certification fee Rs. 500	➤ Online certification generation with QR Code ➤ Performance based certification ➤ Free Access & Certification fee Rs. 1000	➤ Performance based certification with QR code ➤ Free Access & Certification fee Rs. 1000



**MOOCs Introductory session by
Dr D. Thammi Raju, Programme Director**



**Digital content development for different sponsored
programmes**



2.2.4 Policy Advocacy on Education Systems Management

The Academy is actively engaged in the development of policies focusing on the major theme-Education System Management for enriched learning quality & environment regularly. The Academy develops policy briefs by conducting workshops and brainstorming sessions.

National Agricultural Higher Education Project initiated at NAARM

The Technical Committee Meeting-cum-Workshop for the project “**Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education**” under National Agricultural Higher Education Project (NAHEP) was organized. The primary objective of this meeting-cum- workshop was to make the stakeholders and faculty members of NAARM aware about the project and its modalities. Apart from NAARM faculty members, Deans, Directors and Professors of PJTSAU, PVNRTVU, SKLTSHU and ANGRAU attended the meeting. Dr V.P. Sharma, Expert Member and Former Director of MANAGE and Dr Prabhat Kumar, National Coordinator, NAHEP Component 2A also participated in the meeting. It was emphasized that technical committee of the project may start their work with establishment of Career Development Centres at Agricultural universities and Faculty Development Centre at any established university. It was suggested to conduct a study on inbreeding of faculty members at SAUs. It was emphasized that the project’s priority should be student centric, followed by faculty and infrastructure development.

Stakeholders Workshop on Restructuring of PG Curriculum and Syllabus in Community Sciences

Stakeholders Workshop on Restructuring of PG Curriculum and Syllabus Broad Subject Matter Area (BSMA) in Community Sciences prescribed by Indian Council of Agricultural Research was organized by the Academy during 3-4 October, 2018. Sixty participants including faculty members and recruiters from all the five disciplines of Community Sciences viz., Foods &

Nutrition, Human Development and Family Studies, Clothing and Textiles, Family Resource Management and Home Science Extension Education, attended it. Dr Sharda Former Dean, College of Home Science, was the Chairperson of BSMA and Dr Neelam Grewal (Member, Punjab Public Service Commission) was the Convener. DDG (Education), Dr N. S. Rathore, DDG (Education) was the Chief Guest for the valedictory function held on October 4, 2018. This two days’ workshop deliberated on restructuring of the syllabus of Community Sciences with the changing paradigm in order to make it more vocational in nature.

2.3 Enhancing Capacities for Leadership and Governance

Human Resource Management (HRM) division has made their efforts to initiate and infuse competency-based management research system based on strategic objectives for training, performance management and teaching competency framework for achieving organizational goals. Human Resource Management has been contributing significantly to the development of competencies and capacities of scientists and research leaders in the ICAR and Agricultural Universities. In addition to specific research, the HRM division undertake the capacity building of the members of NARES to enhance the capacities for leadership and governance in NARES, update the NARES scientists in the relevant and contemporary issues of human resource management and equip them to deal with stress and emotional intelligence. Human Resource Management focuses on organizational change for promoting innovation through research consortia-change management, Development of methodology for self-assessment, Assessment and development of organizational citizenship behavior for promoting efficiency and effectiveness in NARS. The division is also involved in collection of the baseline data on the understanding about teamwork among ICAR employees, to identify the factors helping and hindering teamwork in ICAR and preparation of case studies of successful teamwork in ICAR. The strategic framework of the thematic area is to enhance capacities for leadership and governance by-

- Institutionalizing a framework for leadership development at all levels of NARES - early, mid-career, senior professional and manager across all functions (research, education, extension).
- Institutionalizing good governance in NARES by building capacities for effective management of institutions, performance assessment, accountability and developing strong organizational value systems and work culture that promote innovation.

To realize the above objectives, the following activities related to research and capacity building programmes were carried out during the year 2018-19:

Emotional Intelligence and Organizational Effectiveness

Emotional Intelligence (EI) has been reported to play a significant role across the workplace having an essential component in determining the leadership effectiveness mainly when leaders are dealing with teams in the workplace. EI leaders inspire team members to work efficiently in order to achieve organizational goal. There has been paucity of study on relations between EI and its positive impact on the organizational effectiveness especially in the scientific organization like ICAR of NARES. Hence, the present investigation was carried out with the following objectives:

- To measure the emotional intelligence of scientific personnel in NARES
- To assess the organizational effectiveness of scientific personnel viz., team work, conflict resolution and leadership effectiveness
- To find out the relationship between emotional intelligence and organizational effectiveness in NARES
- To suggest strategies to improve organizational effectiveness in NARES

During the reporting period (2018-19), psychometric data was collected from a total of 118 scientific personnel regarding their emotional intelligence, team effectiveness (Teamwork skill questionnaire by

University of South Australia), conflict resolution modes (Thomas-Kilmann conflict mode instrument, 2010) and different dimensions of their leadership effectiveness (HRD Press Inc.).

Team Work and its Effect on Organizational Performance in ICAR

The present investigation was carried out with the following objectives:

- To identify the level of teamwork skills and knowledge on teamwork among the stake holders in NARES
- To identify the effects of Teamwork on the organizational performance in the NARES
- To identify the factors effecting teamwork in the organizations

During the reported period (2018-19), data was collected from a total of 853 number of scientific personnel regarding their teamwork skills (Teamwork skill questionnaire by University of South Australia) which indicated that 62.8% respondents are having below average teamwork skills, 34.8% are having average teamwork skills and only 2.4% of respondents have good teamwork skills.

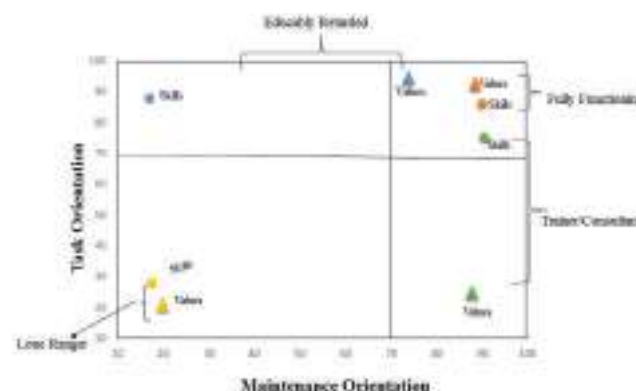


Fig. 2.11: Team work and its effect on organizational performance in ICAR

Regarding the Team orientation behavior inventory (Goldstein *et. al.* 1983), about 100 responses were collected during the period and from the total of 382 respondents, only 225 were having fully functioning team member profile, 75 were having educable retarded profile, 21 were having lone ranger profile while 61 were having trainer consultant profile.



Assessment & Development of Competency Framework for Agricultural Research and Extension Scientists

Project on “Assessment & Development of Competency Framework for Agricultural Research and Extension Scientists” was undertaken with twin objectives to identify the functional and behavioural competencies and to develop a competency framework for scientific cadre for agricultural research and extension. Data were collected from 160 respondents for the relevancy, importance and level of competency on the identified functional and behavioural competencies. Organized a workshop at Indian Institute of Natural Resins and Gums, Ranchi involving scientists from four ICAR institutes located at Ranchi. Held discussions and deliberations to identify attributes of different competencies at Scientist, Senior Scientist and Principal Scientist levels. The major competencies identified were Discipline Orientation, Research attributes (Cognitive), Research Methodologies, Project Management & Governance Skills, Research Communication Skills and Research Application and Extension Attributes under the category of functional competencies. The behavioural competencies included Managing Relationships, Leadership and Networking.

HR Interventions and their Influence on Institutional Innovations

The study was undertaken with the objectives (i) to identify key HR interventions for institutional innovation in ICAR, (ii) to analyze the link among HR interventions, institutional innovation, and (iii) to develop suitable strategies to nurture HR interventions for institutional innovation.

Innovation is indispensable for scientific organizations. Human Resource (HR) can support in the sense of building and buttressing the capabilities, behaviour and culture which can drive innovation. In an era where knowledge is rapidly changing, innovation is critical to sustain and success. The human capital of organization is an issue of increasing importance. Therefore, we need to understand how they can be supported and encouraged to boost innovation and creativity. This means a need to engage with some major aspects of

workplace culture and employment relations which can have a major impact on innovation.

Management Development Programme on Leadership Development

These programmes are planned to catalyze succession planning in National Agricultural Research and Education Systems (NARES). Participants includes Heads of Divisions and other senior functionaries who are in the pre-RMP cadre of the organization and who are responsible for leading various activities at their level. The objectives of this program are:

- To help participants understand their own personality dimensions
- To explore leadership challenges and core competencies needed in present day context
- To develop understanding and skills in research and technology management, information and knowledge management and administration and finance management
- To adapt ideas of diverse skills for sharpening capacities to effectively

During the reporting period, two such programmes were conducted. A total of 52 participants (47 male & 5 female) attended these programmes. Heads of the Divisions and other senior functionaries, who are in the pre-Research Management positions (pre-RMPs) in the National Agricultural Research and Education System (NARES) play a very important bridge role in the organization and they take up leadership responsibilities in many activities that they perform. The programme provided an opportunity for the pre-RMPs, who are in the “middle zone”, to develop smooth and gradual transformation to step into future leadership position. The sessions covered the topics on Leadership and personality development, Research and technology management, Information and knowledge management, Administration, finance, and vigilance management.

Management Development Programme for HRD Nodal Officers of ICAR for Effective Implementation of Training Functions

A programme for HRD Nodal Officers of ICAR for effective implementation of training functions was

conducted with total of 23 participants (19 male & 4 female) with the objective:

- Enhancing competencies of the HRD Nodal Officers to provide support and guidance in the implementation of capacity building programmes and training functions.

The sessions covered during the program were Systematic and Competency Approach to Training, Training Need Analysis (TNA) of employees of all categories, Design and development of training programme, Development and Implementation of Cadre Training Plan and Annual Training Plan for all categories of employees, Evaluation and Monitoring of trainings at the Institute level, Developing training schedule/calendar for the Institute.

The above topics were covered by different pedagogical approaches consisting of presentations, video presentations, interactive discussions, etc. Finally, different practical ways and means of developing leadership characteristics it was suggested in order to enhance one's own personal and organizational effectiveness. Apart from class room sessions, regular early morning yoga sessions for participants were conducted.



MDP for HRD Nodal Officers of ICAR session in progress

Need Based Programs

These Programs are organized for mid-career managerial, scientific, technical and administrative personnel of National Agricultural Research and Education System. They are designed for imparting advanced knowledge and skills in the human resource management based on express demand from different

stakeholders in NAREs like Agricultural Universities, Government Departments and ICAR institutes in enhancing the efficiency of their employees.

A training programme on '**Stress Management**' was conducted with total of 17 participants including 15 men and 2 women scientists from ICAR and SAU's. Majority participants were of Principal Scientist cadre and some were HODs and Directors of ICAR Institutes. The training program facilitated the participants to develop in-depth understanding of personal and psychological causes of stress/anxiety and its effects - physical, social and organizational behaviour. The participants were made to understand the mechanism of occurrence of various stress related disorders - cardiac, hypertension, diabetes and other psycho-somatic diseases. Imparted various techniques to develop constructive means for coping up with distress and achieve productive and enriched life-style.



Stress Management programme

A training program for **Coromandel International Collaborative Capability Building Programme for Young Agribusiness Professionals** was conducted in two phases. The objective of the program is to provide the basic understanding of crop production and crop protection practices, and to upgrade the knowledge and skill in sales and marketing domain. In the second phase, techniques for improving business management skills and to develop personality for taking higher responsibilities in sales and marketing were imparted.

The participants were exposed to present agricultural scenario, fundamentals of production and protection practices for various field and horticultural crops, FABs

of Coromandel Product Portfolio, Organic fertilizers, commercial demonstrations and farmers meetings, value added selling skills. The behavioral aspects such as personality development, time management and emotional management were also covered during training programme. In the second phase, the topics covered were agronomy aspects of major field crops, balanced nutrients landscape, rural marketing, agribusiness and marketing plan, competitor analysis, channel management, market research, field survey, farmers buying behavior and customer feedback and GIS applications. Participants learned target setting, delegation and creativity, inter-team interactions through experiential learning and advanced analytics through hands-on exercises.

A training programme for Agronomists of Coromandel International was also organized to expose them to current trends in agriculture, soil health and nutrition management and to provide knowledge in Coromandel product portfolio. The topics covered were agricultural scenario, basics of plant nutrition and other soft skill management.



Young Agribusiness Professionals of Coromandel International with NAARM Faculty

A program for Retail Store Managers in 5 phases was conducted under **Coromandel Retail Learning Academy Collaborative Skill Enhancement and Employee Development (SEED) Programme** with the objective to impart the required competencies to agri-input retail store managers for taking up higher responsibilities.



Dr S.K. Soam, Joint Director I/c interacting with participants

The programme was designed to provide knowledge and skill in technical, operational and behavioral domains. The modules covered were agricultural policies, developments in agronomic and crop protection practices, supply chain management, customer focused business, competitor analysis, business strategy, business analytics, market research and business communication. The behavioral aspects included positive thinking, personality development and team work.

The off campus **Faculty Development Programme for Improving Efficiency and Effectiveness in Teaching, Research and Extension** at University of Agricultural and Horticultural Sciences, Shivamogga, Karnataka was conducted. The main objectives were: (i) To reflect on and renew the knowledge and skills of the faculty of the University for improving the quality of higher education and human resources management (ii) To impart knowledge and skill in teaching, research and extension. Areas of deliberations during the training programme included Student Centred Education Technology, Effective Teaching-Learning, Evaluation and testing, Positive thinking, Stress Management, Personality Development for Leadership, Motivation, Interpersonal Relationship and Role Perception, Scientific Communication, ICTs in Extension, Research Project Prioritization, IPR and Technology Management.

A program for Faculty members of ICAR-Central Institute of Fisheries Education, Mumbai was conducted on **Emotional Intelligence for Personal**

and Work Excellence. The aim of the training was to:

- Facilitate participants to understand their own self and personality dimensions
- Enable participants to learn and develop emotional intelligence
- Strengthen the emotional orientation of participants for personal and work excellence

A total of 27 faculty (22 male & 5 female) attended the

programme. The participants were introduced to the topics such as overview of types of intelligence (IQ, EQ and SQ), assessment and interpretation of EQ and SQ, effectiveness to greatness: an interactive discussion, assessment and interpretation of organization effectiveness traits, Emotional Intelligence for work performance, layers of personality and self awareness, personal and work excellence through self-actualization, relationship management, stress management and positive thinking.



Program on Emotional Intelligence at ICAR-CIFE

2.4 Extension Systems Management in a Market driven Environment

This theme is one of the core themes of the Academy, which essentially aims to:

- Enhance operational, adaptive and generational capacity of front-line agricultural extension systems to address emerging challenges.
- Enhance capacities for use of ICTs to provide customized knowledge, skills and solutions to farmers, farmer groups, rural communities and for social networking.

In order to advance the objectives of this broad theme, the Academy focused on the following broad areas for its research, academic and capacity building activities:

- 2.4.1 Extension policy, planning and management;
- 2.4.2 Extension information systems;
- 2.4.3 Gender mainstreaming in extension

2.4.1 Extension Policy, Planning and Management

The extension system, which carried the bandwagon of green revolution on its shoulders during the 70s and 80s, is at crossroads. There has been a demand for re-examining the architecture of Agricultural Extension research, so that scientists, administrators and entrepreneurs can together help farmers to bridge the gap between yields that are possible and those that are actually achieved. A strong, vibrant and responsive extension research system can play a pivotal role in enabling the country to achieve the plan objective of faster, sustainable and more inclusive agricultural growth.

Research Role Analysis of Extension Scientists in ICAR

Framework for the study has been conceptualized. Item analysis of 25 Extension research themes have been reviewed and factor loadings have been worked out and presented in three factors by Varimax rotation: item analysis has been done and judges rating on a 10-point continuum of importance and relevance has

been elicited from 21 out of 92 extension scientists of ICAR and the 25 statements scored as above. The coefficient of concordant W revealed a very good agreement among the judges. The reliability was computed by split half method with r value of 0.94 and the Varimax rotation has yielded three factors to be named as extension research roles.

Management and Impact Assessment of Farmer FIRST Projects

Conducted the following need based capacity building programmes at ICAR-NAARM under the project:

1. Innovative Practices in Extension Research and Evaluation and
 2. Socio- Economic Impact Assessment of Research Programmes.
- ❖ Compiled a book on farmers' innovations, farm innovations and success stories across FFP implementing centres.
 - ❖ Identified feasible and successful technologies for implementation at various FFP centres.
 - ❖ ADOPT (Adoption and Diffusion Outcome Prediction Tool) is being administered for assessing the attributes and spillover of technology for scaling up.



Field visit to FFP implemented by IISS, Bhopal

6th Management Development Programme for Newly Recruited Programme Coordinators of Krishi Vigyan Kendras.

The MDP for newly recruited Programme Coordinators of KVKs was organized in three

phases. The programme registered a total participation of 29 heads of KVKs representing 8 ATARI zones of the country. The modules covered in the programme were organization and management aspects of KVKs, personality development, communication skills, time management, positive thinking, media management, statistical tools and impact assessment methodologies.



MDP for Newly Recruited Programme Coordinators of Krishi Vigyan Kendra

2.4.2 Extension Information Systems

Agricultural extension is the primary delivery system for information to farmers. Without relevant and reliable agricultural information, improvement in agricultural production may not be achieved. Extension organizations have the important role of linking farmers to information on improved agricultural technologies. One of the prime ingredients for rural development in developing countries is information access. Although the Information and Communication Technology (ICT) revolution in these countries has gained momentum, most of the farming communities still have no access to value added information. Academy focus on the agricultural research on extension information systems to enhance their knowledge by increased stakeholder participation.

ICAR Winter School on “ICT interventions for Agricultural Development”

The 21-days winter school was conducted to sensitize entry and middle level NARS professionals about the concept of ICT and its use in Agricultural Development, to enhance the competency of NARS professionals in using various tools and techniques of ICTs in agriculture and to impart appropriate skills for ICT use and its application in agricultural development. Participants were from nine states in the cadre of Assistant/Associate Professors hailing from 12 SAUs and two ICAR institutes. Topics on ICTs right from Community Radio to Drones, web designing, blogs, online education, expert systems, mobile apps, IOT and sensors in agriculture, Social media, online tools for data collection, online bidding, flipped classrooms *etc.* were covered. Displaying of practical applications of their ICT technologies were by Indian Blog Writer’s Association, Skykrafts, Kalgudi, AEGIS Consulting Services and Mobitech. The innovation during the program was development of website and demonstration of IOT in Agriculture & Drones by the participants.



Demonstration of Drone by the participants during the programme



Certificate distribution to participants by Director, NAARM

2.4.3 Gender Mainstreaming in Extension

Women are the main farmers or producers, but their roles remain unrecognized. Despite the necessity of utilization of untapped potential of men and women, failure to address gender based differences bring about disparities in development outcomes. Thus, the Academy concentrated towards gender mainstreaming.

Gender based ICT led information management in agriculture

Gender based digital divide needs to be addressed by studying gender related issues for reducing the inequalities and enhancing the access to use of ICTs as both producers and consumers of knowledge. In view of self-help groups, farmers’ organizations engaged in production and marketing in agriculture, it is necessary to understand gender participation and knowledge management. There is a need for ‘Pro-active gender sensitive ICT strategy’ to build the capacities of the rural women to enjoy the fruits of development, sustain livelihoods, and improve their quality of life. During the project, a Web based Geo-Spatial Village Information System (GSVIS) Interactive Maps through Web Interface for S. Lingotam and Tallasingaram was developed and 141 Soil Health cards were linked to the cadastral maps. Database on “Gender disaggregated ICTs use in agriculture” was developed, the first mobile application from NAARM in collaboration with CDAC was developed for prioritizing the agricultural problems namely, “m PRIORITIZE”. The project output 309 Vernacular webpages and 286 vernacular modules on major crops and animal

husbandry developed for information sharing, four Vernacular Video modules on Homestead products were developed for facilitating entrepreneurial development and one Vernacular video module was developed on agriculture schemes for awareness creation on doubling farmer's income.



m-PRIORITIZE Mobile application developed by NAARM

ICTs for empowering farm women

With the onset of new advances in information technology, especially in computing and communications, the support resources created in digital media are gaining ground especially in technology transfer. The ICTs have become very effective in agriculture information sharing and agribusiness. Development of multimedia-based web environment facilitates effective communication that support decision-making. There is, therefore, a need to have a dialogue among all the stakeholders who are involved in technology transfer as well as agribusiness on strategies to empower the farming community – especially the farm women to contribute their might in food and nutritional security and improving and sustaining livelihoods. Considering these needs, a Management Development Programme was organized to provide exposure, enhance knowledge and skills in Information and Communication Technologies (ICTs) for farm women empowerment. The program was conducted with 18 participants including 11 men and 7 women participants. The objectives were:

(i) to sensitize on prevalent ICT models using traditional and modern ICT tools in agriculture and their scope for gender sensitive information sharing and Agri business, (ii) to identify the role of pluralistic media and skill development in agriculture information sharing and entrepreneurial development for farm women, and (iii) to evolve strategies on sharing of experiences on ICTs use for empowering farm women in agriculture.

Gender sensitive ICT models and their scope for information sharing and agribusiness, role of pluralistic media including traditional/ mainstream media and modern digital media in agriculture information sharing and entrepreneurial development for farm women, m-extension and e-extension evolving strategies for empowering farm women in agriculture was discussed.



Sensitization of ICTs to farm women

2.5 Information and Communication Management for Promoting Innovation and Governance

This thematic area covers the objectives such as enhancing the research capacity building and consultancy in the areas of Information Technology based decision support systems; Geographical Information Systems in agricultural management; ICT applications for village knowledge centers, Development of digital multimedia and e-learning resources for agriculture; Geospatial knowledge discovery and management; ICTs in agri supply chains; ICT policy for NARS; Emerging information technology s ad systems; ICTs in participatory

technology development and transfer; Designing e-learning systems and Distance training,

In order to advance the objectives of the broad theme, the Academy focused in the following broad areas for its research, academic and capacity building activities:

2.5.1 Application of GIS, GPS and remote sensing technologies

2.5.2 Development of Information systems and Knowledge Management

2.5.3 Statistical computing and data analytics

2.5.4 Integration of emerging ICT Technologies

2.5.1 Applications of GIS, GPS and remote sensing technologies

Geo-Spatial Knowledge System for Monitoring Climate Change in Agro Climatic Zone-X : Southern Plateau and Hills region

Climate variability significantly impact the agricultural growth, stress, cropping pattern, phenophase and its vulnerability. The main objective of the study was to

find the status of vegetation at village level. It plays an important role to assist policy makers as a decision support feature. Satellite derived data sets were used to study the time series trend of NDVI (Normalized Difference Vegetation Index), NDWI (Normalized Difference Water Index) and NDDI (Normalized Difference Drought Index) for Tallasingaram village, Nalgonda district, Telangana state (NAARM adopted village).

Identifying, delineating and mapping land cover is important for village level monitoring studies, resource management and planning activities. Identification of land cover establishes the baseline from which monitoring activities can be performed. This data set provides land use and land cover classification products at 30-m resolution at village level for a particular cropping session. The data were derived from Landsat 8 and visual interpretation through google earth platform. The data were classified according to the Classification & Regression Tree (CART) algorithm.

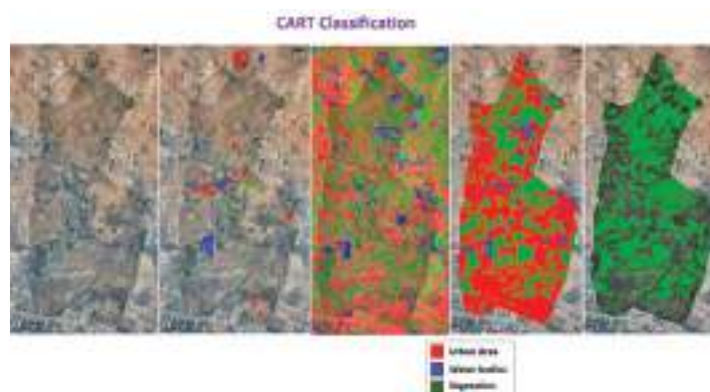


Fig. 2.12: Data classified as per Classification & Regression Tree (CART) algorithm

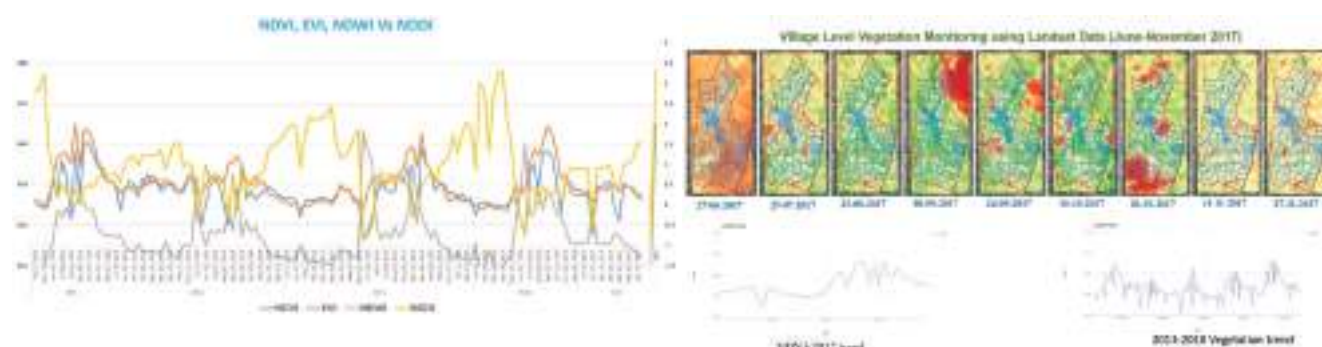


Fig. 2.13: Result of time series NDVI analysis

The study uses Landsat 8 with 30 m resolution 8 days interval images from 2013-2017. The results shows that, a temporal curve that summarizes the various stages that green vegetation undergoes during a complete growing season. This information helps the planners and decision makers to evaluate performance of vegetation growth at village level, in turn assess the crop insurance claims by the farmers. Enhanced Vegetation Index (EVI) was also developed as an alternative vegetation index to address some of the limitations of the NDVI.

2.5.2 Development of Information systems and Knowledge Management

Text Analytics of the Queries raised at Kisan Call Centres in South India

In order to harness the potential of ICT in Agriculture, Ministry of Agriculture and Farmer Welfare has launched the “Kisan Call Centres (KCCs)” scheme on January 21, 2004. Under this scheme, farmers can call to Kisan Call Centre and enquire about various queries/problems related to agriculture and allied sectors. The Farm Tele Advisors (FTAs) answer the call and simultaneously record the query details which has made available a large amount of text data. This project is an attempt to study such data with the objectives: i) to identify the trending queries raised at Kisan Call Centres located in South India, and ii) to group the districts of South India based on the queries raised at Kisan Call Centres. As a part of the study, text analysis was carried out for the Nizamabad district of Telangana state for the agriculture year 2016-17, considering the queries raised in each of the months as text corpora. A total of 1,384 queries were asked by the farmers out of which the highest number of queries were related to weather (47.0%) and plant protection (25.9%), together accounting for 72.8% of the total queries raised. The word cloud (Fig. 2.14) clearly indicates

that the most number of queries were related to ‘weather’, followed by ‘management’, ‘paddy’, ‘weed’, ‘leaf’ and ‘borer’. This underlines the importance of timely communication of weather information to the farmers for timely operations, cost minimization in terms of postponing input application or plant protection operations etc., besides market information for selling the farm produce.



Fig. 2.14: Word Cloud indicating maximum no. of queries

Economic Analysis of Eucalyptus and Subabul based Agroforestry Ecosystem in Telangana and Andhra Pradesh

The study sponsored by M/s. ITC Ltd., was conducted with the objectives (i) To work out the profitability of cultivation of selected trees as sole crop and under agroforestry system, and (ii) To assess the resource use efficiency of various inputs used under different systems of cultivation of trees. Pilot survey and collecting secondary information about agroforestry and tree plantation in selected areas were completed and interview schedule is prepared and collection of data is in progress.



Visit to ITC Khammam and Bhadrachalam Project for Survey

Decision Support System of Maize Inbred Germplasm

The decision support system on maize germplasm was developed based on the technical aspects and various symptoms. The system was hosted at <http://wnciimr.org>. DSS serves the needs of the focused breeder by cutting down the time required for initiating breeding/research programme, preliminary selection of inbred lines to develop prospective hybrids. The key elements of the DSS are identification of the traits/characters that are needed for selection of inbred lines; there are 14 such traits housed in the DSS. The traits which were recorded in the field are supplemented with the images of the tassel and ears of the inbred lines as they form the primary source of information for short listing of lines. The DSS is supplemented with information on the pedigree of the germplasm lines. The strength of the DSS lies in its potential to allow a permutation and combination of the traits and collating the information and displaying the same along with the images. It has a provision for having the information such as Growing Degree Days (GDD) of the inbred lines.

As on date, the database has been updated with the information of over 400 maize inbred lines with full pedigree information. Additional feature of “Image Library” is also enabled to support the breeders in their decision support. The database is being updated with additional features viz. Growing Degree Days (GDD), year and season of evaluation, heterotic group, special traits, etc.

Establishment of Skill Development in Bioinformatics Infrastructure Facility (SKILL-BIF) on Agricultural Biotechnology at NAARM under the BTISnet Programme

The aim of SKILL-BIF centers is teaching of biology and biotechnology through bioinformatics. The scheme is designed to expose scientists, teachers, FOCARS trainers and students to the use of bioinformatics in solving hard core biological problems. The Skill-BIF facility primarily provides educational and training facilities by supporting the bioinformatics and computational needs of the various research and academic programmes conducted in the areas of agricultural biotechnology. The center has provision for use of lecture materials, video clippings, demonstrations, tutorials and online facilities for teaching. It periodically organizes hands-on trainings on various aspects of application of bioinformatics tools and molecular techniques in agricultural research. The facility primarily provides educational and training facilities by supporting the bioinformatics and computational needs of the various research and academic programmes conducted in the areas of agriculture and allied sectors. It will be useful for the biological databases to solve various problems and mysteries to uncover the wealth of hidden biological information and enhance the standard of life for mankind.



ICAR Research Data Repository for Knowledge Management (KRISHI)

ICAR has come up with a policy for research data management in 2014 to streamline and strengthen collection, storage and reuse data. The implementation of this policy has been entrusted to the Academy along with ICAR-IASRI, ICAR-IARI and ICAR-NBSSLUP with ICAR-IASRI being the lead center. The implementation involves creation of necessary information systems and develop relevant policies and initiate activities across ICAR institutes. As a part of this initiative, a portal, KRISHI (<https://krishi.icar.gov.in>) was created with six repositories namely, experimental, observational, survey data repositories, technology repository, publication repository and Geo-portal for spatial data repository and visualization.

Technology Repository

During the year, the technology repository was updated based on the feedback from various ICAR institutes and has been opened for submission by all ICAR institutes. A detailed work flow in consultation with IPR cell has been put in place before submission. Before submission, technologies which can go public and technologies which have to be protected are identified first at the institute level and accordingly, the details of the technology are shown. Few details such as brief uses of the technology are shown for technologies having commercial potential to attract potential buyer. The technology repository holds more than 400 technologies at present in open access. The repository was formally opened during the 3rd National Workshop held during December 4-5, 2018 by Secretary, DARE and DG ICAR.

Data Repository

The information system developed for AICRPs by the Academy has now been used by Sorghum, Maize and Vegetable Crops. The same will be used by all varietal improvement AICRPs. During the year, website of AICRP on Farm Implementation and Machinery was hosted at ICAR Data Center as the part of the project activity. The website

provides information about the AICRP, technologies developed, audio and video gallery.

Audio/Video/App Gallery

As a part of the consolidating the digital resources developed by ICAR institutes, Audio and video Gallery was developed. The gallery provides a convenient way to search different audio/video resources of ICAR institutes. At present more than 1340 videos are available which can be searched, browsed through key words, institute-wise. The video gallery is accessible through URL, <https://krishi.icar.gov.in/KRISHISite/Video/> and audio gallery through <https://krishi.icar.gov.in/KRISHISite/Audio>.

Dashboard

To monitor various stages of progress under a project, a centralized dashboard has been created (<https://krishi.icar.gov.in/KRISHISite/KRISHIreport.jsp>). The dashboard provides both tabular and visual reports. The reports are available at different levels such as SMD-wise, Institute-wise, repository-wise, etc. The dashboard is developed using open source software.

Policy Support

Data management policy of ICAR released in 2014 delved around a managing research data for long-term storage and retrieval. Quality of research data is important, particularly when the data is to be stored and reused for other purposes. A quality data acquisition guidelines was prepared with inputs from different subject matter specialists and submitted to the council. The guidelines suggested sweeping reforms in data collection and storage. Some of the suggestions involve geo-tagging, standard operating procedures, periodical calibrations of equipment, electronic data captures through digital devices. The draft was presented to Senior Officers Committee of ICAR. After deliberations, it was decided to bring a comprehensive policy document incorporating data management policy, IPR policy, Open Access Policy, Data licensing policy to bring uniformity and clarity. This policy document is in the advance stage of completion.

2.5.3 Statistical Computing and Data Analytics

Training Programme on “**Developing Winning Research Proposals**” was conducted in NAARM as well as in off campus mode at Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya (RVSKVV), Gwalior, Madhya Pradesh; National Institute of Food Technology Entrepreneurship and Management (NIFTEM), Sonapat, Haryana; University of Agricultural Sciences, Bangalore; College of Agriculture, Indore and Sharda University, Greater NOIDA. A total 169 (117 male and 52 female) participants attended these meetings. The broad objectives was to give practice in writing various components of a research proposal, explain the use of log frame, priority setting, and programme evaluation and review techniques in research programme planning and to develop a good project design and estimate that is rationally accepted.



DWRP program participants

Participatory Rural Appraisal (PRA) and Communication Strategies

The sponsored program by Indira Gandhi National Forest Academy (IGNFA) was designed for 87 IFS probationers. The training comprised of hands-on modules on various PRA tools, mobile-based digital communication techniques, use of analytical hierarchy process (AHP) for prioritization of decisions and an experiential learning module on panel discussion. The probationers were introduced to the emerging issues in natural resource management with respect to climate change.



IFS Probationers during Participatory Rural Appraisal (PRA) and Communication Strategies training program

Competence Enhancement Programme on Soft Skills and Personality Development for Technical Staff of ICAR

A competency enhancement programme on soft skills and personality development for technical staff of ICAR was held at Academy, exclusively for the technical staff of ICAR. The program aimed to enhance the soft skills and personality of participants. Twenty-one participants from different ICAR institutions across the country participated in the program.



Technical Staff of ICAR participating in training

The program consisted of both theoretical and practical aspects on various topics. The trainees were exposed to Communication skills, Presentation skills, Interpersonal relationships at work, Team work, Personality development, time management, inter-team interaction, Official language policies and Hindi word processing, MS Word, Power Point, Excel, Office communication & drafting, ICAR - ERP etc.

Training programmes on “Koha for Library staff of ICAR”

Academy conducted the programmes with the objectives: to make participants understand the application of Koha in library automation, to provide hands on training on various modules of Koha related to acquisition, cataloguing, OPAC, etc., and to provide the knowledge of broad aspects of Koha installation and customization as per institute requirements. The program was designed to cover both theoretical and practical aspects on various topics. The trainees were exposed to introduction to linux, installation of Ubuntu operating system, installation and configuration of Koha and hands on training by our internal faculty. Introduction to Koha modules, administration, acquisition, patron management, cataloguing, circulation, serials, OPAC and hands on training were covered by the guest faculty Mr. Indranil Das Gupta who is a “Member, Koha Community Release Team”, “Database Documentation Manager, Koha Community Project”.



Participants of training on “Koha for ICAR Library Staff”

Training Programme on ICTs and its Applications in Agriculture

A training programme was conducted to upgrade the knowledge and skills of agricultural students for sharing of knowledge management process in education. The training programme was designed by the Academy in the present era of digital technology led communication, the dissemination of agriculture knowledge faces certain challenges due to involvement of variety of stakeholders, etc., The topics covered were

basic computers applications in agriculture; blogs, database creation and databases and internet resources, ICT role in agriculture, ICT applications in agricultural extension, geospatial knowledge management, CMS content management and E-learning, virtual learning, expert system language, statistics and Data analysis, Artificial Intelligence and IoT applications agriculture, ICT applications in rural development, ICTs in Agro advisory Services, ICT application in Animal Science and Fisheries, ICTs in agriculture marketing, Bioinformatics and big data and cloud computing in agriculture, video production theory and practical and challenges and opportunities in agricultural research, etc. Library information system, OERs and on line collaboration tools. Open source tools, social media and their applications. The training covered encompasses a range of ICT mediated applications, data management tools, knowledge management initiative in ICAR and web based applications in a networked environment for the data sharing through open source knowledge management tools which will be very much useful to the agriculture students. Twenty five students from ANGRAU participated in the training programme.



Dr A.K. Srivastava, Member ASRB giving the certificates to participants

Training programme on Advances in Web and Mobile Application Development

Training programme on advances in web and mobile application development for the participants of ICAR, SAUs was conducted at the Academy. The programmes aimed to create awareness among

the participants about the advances in web and mobile application development and to educate the participants in various existing and advanced software tools for web and mobile application development. Nineteen participants from different ICAR institutions, KVKs and SAUs across the country participated in the programmes. The programmes consisted of both theoretical and practical aspects on various topics. The trainees were exposed to Basics of core java, introduction to android development, installing android studio, user interfaces, visual components, working with resources, interacting with UI, storing and retrieving data through SQLite database, user interface controls, integrating with web services, activities and intents testing, debugging, publishing, latest tools in mobile apps development, etc.



Participant receiving certificate

National Dialogue on Artificial Intelligence (AI) and Internet of Things (IoT) Applications in Agriculture

The Academy organized a special programme on 'National Dialogue on Artificial Intelligence (AI) and the Internet of Things (IoT) Applications in Agriculture'. The purpose was to sensitize the participants about the enormous potential of data/knowledge management, AI and IoT in the area of agriculture and allied sectors and prepare a base paper for the next higher-level policy workshop. About 30 participants from 20 ICAR institutes situated all over India participated in the dialogue. The programme was designed to cope with the challenges faced by the agriculture sector due to the involvement of a variety of stakeholders such as scientists, farmers, extension functionaries, inputs

suppliers, etc. The participants deliberated on the topics like AI in agriculture, agricultural robots; Crop, soil monitoring and predictive analytics; and IoT, methods like precision farming, application of drones, livestock monitoring and smart greenhouses. Other areas of group discussions were AI and IoT applications in crops, natural resource management and animal science and fisheries.



National Dialogue on Artificial Intelligence (AI) and Internet of Things (IoT) Applications in Agriculture

Training Programme on Advanced Bioinformatics Tools and its Applications in Agriculture

The programme is conceived with an idea to provide an insight to the participants about the enormous potential of bioinformatics and its tools management to explore it in the area of agriculture. Twenty eight participants from different parts of the country participated to upgrade their knowledge and skills for sharing of knowledge management process in research and education.



Participant receiving certificate from Director, NAARM

Bioinformatics is an innovative developing field that helps in collecting, linking, and manipulating different types of biological information to discover new

biological insights. This specific genetic knowledge could then be used to produce stronger, more drought resistant, disease resistant, insect resistant and more productive crops. Keeping in view the changed scientific scenario in the post-genomic era and to expose the agricultural scientists to the latest computational biology tools for biological research. The training include as theory and hands-on sessions on different aspects.

Geospatial Analysis using QGIS & R

The main objectives of this training programme was to enhance competency levels of NARES professionals in understanding of GIS, GPS and RS by using QGIS and R software applications in Agricultural Sciences. The programme consisted of a blend of lectures, self-exploration instruments, case analysis and hands on exercises. The training covered mainly introduction to geospatial technologies in agriculture, relational database and geodatabase concepts for GIS, Introduction to GIS, GPS and remote sensing, exploring data from open source platforms for geodata, spatial database design and implementation using quantum GIS open source software, spatial analysis like clipping, joining, buffering, interpolation, krigging etc., image processing and land use land cover using R open source software.



Participants of training program on Geospatial Analysis using QGIS & R

Spatial Data Analysis and Mapping using GIS Tools for PJTSAU Faculty Members

The topics covered under the programme were geospatial technologies in agriculture, relational database and geodatabases concepts for GIS,

Introduction to GIS and GPS exploring data from open source platforms for spatial database design and implementation using quantum GIS open source software, Spatial analysis like clipping, joining, buffering, interpolation, krigging etc. and hands on practical using GPS. A total of from PJTSAU 14 participants attended this programme consisting of 9 male and 5 female participants.



Participants of training program on Spatial Data Analysis and Mapping using GIS Tools

Analysis of Experimental Data

The objective of this program was to orient the researchers with the basics of statistical tools and techniques mostly utilized in agricultural research. To train the participants in performing statistical analysis of data generated from agricultural research experiments using Statistical Analysis System (SAS) and other software packages. The training consisted of both lecture cum practical demonstration followed by hands-on practical. Communication of research results wherein research papers published using various statistical techniques were discussed. Data analysis using SAS was performed to the participants using SAS foundation 9.3 and SAS Enterprise Guide (EG). The topics covered include “Principle of Experiments for Conducting Research; Exploratory Data Analysis; Introduction to SAS and Data Management, Correlation, Regression and Diagnostics; Analysis of Experimental Data using Basic as well as advanced designs, Pooled Analysis, Multivariate Data Analysis using Principal Component Analysis (PCA), Factor Analysis, Response Surface. Emphasis was given to analysis of data using appropriate statistical techniques and interpretation of the results. Data from real-life studies and simulated data were utilized to explain various statistical techniques.

Senior Executive Development Programme on Developing Effective Organizational Leadership for Senior Officers of ICAR

Senior officers were trained for the domestic component (for both the batches) at Administrative Staff College of India (ASCI), Hyderabad and International component for Batch-I at China, Malaysia and Thailand; and for Batch-II at Australia, New Zealand and Singapore. 20 participants participated in this programme. The objective of the programme was to expose the participants in the key areas under the theme of the programme, i.e., Effective Organizational Leadership. Both the batches made presentations at ICAR on 19th June, 2018 for mutual learning and on 23rd July, 2018 presentations were done in front of Director General, ICAR with the purpose of strategic implementation of the knowledge gained.



Senior Executive Development Programme on Developing Effective Organizational Leadership for Senior Officers of ICAR at ASCI, Hyderabad

Motivation, Positive Thinking and Communication Skills for Technical Officers of ICAR

An off-campus training programme on 'Motivation, Positive Thinking and Communication skills' for Technical Officers (T5 & above) of ICAR Institutes was held at Indian Institute of Soil and Water Conservation, Dehradun which was attended by a total of 24 participants including 5 females from 17 ICAR institutes representing 13 different states. The main objective of this training programme was to improve the skills and efficiency of the participants in discharging functions and providing constructive technical support to meet the organizational goals and targets. The training programme included

topics on role & importance of technical officers, communication skills, motivation, time management, personality profiling, socio-psychological issues and use of tools like MS Excel & MS Power Point for better communication. The programme also included experiential learning via visits to Sainji Watershed, Mussorie and Indira Gandhi National Forest Academy.



Training programme for Technical Staff of ICAR institutes at IISWC, Dehradun

Training Workshop for Vigilance Officers of ICAR Institutes

Two days' workshop for the Vigilance officers was conducted with the main purpose to create awareness with several provisions of vigilance and also to develop a platform to discuss matters related to preventive and punitive vigilance. Total of 35 participants attended the workshop including 07 female officers. Following are the key recommendations:

- Setting up Vigilance Cell in each institute with committed staff.
- Training and awareness among Vigilance Officers through structured training programme at ICAR-NAARM.
- Opportunities of exposures to Vigilance Officers in other organizations too.
- Identifying few Vigilance Officers as Master Trainers.
- Creating a platform for professional discussion and experience sharing among ICAR institutes Vigilance Officers.

During the workshop it was highlighted that emphasis has to be given on Preventive Vigilance through proactive actions.



Director, NAARM interacting with Vigilance Officers of ICAR

2.6 Mobilizing Science and Technology for Innovation and Sustainable Development

The major focus of this theme has been on catalyzing agricultural innovation and good governance in NARES, conducting research and consultancy in the areas of agricultural policy, agri-innovation, technology management, and use of science based interventional tools for policy support. Efforts have been initiated to infuse innovative ideas and bringing in a new milieu in the entire agricultural research system with the help of online management system and digital framework in order to increase the productivity in research, reduce the drudgery, avoid unnecessary overlapping, and enhance the overall efficiency in research. Special focus is accorded to agricultural technology management by undertaking research and training activities in the domain of technology management with the application of management fundamentals for efficient management of research outputs and for reaping competitive advantage in the organization.

In order to advance the objective of this broad theme, the Academy focused on the following broad areas for its research, academic and capacity building activities

2.6.1 Research Management

2.6.2 Technology and Innovation Management

2.6.3 Research in the context of food security, climate change and globalization

2.6.1 Research Management

Developing a Framework for Enhancing Efficiency of Agricultural Research Management in ICAR

The study envisages to develop frameworks to enhance efficiency and infuse transparency in research management in ICAR. A schematic of the revised scope of the project is depicted below.



Fig. 2.15: Schematic of the Project Components

The Priority Setting, Monitoring and Evaluation (PME) Cells established in all the institutions cater to coordination of the research efforts at the institute level. In order to ensure that the Scientists associated with the Cell are equipped with the most updated knowledge and skill required to effectively execute their roles, the training content was revised and during 2017-18, it was offered as a Management Development Program. About 30 In-charges of PME Cells of various ICAR institutes, nominated by the Director General, ICAR were sensitized on the various project management guidelines and approaches.

The web-applications developed for project management viz., Research Concept Writer (RCW) and Project Logframe writer (PLW) were integrated in the training programmes of the Academy (Module on Developing Winning Research Projects - DWRP,

Foundation Courses, MDPs, etc), through which over 250 participants were provided hands-on training during 2018-19, under various programmes. Over 1125 researchers from India and 18 different countries including Bangladesh, New Zealand, Singapore, USA

and Germany had used these online applications for project proposal development during 2018-19. A Project Management Application is being developed for research proposal development, a schematic of which is provided hereunder:



Fig. 2.16: Schematic of the Project Management System

A comprehensive scientometric analysis is underway on the research productivity, citation trends and collaboration status of all ICAR institutions using Web of Science (WoS) platform of Clarivate Analytics, USA, by analyzing the records published during 2008-2018.

During the said period, all the constituent institutes and centres of ICAR had published over 22,700 research articles and 900 review articles. A critical analysis of the publications made by individual ICAR institutes in collaboration with various countries showed that USA with 735 publications, is the most productive collaborator of ICAR, followed by Australia (259), England (216), Peoples Republic of China (142) and Germany (137).

The Indian Journal of Animal Sciences was found to be the most popular journal carrying maximum research papers (1521) of researchers from ICAR, which was followed by Indian Journal of Agricultural Sciences (1419) and Indian Journal of Fisheries (559).

The Academy envisions to develop a comprehensive dashboard for evidence-based research governance by integrating various metrics of research inputs (human resource and finance) and outputs (publications, citations, technology and varieties produced).

Evidence-based Research Governance

India's agricultural research system is the largest and institutionally most complex agricultural research systems which has contributed in a large way to increasing agriculture production and productivity.

The information related to different aspects of agricultural research is important to minimize the research expenditure on less priority areas and to facilitate farmers' friendly research.

ICAR-NAARM have been associated since 2014 with International Food Policy Research Institute (IFPRI), Washington on its Agricultural Science and Technology Indicators (ASTI) initiative, which compiles, analyses, and publishes primary data on institutional developments, investments, and human resources in agricultural R&D in low- and middle-income countries in the world. Currently, the research performance data engaged with them in the fourth round of data collection for the period from 2014-15 till 2017-18 for the whole agricultural sector on the country, covering both the public institutions and private organizations involved in R&D.

In order to avoid collection of R&D performance data by multitude of agencies, NAARM held discussions with the IFPRI-ASTI and the National Science and Technology Management Information System (NSTMIS) team of Department of Science and Technology, Government of India during

April - May, 2018 for synchronizing the data collection efforts. Consequently, DST awarded a new study, to complement the ASTI project with the objective of (i) profiling the human and financial resources devoted to agricultural R&D in India and map their performance based on select indicators and (ii) developing a national framework for evidence-based S&T governance and build a dashboard to aid in its institutionalization.

The study derives its relevance on the renewed focus by the Govt of India for data-driven decision making across sectors; mainstreaming Digital India initiative, and to democratize access/use of S&T data to all stakeholders so as to avoid duplication of efforts by multitude of agencies in assessing performance. An application has been built for data collection with innovative features like real-time monitoring of survey, collaborative survey, auto-generation of institute fact sheets, etc. ICAR-NAARM has developed a prototype for the Evidence-based S&T Governance as shown in the schematic. The survey covers about 300 institutions, which represent over 2400 research units spread across ICAR, agricultural universities and other public sector organizations in the agricultural sector.



Fig. 2.17: Screen shot of the Survey Application



Fig. 2.18: Schematic of the Framework for Data-Driven S&T Governance

Agricultural Research and Development Infrastructure in select Indian States

The study was taken up with the objectives:

- To assess the status and gaps of infrastructure and resources available for agriculture research in the study states,
- To analyze the productivity and efficiency of available resources and infrastructure, and
- To assess the impact of agricultural research infrastructure on farmers.

Interim findings of the project indicate that the research productivity of four ICAR institutes two each from the two study states i.e. Andhra Pradesh and Telangana was measured by two indicators viz. budget spent per technology and number of scientists per technology. In terms of budget the least budget spent (Rs.31.5 lakhs) for bringing out one technology was recorded by Indian Institute of Millets Research taking the past five years' period for this analysis. On the other hand, in the case of Central Tobacco Research Institute (CTRI), which needed the highest budget per technology (Rs. 379 lakhs). This is understandable as the clientele for improved crop cultivation technologies as also the technologies for value added millet products is in great

demand created due to health and diet consciousness among consumers. The research publication intensity of four sample ICAR institutes (two in each state) were estimated by taking into account the number of research articles, bulletins, policy papers etc. published in the last four years. It may be noted that CRIDA and IOPR had the higher publications intensity (0.55 and 0.54 papers per scientist per year). Paddy is the dominant crop in West Godavari district with Swarna as the leading variety with almost 40 per cent area among sample farmers. MTU-1121 is another variety which was tested for its technology impact in the present study since it is promising and is likely to replace older varieties in due course of time. Groundnut is the dominant crop in Ananthapur district with K6 as the leading variety with almost 60 per cent area among sample farmers followed by TAG variety with 17 per cent.

Foundation Course for Agricultural Research Service (FOCARS)

FOCARS is the flagship programme of the Academy and is offering the same twice a year. The Foundation course aims at building the competencies of the scientist probationers of Agricultural Research Service so as to nurture and develop the skills required to

meet the professional challenges. The competency framework of FOCARS comprises the development of core discipline research skills, personal effectiveness, information management and business orientation.

108th FOCARS

The 108th FOCARS started on July 2, 2018 and completed on September 29, 2018. A total of 135 ARS probationers attended this course, consisting of 85 men and 50 women scientists representing 24 states and 32 disciplines. The course was inaugurated by Dr Ashok Dalwai, CEO, National Rainfed Area Authority (NRAA) and Chairman, Committee on Doubling of Farmers' Income. He motivated the young scientists with the examples of all-time great scientists like Dr Norman Borlaug, Dr MS Swaminathan and Dr Verghese Kurien. During the First phase of 39 working days, there were total 82 sessions. Besides, there were practical sessions in the field of video production, digital presentation, team building and personality development. The participants also visited International Crops Research Institute for Semi-Arid tropics (ICRISAT) and ICAR-Central Research Institute for Dryland Agriculture (CRIDA) apart from exposure to different eminent guest speakers on different aspects of Indian agriculture and allied subjects. During the Second phase, participants were exposed to 21 days Field Experience Training (FET) program at 22 locations in 11 states of India. During the Third phase of 14 working days, there were 20 sessions and two practical sessions each in the field of Data Analytics and IP-Informatics. This phase offers more of experiential learning through several

presentations like FET report and scientific writing workshop.

Dr T Mohapatra, Secretary, DARE & DG, ICAR, Dr AK Srivastava, Chairman, ASRB, Dr Chhabilendra Roul, Special Secretary, DARE, Dr J. K. Jena, DDG, Fisheries, Dr AK Singh, DDG, Extension, Dr Anupama Verma, NAAS Fellow, Dr SA Patil, Ex. Director, IARI, Dr S Senthil Kumar, IGNFA, Dehradun and other prominent dignitaries addressed the probationers. Dr Misha Madhavan received the best FOCARS trainee award. This course ended with the valedictory address by Dr Manju Sharma, Ex-Secretary, DBT, GOI on September 29, 2018. A video story of 3 months FOCARS period prepared by the FOCARS probationers was also released during the valedictory session.

109th FOCARS

The Academy organized 109th FOCARS for 32 ARS Probationers during January 4 - April 3, 2019. The programme was organized in three phases for 73 working days. The probationers belonged to 16 ARS disciplines. The sessions were organized on core research skills, personal effectiveness, awareness and use of IT tools, database management, awareness of research environment, orientation of organization and other skill components. Innovations brought in this FOCARS training included module on science communication and encouraged the probationers to prepare book chapters and this initiative has led to bringing out a book on 'Challenges and Emerging Opportunities in Indian Agriculture'.



DG addressing FOCARS



108th FOCARS participants



109th FOCARS participants with Dr T. Mohapatra, DG, ICAR and NAARM faculty members



Dr Panjab Singh, Ex-DG, ICAR address during valedictory session

2.6.2 Technology and Innovation Management

Commercialization of Agricultural Technology Scheme - National Agriculture Innovation Fund (NAIF) (Component-I) (IPTM - ZTMC)

NAARM acted as “help desk” services for Agri Business Incubators/ Start Ups/ SAUs for organizing capacity building programmes and training on technology transfer and IP management. NAARM took a lead in helping the ICAR institutions such as IIOR, PDP, NRC Meat, Hyderabad in conducting prior art, filing their intellectual property (patent/trademark) and liaisons with patent agents/lawyers. Copyright applications for five in-house software was filed through NAARM. Research studies on Intellectual Property Landscaping of emerging technology such as Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) for sustainable value-chain development was carried out.

Genome editing of food crops is emerging as a new tool to meet food and nutritional security in present scenario of disproportionate balance within rising populations and diminishing resources, aggravated by cycles of climatic changes. Among several gene editing techniques ‘CRISPR along with CAS proteins’ is considered to be the most versatile tool, representing tremendous potential and promise. Rapid pace of development of this technology challenges policymakers and R&D investors for maximizing investments in a more pragmatic manner. The purpose of the study is to assess trends of this emerging technology in agricultural production-consumption systems. Use of indicators like research publications (to collaborate early phase research activity), patents (to assess technology development) and commercialized products in market (for assessment of market need), in standardized matrix to analyze and forecast trends of emerging technology.

Timeline analysis was undertaken using priority year as indicator of inventive activities for a period of 15 years i.e. from 2003-2017. It was observed that there was major increase in filing from 2014 onwards. Maximum number of patents were filed by government funded institutions (48%) indicating the level of R&D investments by these entities as compared to private firms (34%). Top ten assignees include Universities in the US and China, whereas DuPont and Dow are the major players from industry.

An attempt was made to determine technology life cycle stage of CRISPR in agricultural PCS. Using cumulative number of patent filings over a period of six years, the S-curve analysis was traced. It was observed that there was an increase in cumulative number of patents related to CRISPR technology in agricultural PCS. These initial results indicate that this technology is still in growth phase.

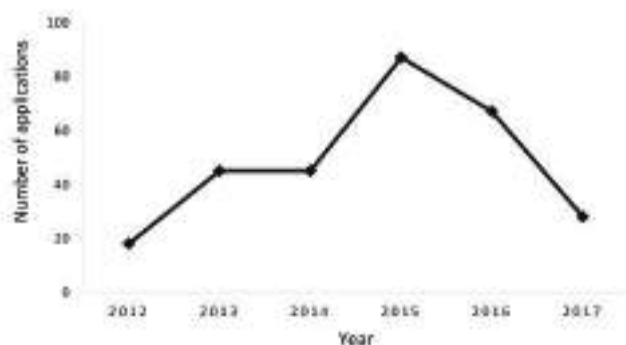


Fig. 2.19: Patent timeline analysis using priority year (n=290)

It is significant to note trends of concurrent efforts in scientific discoveries, technology development and its deployment in new product development. Regulatory processing at early-stage R&D certainly indicate new trends; quite unlike the past instances of delays especially during GMO technologies approvals. Regulatory framework, anticipatory engagement with stakeholders across entire agricultural PCS, ethical concerns and ecosystem impacts need more attention for successful deployment of CRISPR technology. Investments from public grants, industry or even agri-investors can be a reality only if there are strong indications for more forthright frameworks on these dimensions.

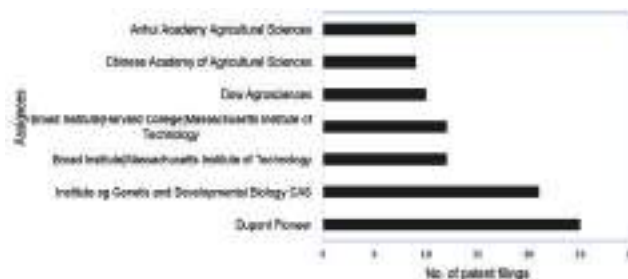


Fig. 2.20: Major assignees with their respective number of patent filings

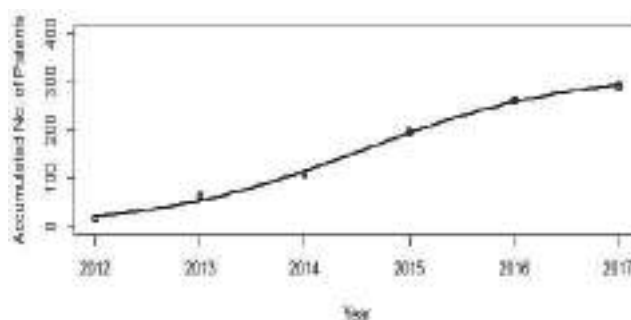


Fig. 2.21: S-Curve of CRISPR technology based on accumulated number of patents

Post Graduate Diploma in Technology Management in Agriculture

NAARM offers a one-year Post graduate Diploma Programme on Technology Management in Agriculture (PGD-TMA) since in 2011 on distance education mode in collaboration with the University of Hyderabad (UoH) as a stepping stone towards better management of agricultural technologies within the changing landscapes of IP regime. The course is specially designed to cater to the needs of building a strong IP portfolio along with efficient technology management, diffusion and commercialization skills among the scientists, teachers, students and professionals engaged in agriculture and related industries. The number of students admitted in 2018 was 34 with 7 female and 27 male students. These students represent 13 domain states. Most of the students are in-service candidates representing various Governmental and Non-Governmental Organizations such as Agricultural Universities, ICAR, ICRISAT, Coromandel, Du Pont, Food Corporation of India (FCI), Steel Authority of India, GVK Group, Syngenta, Bharat Petroleum, NABARD and State departments.

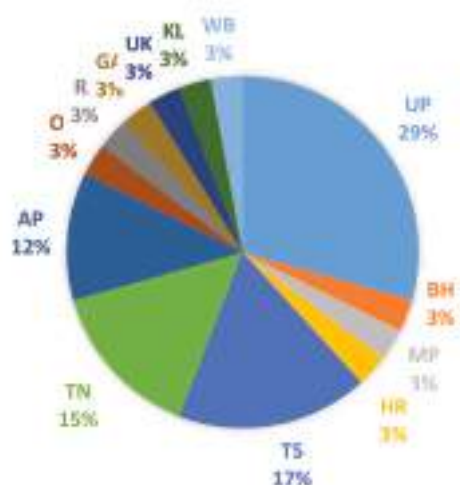


Fig. 2.22: States representation in 7th batch of PGDTMA

Two contact sessions of 3 days duration were conducted at the Academy in September 18 and January 19 covering the entire course. Resource persons from NAARM, Industry and other organizations contributed towards the lectures.



PGDTMA 2018 batch students

2.6.3 Research in the context of food security, climate change and globalization

International Market Signals for Indian Agricultural trade during WTO Regime: R&D Strategies and Policies

Exports coefficients of major agricultural commodities were estimated, which would enable for furthering for R&D strategies and policies. For cereals; import price and for pulses, import quantity had a negative impact on production. For oilseeds, export quantity and export price had a negative impact on production.

As far as total agricultural exports are concerned, trade balance for pulses had a negative impact whereas for oilseeds, both trade balance as well as production had negative impact on total agricultural exports. On the other hand, for pulses, production, export quantity, export price had a positive effect on total agricultural exports, and for oilseeds, export quantity as well as export price had positive effects. The results have shown that cereals' production, export quantity-price, trade balance had a positive effect on total agricultural exports. Total agricultural imports for cereals, import quantity and import price whereas for pulses, trade balance had a negative impact on total agricultural imports. For oilseeds, production, import quantity and trade balance had a negative impact on total agricultural imports. It was suggested that both agricultural output exporting and importing countries in the world need to look at international market signals such as, world trading prices, trends, growth rates, elasticities, market destinations etc.

Environmental Impact Assessment of Tobacco Curing in Andhra Pradesh and Karnataka

This study was conducted with the following objectives;

- To survey the sources and types of wood used for curing of tobacco and to determine wood consumption for flue curing;
- To assess the impact of tobacco curing on loss of forest wood and deforestation and to make an assessment of environmental impact of tobacco curing;
- To study the supply chains of wood fuel for curing, demand and supply patterns, price sensitivity and purchase patterns of wood fuel.
- To explore the feasibility of identifying energy efficient tobacco curing to reduce the dependence on fuel wood.

Survey results in Andhra Pradesh and Karnataka states revealed that Eucalyptus (33.8%), old felled trees of cashew (22.5%) and mango (14.8%) are the three dominant species for the fuel wood used in the barns of West Godavari. Most of the fuel



wood (88%) used for tobacco curing in Prakasam district is obtained from *Prosopis juliflora*. The tobacco barns in Mysuru district used Eucalyptus (24%) followed by Tamarind (20%), *Prosopis cineraria* (14%) and other fuel material like coffee husk and coffee stems. The average specific fuel consumption (SFC) of traditional/ conventional barns ranged between 5.06 kg of fuel wood/ kg of cured leaf (West Godavari) and 5.27 kg of fuel wood/ kg of cured leaf (Hassan).

The SFC in respect of modern barn technologies like venturi furnace, roof insulation with turbo vents ranged between 3.95-4.45 kg fuel wood/ kg of cured leaf in West Godavari; 4.02-4.43 kg in Prakasam; 4.18-4.70 kg in Mysuru and from 4.20-4.65 kg / kg of cured leaf in Hassan district. The total fuel wood consumed for curing FCV tobacco produced in the country during the year 2017-18 was estimated to be 1,142 thousand tonnes. Considering the leaf production in the last five years, this figure works out to 1,203 thousand tonnes per annum. Fuel wood is sourced from several locations or lands with varied legal/ ownership status such as farm lands (agro-forestry), community lands (weeds, sporadic trees) and forest plantations. Burning of fuel wood is net neutral in terms of Carbon footprint. Because, there is always regeneration of wood from trees i.e. the trees capture almost same amount of C in them both in above and below the soil biomass and also push in to the soil through the process of building up of organic carbon by recycling (Singh *et al*, 2017). Concerted and collaborative efforts by the Board in association with tobacco traders, line departments and farmers' associations are needed for speedy and complete adoption of technologies such as barns with energy saving technologies.

Regional Consultation on Climate Smart Agricultural Policies, Strategies and Agricultural Development Programs towards Climate Change Adaptation and Mitigation

Academy organized two days regional consultation on, "Climate smart agricultural policies, strategies

and agricultural development programmes towards climate change adaptation and mitigation", jointly with SAARC Agriculture Centre (SAC), Bangladesh; Asia-Pacific Network for Global Change Research (APN), Japan; Ministry of Environment, Forest and Climate Change (MoEFCC), India at NAARM, Hyderabad. The consultation was inaugurated by Shri Shailendra Kumar Joshi, Chief Secretary of Telangana, who called for sharing mutual experiences towards effective adaptation to climate change, based on the common characteristics in agricultural practices and food systems among these countries. Dr P. Appa Rao, Vice Chancellor, University of Hyderabad and Dr K.J. Ramesh, Director General, Indian Meteorology Department spoke on the occasion. Dr Kirit Parikh, Former Member, Planning Commission, GoI, in his key note address, emphasized the need to undertake district level analysis to develop climate resilient agrarian solutions. During the three-days consultation, the delegates from the SAARC countries presented the state of knowledge, adaptation strategies, scope for knowledge and experience sharing with special reference to building climate resilient agrarian systems in the region. The roadmap prepared as a fallout of this regional consultation through active involvement of all member countries is envisaged to take forward the decisions on "Koronivia Joint Work on Agriculture", by the 23rd Conference of Parties (CoP) to the United Nations Framework Convention on Climate Change (UNFCCC). The workshop was attended by senior officers and agricultural experts from Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan and Sri Lanka. Scientists and senior officials from ICAR, NRSA, MoES, MoEFCC and NABARD also participated in the meeting.



Director, NAARM addressing during Regional Consultation on Climate Smart Agricultural Policies, Strategies and Agricultural Development Programs towards Climate Change Adaptation and Mitigation

Need Based Program

MDP on “Intellectual Property Valuation & Technology Management”

The Academy conducted the MDP on Intellectual Property & Technology Management which was attended by 14 senior level scientists including 11 men and 3 women scientists. The sessions were held on the S & T in emerging economy, IP valuation and monetization, technology management, transfer and commercialization. Concepts on marketing and

business models were also discussed. To have a wider exposure on incubation and innovation, the trainees visited Atal Incubation Centre-CCMB, Hyderabad. The participants gained the skills necessary for critically assessing both established and emerging approaches to business plan analysis and IP valuation and developing their own approaches to managing IP enabled businesses. The programme also dealt with the institutional innovations within NARS towards technology commercialization, entrepreneurship development and IP management.



Participants visit to Atal Incubation Centre-CCMB, Hyderabad during MDP on “Intellectual Property Valuation & Technology Management”

MDP on Priority Setting, Monitoring & Evaluation (PME) for Agricultural Research Projects

The 6th Management Development Programme (MDP) on Priority setting, Monitoring and Evaluation (PME) of agricultural research projects for the In-charges of PME Cell of ICAR institutions was organized by the Academy. The Heads of PME Cell of 30 ICAR institutions participated in the program. The MDP comprised of modules on concepts and principles

of prioritization, monitoring, evaluation and impact assessment of agricultural research projects. The participants were given hands-on training on the various software tools developed by ICAR-NAARM for project management. They were sensitized on the various scientometric tools, global and national frameworks for measuring research efficiency and ICAR guidelines pertaining to the various functions of PME Cell like data policy, publications, project management, technology management, etc.



Participants of the MDP on PME



A participant receiving the certificate from FA, ICAR

Off-campus Training Program on Scientific Communication & Developing Winning Research Proposals

A tailor-made off-campus program for University Campus, Academy of TNJFU and College of Fisheries, Ponneri, Chennai was conducted for 53 newly recruited faculty members. The participants had specialization in the area of Fisheries Resource Management, Aquaculture, Fish Processing Technology, Fisheries Engineering, Fish Health, Biotechnology and Genetics, Social Sciences, etc. The core content of the program included elements of scientific writing and oral communication; tools and tips for communicating research ideas; research concept writing; preparation of project log frames and making winning research proposals. The participants were introduced to the online applications developed by NAARM viz., AHP Analyser, Research Concept Writer and Project Log frame Writer. The participants were also sensitized on the issues confronted by the young researchers and tips for self-motivation and positive thinking. The programme comprised of a

blend of lectures, discussions, group activity, hand-on exercises and presentation.



39th Training Program on Developing Winning Research Proposals (DWRP)

NAARM empanels IT firms for co-development of IT solutions

The Academy has taken a pioneering initiative to empanel private firms to co-develop IP-enabled IT solutions to address issues related to agricultural research management based on mutual expertise. The initiative is believed to aid in establishing a framework

for public-private collaboration and institutionalizing innovation through knowledge products development.

Four firms were empanelled and a launch meeting was held wherein the Academy signed agreements for Joint Product Development with the empanelled firms. Dr P Krishnan, Member Secretary of the Empanelment Operationalizing Committee gave an overview of

the arrangement, following which each of the firms provided a glimpse of their strengths and potential areas of partnership. Dr Ch. Srinivasa Rao, Director, NAARM highlighted the key roles of the Academy and expressed hope that the partnership will provide an opportunity for advancing the interests of the Academy, through mutually beneficial collaboration.



Empaneled firms exchanging agreements with the Academy

2.7 List of Ongoing Research Projects

2.7.1 Institute Projects

No.	PME Code	Title of Project	Team	Duration	
				From	To
1.	AGED/NAARM/SIL/2017/010/00130	Financial inclusion of smallholders and landless households in the underdeveloped state	PC Meena Ranjit Kumar Sanjiv Kumar	April 17	March 19
2.	AGED/NAARM/SIL/2017/011/00131	Structural Changes in Agricultural Marketing in India: Effect on Inclusiveness, Efficiency and Profitability	Ranjit Kumar B Ganesh Kumar PC Meena N Sivaramane Sanjiv Kumar	April 17	March 19
3.	AGED/NAARM/SIL/2018/011/00150	Farmer Producer Companies in India: A Study on their Management Practices and Business Potential	N Sivaramane Ranjit Kumar PC Meena Bharat S Sontakki K Srinivas P Venkatesan	October 18	September 21
4.	AGED/NAARM/SIL/2018/009/00148	Mainstreaming the Small Ruminant Farmers into Efficient Value Chains in India	B Ganesh Kumar Sanjiv Kumar D Thammi Raju PC Meena	October 18	September 20
5.	AGED/NAARM/SIL/2017/012/00132	Team work and its effect on organizational performance in NARES	RVS Rao Alok Kumar KH Rao	April 17	March 20
6.	AGED/NAARM/SIL/2017/013/00133	Competency management system for agricultural research and extension	KH Rao Alok Kumar RVS Rao P Ramesh	April 17	March 20

7.	AGED/NAARM/SIL/ 2018/010/00149	HR Interventions and their influence on Institutional Innovations	Alok Kumar RVS Rao P Ramesh	October 18	September 20
8.	AGED/NAARM/SIL/ 2018/014/00153	Emotional Intelligence and Organizational Effectiveness	P Ramesh RVS Rao	April 18	March 20
9.	AGED/NAARM/SIL/ 2018/008/00147	Decision Support Database of Maize Inbred Germplasm	N Sunil JC Shekhar N Srinivasa Rao	August 16	July 20
10.	AGED/NAARM/SIL/ 2017/014/00134	Geo-Spatial knowledge system for monitoring climate changes in Agro-climatic Zone X: Southern Plateau and Hill Region.	PD Sreekanth, BS Yashavanth P Venkatesan	April 17	March 19
11.	AGED/NAARM/SIL/ 2018/004/00143	Text Analytics of the Queries raised at Kisan Call Centres in South India	BS Yashavanth PD Sreekanth Jabir Ali	April 18	March 20
12.	AGED/NAARM/SIL/ 2017/009/00129	Developing a framework for enhancing agricultural research management efficiency	Ch Srinivasa Rao P Krishnan I Sekar B Ganesh Kumar SK Soam K Kareemulla VV Sumanth Kumar	April 17	March 21
13.	AGED/NAARM/SIL/ 2017/008/00128	IP landscaping of emerging agricultural technologies for sustainable value chain development	Manju Gerard K Srinivas	April 17	September 18
14.	AGED/NAARM/SIL/ 2018/005/00144	Assessment of Technological Interventions in Teaching-Learning Process	S Senthil Vinayagam GRK Murthy D Thammi Raju	April 18	March 20
15.	AGED/NAARM/SIL/ 2018/003/00142	Digital Teaching Evaluation in Agricultural education using TPACK Model	GRK Murthy Senthil Vinayagam M Balakrishnan	April 18	March 20
16.	AGED/NAARM/SIL/ 2018/002/00141	Ag-Academy: A framework for scaling up e-learning Platform	VV Sumanth Kumar M Krishnan P Krishnan SK Soam GRK Murthy N Srinivasa Rao	April 18	March 21
17.	AGED/NAARM/SIL/ 2018/012/00151	Students' Learning Approaches (SLAs) for enhanced learning outcomes in Agriculture Education: A Critical Analysis	D Thammi Raju P Ramesh Bharat S Sontakki P Krishnan	October 18	September 20
18.	AGED/NAARM/SIL/ 2016/009/00119	Gender-based ICT led knowledge management in agriculture	N Sandhya Shenoy VKJ Rao PD Sreekanth N Srinivasa Rao BS Sontakki Surya Rathore Manju Gerard	April 16	March 19
19.	AGED/NAARM/SIL/ 2017/007/00127	Research role analysis of extension scientists in ICAR	VKJ Rao Barat S Sontakki Surya Rathore Alok Kumar	April 17	March 19

(Characters in Project Codes: S-Intra Institutional, C-Inter Institutional, I-Institute Funded, O-Externally Funded, C-Consultancy, L-Institute is Leader, P-Institute is Partner)

2.7.2 Externally Funded Contract Research (Sponsored/Collaborative/Grant-in-Aid Projects)

No.	PME Code	Title of Project	Team	Duration From To		Budget (in Lakhs)	Sponsored by
1.	AGED/ NAARM/SOL/ 2008/003/00067	Commercialization of agricultural technology scheme (IPTM-ZTMC)	Manju Gerard K Srinivas	April 14	March 20	19.15	ICAR, New Delhi
2.	AGED/ NAARM/SOL/ 2017/003/00123	Agricultural Research and Development Infrastructure in select Indian States	K Kareemulla S Ravichandran	July 17	June 19	24.00	NSMIS, DST, GoI
3.	AGED/ NAARM/SOL/ 2017/004/00124	International market signals for Indian agricultural trade during WTO Regime: R & D strategies and policies	MB Dastagiri	April 17	February 19	12.42	DST, GoI
4.	AGED/ NAARM/SOL/ 2016/002/00112	Establishment of Agri-Business Incubation (ABI) Centers under XII plan Schemes for National Agriculture Innovation Fund	K Srinivas Manju Gerard	January 16	March 20	79.15	ICAR, New Delhi
5.	AGED/ NAARM/ SOL/2018/013/ 00152	Profiling the Resources Devoted to Agricultural R&D and Development of National framework for Evidence -based S&T Governance in India	P Krishnan B Ganesh Kumar BS Yashavanth M Balakrishnan SK Soam	October 18	September 19	29.00	DST, New Delhi
6.	AGED/ NAARM/SOL/ 2015/007/00110	KRISHI: ICAR Research Data Repository for Knowledge Management	A Dhandapani SK Soam	November 15	March 20	76.00	Data Management, ICAR
7.	AGED/ NAARM/ COP/2019/001/ 00154	Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education	SK Soam (CC-PI) D Thammi Raju N Srinivasa Rao Alok Kumar VV Sumanth Kumar Sanjiv Kumar Surya Rathore	February 19	March 21	1704.17	World Bank
8.	AGED/ NAARM/SOL/ 2014/009/00100	Technology Enhanced Learning in Agriculture Education (TELAge)	GRK Murthy S Senthil Vinayagam D Thammi Raju P Ramesh	January 14	March 20	425.00	ICAR, Niche Area of Excellence (NAE) Scheme
9.	AGED/ NAARM/CCP/ 2017/017/00137	Adoption of Insect Resistance Management Practices in Bt Cotton Cultivation in India	Ranjit Kumar N Sivaramane A Dhandapani PC Meena Sanjiv Kumar	February 18	May 19	30.68	Mahyco Monsanto Biotech (India) Pvt. Ltd.

10.	AGED/ NAARM/CCL/ 2019/004/00157	Adoption of Natural Farming and its effect on crop yield and Farmers Livelihood in India	Ranjit Kumar P Ramesh PC Meena Sanjiv Kumar BS Yashavanth Indoria Sumanta Kundu Manjunath	January 19	December 19	35.00	NITI Aayog
11.	AGED/ NAARM/SCP/ 2018/001/00140	Strengthening and Integrating Agricultural Science and Technology Indicators ASTI) in Indian NARES	Ch Srinivasa Rao B Ganesh Kumar P Krishnan SK Soam	October 17	June 19	30.15	IFPRI
12.	AGED/ NAARM/SCP/ 2017/002/00122	Environmental Impact Assessment of Tobacco Curing	K Kareemulla I Sekar S Ravichandran P Krishnan B Ganesh Kumar BS Yashavanth	April 17	June 19	20.87	Tobacco Board
13.	AGED/ NAARM/CCP/ 2016/010/00120	Socio-economic Assessment of Community Based Seed Producers (CBSP) Groups by Women	K Srinivas SP Subash	September 16	August 18	2.50	RGMVP, Rai Bareilly
14.	AGED/ NAARM/CCP/ 2017/005/00125	Developing guidelines and methodologies for IP policies of PJTSAU	Manju Gerard SK Soam	April 17	May 19	5.00	PJTSAU, Hyderabad
15.	AGED/ NAARM/ SOL/2018/007/ 00146	Development of digital Content for PGD in Agri-warehousing Management of MANAGE Hyderabad	S Senthil Vinayagam GRK Murthy	November 18	March 20	9.90	MANAGE, Hyderabad
16.	AGED/ NAARM/ SOL/2018/006/ 00145	Development of digital Content for Plant Biosecurity through MOOC Platform	GRK Murthy S Senthil Vinayagam	July 18	May 19	7.20	NIPHM, Hyderabad
17.	AGED/ NAARM/SCL/ 2019/005/00158	Economic Analysis of Eucalyptus and Subabul based Agroforestry Ecosystem in Telangana and Andhra Pradesh	SK Soam P Venkatesan N Sivaramane Ch Srinivasa Rao	September 18	August 19	11.28	ITC (P) Ltd.
18.	AGED/ NAARM/SOL/ 2019/006/00159	Establishment of Skill Development Bioinformatics Infrastructure Facility (Skill-BIF) for Agricultural Biotechnology	M Balakrishnan SK Soam	December 18	March 21	44.46	DBT, New Delhi
19.	AGED/ NAARM/COP/ 2017/001/00121	Management and Impact Assessment of Farmer FIRST Project	P Venkatesan BS Sontakki N Sivaramane	March 17	March 20	144.05	ICAR, New Delhi

(Characters in Project Codes: S-Intra Institutional, C-Inter Institutional, I-Institute Funded, O- Externally Funded, C- Consultancy, L- Institute is Leader, P- Institute is Partner)

2.8 Capacity Building programmes Organized by ICAR-NAARM for the period 2018-19

S. No.	Title of the Program	Starting Date	Ending Date	Programme Director/s	No. of Participants
1	6 th Foundation Course for Faculty of Agricultural Universities (FOCFAU)	March 12, 2018	April 10, 2018	PD Sreekanth P Krishnan	48
2	Regional Consultation on Climate Smart Agricultural Policies, Strategies and Agricultural Development Programmes towards Climate Change Adaptation and Mitigation	April 17, 2018	April 19, 2018	Ch Srinivasa Rao SK Soam P Krishnan	42
3	Training Programme on Agricultural Research and Education Management (Induction course for newly recruited Assistant Professors of PJTSAU)	April 19, 2018	April 28, 2018	NS Shenoy PD Sreekanth	69
4	Training Programme on Hospitality Management (In Collaboration with National Institute of Tourism and Hospital Management (NITHM) Hyderabad	April 20, 2018	April 25, 2018	Zameer Ahmed Sham Bahadur	29
5	Training Programme on Agricultural Engineering Research Management for the faculty of JNKVV	April 25, 2018	April 28, 2018	SK Soam P Krishnan Ch Srinivasa Rao	29
6	Capability building programme for Young Agri-Business professionals of Coromandal International Ltd, (CIL) (Batch-I, Phase-I)	May 14, 2018	May 18, 2018	KH Rao Ranjit Kumar	32
7	Capability building programme for Young Agri-Business professionals of Coromandal International Ltd, (CIL) (Batch-II, Phase-I)	May 21, 2018	May 25, 2018	KH Rao Ranjit Kumar	28
8	National Dialogue on AI and IoT Applications in Agriculture	June 01, 2018	June 02, 2018	SK Soam M Balakrishnan VV Sumanth Kumar	30
9	MDP on Leadership Development (a pre-RMP Programme)	June 04, 2018	June 15, 2018	KH Rao P Ramesh	17
10	Training Programme on Agricultural Research and Education Management (Induction Course for Newly Recruited Assistant Professors of PJTSAU)	June 06, 2018	June 15, 2018	SK Soam Manju Gerard Sanjiv Kumar	69



S. No.	Title of the Program	Starting Date	Ending Date	Programme Director/s	No. of Participants
11	Scientific Communication and Developing Winning Research For Proposals for the faculty members of TNJFU (Off Campus at TNJFU, Nagapatnam)	June 21, 2018	June 23, 2018	P Krishnan BS Sontakki SK Soam	25
12	Enhancing Efficiency and Behavioral Skills of Stenographers Grade-III, PA, PS, PPS and Sr. PPS of ICAR	June 21, 2018	June 26, 2018	N Raghunath KR Ghansyam	24
13	Training Programme on Motivation, Positive Thinking and Communication Skills for Technical Officers (T-5 and above) of ICAR Institute	June 21, 2018	June 27, 2018	V Murali LM Ahire	29
14	Training Programme on Developing Winning Research Proposals in Agricultural Research for Scientists (Off-Campus at IISS, Bhopal)	June 26, 2018	June 29, 2018	SK Soam	25
15	Training Programme on Spatial Data Analysis and Mapping using GIS Tools for PJTSAU Faculty Members	June 27, 2018	June 29, 2018	PD Sreekanth	14
16	108 th Foundation Course for Agricultural Research Service (FOCARS)	July 02, 2018	September 29, 2018	P Ramesh M Krishnan Alok Kumar	135
17	Refresher Course on Administration and Finance Management to SAs/ AAOs/AFAOs/Assistants of ICAR (Off-Campus at ICAR-CCARI,Goa)	July 05, 2018	July 10, 2018	JNL Das WS Bhat	41
18	Brainstorming Workshop on Development of Proforma for Ranking of ICAR Institutes (NAAS, New Delhi)	July 20, 2018	July 20, 2018	SK Soam	36
19	Training Programme on Motivation, Positive Thinking and Communication Skills for Technical Staff (T-1 to T-4) of ICAR (Off Campus at ICAR-CIAE, Bhopal)	August 01, 2018	August 07, 2018	KV Kumar M Shekhar Reddy	36
20	Training Programme on Scientific Communication & Developing Winning Research Proposals (Off-campus at TNJFU, Chennai)	August 02, 2018	August 04, 2018	P Krishnan BS Sontakki SK Soam	28
21	Developing Business Proposal for Farmers Producer Companies	August 06, 2018	August 10, 2018	Ranjit Kumar K Srinivas	19



S. No.	Title of the Program	Starting Date	Ending Date	Programme Director/s	No. of Participants
22	Training Programme on ICT and its Applications in Agriculture for ANGRAU Students	August 21, 2018	September 19, 2018	M Balakrishnan GRK Murthy VV Sumanth Kumar	25
23	Training Programme on Developing Winning Research Proposals	August 23, 2018	August 28, 2018	SK Soam P Krishnan	19
24	Training Programme on Intellectual Property Valuation and Technology Management	August 24, 2018	August 29, 2018	Manju Gerard K Srinivas	14
25	Training Programme on Analysis of Experimental Data	September 06, 2018	September 11, 2018	S Ravichandran A Dandapani	27
26	Training Programme on Stress Management	September 17, 2018	September 20, 2018	P Ramesh A Debnath	17
27	Capability Building Programme for Young Agri-Business Professionals of Coromandal international (Batch-III, Phase-I)	September 17, 2018	September 20, 2018	KH Rao Ranjit Kumar	35
28	Training Programme on Soft Skills and Personality Development for Technical Staff of ICAR	September 18, 2018	September 27, 2018	N Srinivasa Rao S Raju Kumar	21
29	Training Programme on Enhancing Efficiency and Behavioral Skills of Stenographers Grade- III, PA, PS PPS and Sr. PPS of ICAR (off campus at CIFE Mumbai)	September 24, 2018	September 29, 2018	WS Bhat PG Kohad	27
30	Training Programme on Innovative Practices in Extension Research and Evaluation	September 24, 2018	September 29, 2018	BS Sontakki P Venkatesan	19
31	Training Programme on Developing Winning Research Proposals Agricultural Research (Off Campus at RVSKVV Gwalior)	September 25, 2018	September 27, 2018	SK Soam P Krishnan Ch Srinivasa Rao	25
32	Training Programme on Advanced Bio-informatics tools and its Applications in Agriculture	September 25, 2018	September 29, 2018	M Balakrishnan N Srinivasa Rao	28
33	Training Programme on Motivation, Positive Thinking and Communication Skills for Technical Staff (T-1 to T-4) of ICAR	October 04, 2018	October 10, 2018	MA Basith G Aneeja	24

S. No.	Title of the Program	Starting Date	Ending Date	Programme Director/s	No. of Participants
34	Training Programme on Advances in Web and Mobile Application Development	October 05, 2018	October 10, 2018	N Srinivasa Rao M Balakrishnan	19
35	PRA and Communication Strategies to the Probationers of Indian Forest Services	October 06, 2018	October 07, 2018	SK Soam BS Sontakki P Krishnan	86
36	Training Programme on skill enhancing and employees development (SEED) for Retail Sales Managers of Coromandel International (Phase-I)	October 09, 2018	October 11, 2018	KH Rao Ranjit Kumar	30
37	MDP on Administrative & Financial Management for DS/CAO/CFAO/SAO/US/SFAO (Off Campus at	October 09, 2018	October 12, 2018	S George Ashish Roy	12
38	Developing Winning Research Proposals in Food Technology and Allied Disciplines (Off Campus at NIFTEM, Sonapat)	October 13, 2018	October 15, 2018	SK Soam Ch Srinivasa Rao	27
39	Training Programme on Socio-Economic Impact Assessment of Research Programmes	October 25, 2018	November 27, 2018	N Sivaramane P Venkatesan	23
40	Training Workshop for Vigilance Officers of ICAR Institutes	October 31, 2018	November 01, 2018	SK Soam Ashish Roy	35
41	Workshop on Krishi Kala Utsav - II	October 27, 2018	November 02, 2018	P Namdev M Krishnan S Senthil Vinayagam	26
42	Induction Training Programme to newly recruited Assistant Professors of PJTSAU (Batch - 3)	November 05, 2018	November 14, 2018	SK Soam PC Meena V V Sumanth Kumar	45
43	Training Programme on skill enhancing and employees development (SEED) for Retail Sales Managers of Coromandal International (Phase-II)	November 13, 2018	November 15, 2018	KH Rao Ranjit Kumar	30
44	Establishment and Financial Matters for Assistants/AAO/Aos/JAO/AF&AO/F&AO/Section Officers of ICAR (Off-Campus at ICAR-CPRI, Shimla)	November 15, 2018	November 20, 2018	JNL Das WS Bhat	26



S. No.	Title of the Program	Starting Date	Ending Date	Programme Director/s	No. of Participants
45	ICAR Winter School on ICT Interventions for Agricultural Development	November 28, 2018	December 18, 2018	Surya Rathore Alok Kumar VV Sumanth Kumar	26
46	Impact Assessment of Agriculture Research and Technology	December 04, 2018	December 07, 2018	B Ganesh Kumar I Sekha	16
47	MDP for Newly Recruited Programme Coordinators of KVKs	December 04, 2018	December 18, 2018	Surya Rathore P Venkatesan BS Sontakki	29
48	Establishment and Financial Matters for Assistants/AAO/Aos/JAO/AF&AO/F&AO/Section Officers of ICAR (Off Campus at ICAR-NIASM, Baramati)	December 04, 2018	December 14, 2018	JNL Das PG Kohad	17
49	Workshop for the Consultancy Project Developing Guidelines & Methodologies for IP Policy of PJTSAU	December 17, 2018	December 17, 2018	Manju Gerard	46
50	Training Programme on Skill Enhancing and Employees Development (SEED) for Retail Sales Managers Coromandel International (Phase-III)	December 17, 2018	December 19, 2018	KH Rao Ranjit Kumar	26
51	MDP on Priority Setting, Monitoring and Evaluation (PME) of Agriculture Research Project	December 17, 2018	December 22, 2018	P Krishnan B Ganesh Kumar	30
52	MDP on Leadership Development (a pre-RMP Programme)	December 18, 2018	December 29, 2018	RVS Rao Alok Kumar	30
53	MDP on Administrative & Financial Management for DS/CAO/CFAO/SAO/US/SFAO	December 19, 2018	December 22, 2018	Ashish Roy S George	20
54	Training Programme on Developing Winning Research Proposals (DWRP) in Agriculture Research for Faculty Members (Off Campus at University of Agriculture, Bangalore)	December 26, 2018	December 29, 2018	SK Soam P Krishnan	29
55	109 th Foundation Course for Agricultural Research Service (FOCARS)	January 04, 2019	April 03, 2019	S Senthil Vinayagam PC Meena	32



S. No.	Title of the Program	Starting Date	Ending Date	Programme Director/s	No. of Participants
56	Training Programme on Enhancing Efficiency and Behavioral Skills of Stenographers Grade- III, PA, PS PPS and Sr. PPS of ICAR (Off Campus at NBSS & LUP-Regional Station, Kolkata)	January 07, 2019	January 12, 2019	WS Bhat PG Kohad	24
57	Training Programme on Skill Enhancing and Employees Development (SEED) for Retail Sales Managers Coromandal International (Phase - IV)	January 21, 2019	January 23, 2019	KH Rao Ranjit Kumar	24
58	Entrepreneurship Development Programme for students (a-IDEA and CII)	January 21, 2019	January 25, 2019	K Srinivas Ranjit Kumar	25
59	Training Programme on Geospatial Analysis using QGIS & R	February 01, 2019	February 06, 2019	PD Sreekanth BS Yashavanth	14
60	MDP on Information and Communication Technologies for Empowering Form Women	February 01, 2019	February 06, 2019	NS Shanoy VKJ Rao	11
61	Faculty Development Programme for Increasing Efficiency, Effectiveness in Teaching, Research & Extension (Off Campus at UAHS, Shivamogga)	February 04, 2019	February 08, 2019	KH Rao BS Sintakki Alok Kumar	31
62	Hindi Training Programme	February 05, 2019	February 05, 2019	S Pradeep Singh	35
63	Training Programme on Emotional Intelligence for Personal and Work Excellence (Off Campus at CIFE, Mumbai)	February 11, 2019	February 13, 2019	P Ramesh RVS Rao	22
64	Training on Entrepreneurial Skills Development for Agri-Graduates of ANGRAU	February 14, 2019	March 06, 2019	Ranjit Kumar N Srivaramane	45
65	Capability building programme for agronomists of Coromandal International Ltd. (CIL)	February 18, 2019	February 21, 2019	KH Rao Ranjit Kumar	53
66	Training Programme on Analysis of Experimental Data	February 21, 2019	February 26, 2019	A Dhandapani S Ravichandran	17
67	Training Programme on KOHA for Library Staff of ICAR	February 21, 2019	February 26, 2019	N Srinivasa Rao SK Soam	37



S. No.	Title of the Program	Starting Date	Ending Date	Programme Director/s	No. of Participants
68	Training Programme on Digital Teaching Techniques for the faculty (off campus at College of Agriculture, UAS, Bangalore)	February 26, 2019	March 02, 2019	VV Sumath Kumar GRK Murthy	30
69	Capability Building Programme for Young Agribusiness Professionals of Coromandel International (Batch-I, Phase-II)	March 04, 2019	March 08, 2019	KH Rao Ranjit Kumar	25
70	Training Programme Skill Enhancing and Employees Development (SEED) for Retail Sales Managers of CIL (Phase-V)	March 06, 2019	March 08, 2019	KH Rao Ranjit Kumar	28
71	Developing Winning Research Proposals (DWRP) for Faculty of Rajmata Vijayaraje Scindhia, College of Agriculture (Off Campus at Indore)	March 07, 2019	March 09, 2019	SK Soam P Krishnan Ch Srinivasa Rao	25
72	Capability Building Programme for Young Agribusiness Professionals of Coromandel International (Batch-II, Phase-II)	March 11, 2019	March 15, 2019	KH Rao Ranjit Kumar	21
73	Training Programme on Motivation, Positive Thinking and Communication Skills for Technical Officers of ICAR (Off Campus at ICAR-IISWC, Dehradun)	March 13, 2019	March 19, 2019	BS Yashavanth P Vijender Reddy PD Sreekanth	24
74	MDP for HRD Nodal Officers of ICAR for effective implementation of Training Functions	March 14, 2019	March 16, 2019	RVS Rao KH Rao P Ramesh	23
75	Digital Teaching Techniques	March 14, 2019	March 20, 2019	VV Sumath Kumar GRK Murthy	25
76	Capability Building Programme for Young Agribusiness Professionals of Coromandel International (Batch-III, Phase-II)	March 17, 2019	March 21, 2019	KH Rao Ranjit Kumar	33
77	Training on Entrepreneurial Skill Development of ANGRAU Agri-Graduates	March 20, 2019	April 09, 2019	Ranjit Kumar K Srinivas Sanjiv Kumar	45
78	Massive Open online course (MOOC) on Teaching Excellence	November 01, 2018	November 13, 2018	D Thammi Raju GRK Murthy	1119
Total No. of Participants					3417

Educational Programmes (ongoing)

Post Graduate Diploma in Management Agriculture June, 2017 (9 th Batch)	PGDMA 2017-19	June 2017	April 2019	S Ravichandran P Krishnan	34(19)	Students
Post Graduate Diploma in Management in Agriculture (10 th Batch)	PGDMA 2018-20	June 2018	April 2020	D Thammi Raju (upto March 19, 2019) Sanjiv Kumar (March 20, 2019 on wards) M Dastagiri	27(12)	Students
Post Graduate Diploma in Technology Management in Agriculture	PGD-TMA 2018	June 2018	May 19	Manju Gerard	34(7)	Students
Post Graduate Diploma in Educational Technology and Management	PGD-ETM 2019	March 19	February 20	S Senthil Vinayagam	48(19)	Students

Within parenthesis-no. of girl students

2.9 PGDMA Students' Guidance by NAARM Faculty

2.9.1 Project report submitted by the 9th batch of PGDMA students

SI No.	Roll No.	Student Name	Title of the project	Academy Mentor/ Guide
1	1701	Adapa Teja	Development of automatized logistics model	Dr S Ravichandran
2	1702	Ankita Kala	Impact assessment of Arize hybrid rice in Nepal and Chhattisgarh	Dr SK Soam
3	1703	Anushree	Credit risk analysis for Rallis in Secunderabad and Guntur Zone	Dr KH Rao
4	1704	Anu Choudhary	Rasi seeds business opportunity ,product mapping of field crops and sales strategy for Himachal Pradesh	Dr D Thammi Raju
5	1705	Balambika .M	Enabling Central Know Your Customer for Sammunati Financial Intermediaries and Services Private Limited	Dr PD Sreekanth
6	1706	Bhargavi .D	Assessment of capacity development needs of Agri startups	Dr I Sekar
7	1708	Chandra Shekar Cheruku	Study on value chain of agroforestry systems in Mahabubnagar & Medak Districts of Telangana	Dr P Krishnan
8	1709	Damini	Distribution adequacy analysis for agri inputs: Benchmarking Rallis with key competitors.	Dr Ranjit Kumar
9	1710	Dasari Harika	Identification of potential districts in Telangana for Samunnati	Dr N Sivaramane
10	1711	Deny Devassy	Assessment of capacity development needs of Agri startups	Dr BS Sontakki
11	1712	Dharshana V	Scope and potential repositories in commodity market in Tamil Nadu	Dr Surya Rathore
12	1713	Dhivagar .B	Assessment of Direct Benefit Transfer of fertilizer industry in key districts of Tamil Nadu.	Dr N Srinivasa Rao
13	1714	Dhivya B	Status, Prospects and strategies for business development of SamAgro	Dr M Krishnan



14	1715	Goundeboyina Sheshank Yadav	Bringing operational excellence in trade of Samunnati	Dr A Dhandapani
15	1716	Karthika R	Modules in loan origination system and profiling of startups	Dr K Kareemulla
16	1717	Mallikarjun G	Identification and categorization of organic farmers in Dharwad catchment, Bidar and Gulbarga, Karnataka	Dr Alok Kumar
17	1718	Manjushree S N	A study on market potential and output marketing strategies of FPOs in fruits and vegetables in Karnataka	Dr VKJ Rao
18	1719	Mankala Santhosh	Study on <i>Rabi</i> hybrid paddy markets of Telangana and Odisha States	Dr M Balakrishnan
19	1720	Marigouda	Study of factors influencing adoption of eNAM	Dr N Sivaramane
20	1721	Merin Maria Thomas	Leveraging data analytics to enhance supply chain efficiency and Implementation of ISO 22000: 2018	Dr N Senthil Vinayagam
21	1722	Minati Majhi	Scope of improvement of farm profitability and productivity of pulpwood plantation in Malkangiri district of Odisha	Dr P Venkatesan
22	1723	Pasula Vinaykumar	Sulphur based fertilisers utilization strategy for high sulphur demanding crops in India and its socio-economic impact	Dr Ch Srinivasa Rao
23	1724	Pooja N	Market research analysis of hybrid paddy seeds in Bihar	Dr B Ganesh Kumar
24	1725	Prakhar Rathi	Scope of repository ecosystem in commodity sector	Dr S Ravichandran
25	1726	Rahul Kumar Singh	Channel diagnostics and channel segmentation of pesticide dealers of Rallis India Ltd for Kolkata Zone.	Dr Sandhya Shenoy
26	1727	Rajratan	Assessment of effectiveness of e-NAM in Parbhani APMC market.	Dr B Ganesh Kumar
27	1728	Romit Kumar Singh	Agri extension impact assessment of ITC-Wheat Development Programme (WDP) in Uttar Pradesh	Dr Ch Srinivasa Rao
28	1729	Sayyeda Salima Asra	Identification of risk drivers and laying a mitigation framework w.r.t retail sector	Dr RVS Rao
29	1730	Shailendra Nageshwar	Understanding institutional lending opportunities for FPOs and developing a scorecard for creditworthiness	Dr PC Meena
30	1731	Sharuti Dogra	Develop a crop acreage prediction model and the expected acreage of key crops for Rasi Seeds for the year 2019	Dr P Ramesh
31	1732	Sruthy Antony	Assessing credit worthiness of farmers and development of credit score card	Dr Sanjiv Kumar
32	1733	Sivaranjani K	Lead value addition in the product line operations of cut vegetables and packaging	Dr S Ravichandran
33	1734	Usha B	Study of CSR landscape and development of CSR collaboration strategy for Samfins	Dr B S Yasavanth
34	1735	Yogesh R	Analyze the problems in meal transportation & ways to reduce the freight charges for the meal transportation from Latur to Tamil Nadu	Dr M Balakrishnan

2.9.2 Project report submitted by the students of PGD-TMA during 2018-19

1	33-TMA-426-17	Mr. K. Madhava Rao	Agriculture finance and crop insurance in Gajapati district and its major constraints
2	33-TMA-421-17	Mr. B Hemanth	Hybrid seed production technology management in rice, maize and cotton
3	33-TMA-439-17	Mr. NRVN Gowripathi Rao	A study on role of product development in technology management of agricultural machines
4	33-TMA-456-17	Dr Sanjay Kumar Singh	Impact of indigenously developed Peste des Petitis Ruminants (PPR) and Goat pox vaccine through Academia-Industry interventions
5	33-TMA-405-17	Dr Amit Gautam	Morphological characterization and diversity analysis of yield contributing traits of reference wheat varieties under DUS testing in PPV & FR Project
6	33-TMA-450-17	Mr. Prem Shanker	DUS Testing & PVP application for Okra Hybrid TMOH 2331
7	33-TMA-412-17	Mr. Balram Singh Yadav	Study & analysis of patent valuation methodologies used by a selected enterprise (QUALCOMM) for selected technology domain (SEPs of Digital Communication)
8	33-TMA-419-17	Mr. Gaurav Kumar	Impact of vermicompost on fertility of soil; a case study of Nalanda, Bihar.
9	33-TMA-437-17	Ms. Anusha Muthyala Kasu	Corn-Chemical Treatment and AI analysis in APAC (Asia-Pacific)
10	33-TMA-461-17	Mr. Surajit Malik	Gathuwan Rice of Chhattisgarh: Empirical evidences as potential geographic indication
11	33-TMA-406-17	Mr. Er. Amit Lohan	Self Help Groups (SHGs) Impact on women empowerment through microfinance: a field survey of Jind District, Haryana (India)
12	33-TMA-430-17	Mr. Manash Kumar Behera	An overall scenario of organic farming in India and its adaptability among farming community of India
13	33-TMA-427-17	Mr. Kunwar Zeeshan Khan	Supply chain management and post harvest losses of tomato in Bulandshahr district of Uttar Pradesh
14	33-TMA-433-17	Ms. Manisha Pandey	Estimation of supply chain management and post harvest losses of green chillies in Allahabad District, Uttar Pradesh

2.9.3 NAARM Faculty Member as Major Guide/Supervisor for other Research Students

Name of the faculty	Student	Title of the Thesis	Program MSc/PhD	University/ Institute
B. Ganesh Kumar	B Praveen Reddy	Marketing strategies of different local, domestic and multi-national brands of pesticides in Nizamabad district of Telangana	MBA (Agribusiness Management)	PJTSAU, Hyderabad
B. Ganesh Kumar	D Mounisha	A study on consumer behavior and consumption pattern on frozen non-veg food products in Hyderabad, Telangana	MBA (Agribusiness Management)	PJTSAU, Hyderabad
Ch. Srinivasa Rao	KC Nataraja	Potential of Biochar in Improving Soil Health and Crop Productivity under Maize - Groundnut Cropping System in Semi - Arid Alfisols of Southern Andhra Pradesh	Ph.D	S.V. Agricultural College (ANGRAU), Tirupati
M. Balakrishnan	M Naveen Kumar	Development of Hybrid Algorithms with an Interface for Accurate Prediction of Sugarcane Yield In Telangana	Ph.D	Bharathiar University, Coimbatore
	Md. Atheeq Sultan Ghor	Development of Hybrid Neuro-Fuzzy System for Forecasting of Turmeric Yield in Telangana	Ph.D	Bharathiar University , Coimbatore
	D Karunakaran	Development of Algorithm for prediction of fish yield in reservoirs using Artificial Neural Network	Ph.D	Bharathiar University, Coimbatore
Manju Gerard (Co-Guide)	Shyla M Haqq	Synthesis and characterization of silver nanoparticles biofabricated from medicinal plants: <i>Jasminum auriculatum</i> and <i>Chrysanthemum coronarium</i>	Ph.D	SHUATS, Allahabad
M Krishnan	Neha W Qureshi	Amplified Economics of an Intrinsic Common Use Resource – The Dal Lake	Ph.D	ICAR - CIFE, Mumbai
	Naorem Dinesh Singh	Propagative Price Transmission in Indian Shrimp Exports Markets	Ph.D	ICAR - CIFE, Mumbai
	Piyashi DebRoy	Resource Use Patterns, Trade-offs and Governance Strategies for Fisheries Development in Lake Rudrasagar, Tripura	Ph.D	ICAR - CIFE, Mumbai
N Sivaramane	Monalisha Kispotta	Economics of Cashew Processing Industries and Farmer's Linkages in Odisha State	MBA (Agribusiness Management)	PJTSAU, Hyderabad
P Krishnan	Dasari Bhoomaiah	Mapping the fisheries research base and productivity of fisheries scientists in India	Ph.D	PRIST University
S Senthil Vinayagam	Seema Kujur	A study on adoption of educational technology in teaching-learning process in Agricultural Universities.	M.Sc (Ag)	IGKV, Raipur
S Senthil Vinayagam	Payal Dewangan	Study on group dynamics of women's groups towards entrepreneurial development in agriculture	M.Sc (Ag)	IGKV, Raipur



Chapter - 3

Centres of Excellence

Centres of Excellence

The Academy has successfully set up two Centres of Excellence, viz Centre of Agri - Innovation (CAI) and Centre for Open and Lifelong Learning in Agriculture (COLLAge).

3.1 Centre for Agri-Innovation (CAI)

The current challenges facing agriculture include not only enhancing productivity and production of agriculture produce but also building opportunities for agricultural goods in the production-consumption continuum. These call for the skill of management of research output towards technologies and up-scaling these to processes and products of commercial value. Therefore, Centre for Agri-Innovation was set up where in technology management, entrepreneurship and incubation, acceleration and investment with strong foresight study are promoted. The initial hand-holding attempts made by this centre including ICAR-funded Agri Business Incubator (ABI) has led towards creating institutional culture for creating value chains through technology transfer from ICAR institute onwards two entrepreneurship channels. The Centre provides hand holding support to the Agri Business incubators by inter-alia developing modules, guidelines and other form as of learning material for capacity building for the startups; and also provides need based support for analysis/valuation of the commercial potential of technologies; negotiation skills; monitoring of IP-rights and license agreements and marketing instruments. The main objectives of the Centre are: to organize training & capacity building programmes on technology transfer, IP management, business incubation, entrepreneurship development and investment; to develop foresight and policy framework for governance and coordination of technology development and commercialization in the NARES and incubation development for agribusiness entrepreneurs and to provide a platform for up-

streaming and down-streaming of technologies through advisory services for IP and technology management leading to entrepreneurship development.

The Centre for Agri-Innovation of the Academy functions through the technology business incubators from ICAR-NAIF (National Agriculture Innovation Fund), Department of Science & Technology (DST), Government of India (GoI) supported Business incubator a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture). It also supports activities of BioNest, an Agri-Bio-incubator for Agri-Biotechnology Startups. BioNest has received the financial support from BIRAC, DBT, GoI, which runs in partnership with Agri-Biotech Foundation (ABF), Rajendranagar, Hyderabad.

The a-IDEA is presently focusing on different sub-sectors of agriculture innovations across the value chain in agriculture:

(i) Sustainable inputs	(viii) ICT & IoT in agriculture,
(ii) Precision/smart agriculture	(ix) Animal husbandry
(iii) Innovative food technology	(x) Urban/ vertical farming
(iv) Supply chain technology	(xi) Agri-biotech
(v) Fisheries	(xii) Post-harvest technology
(vi) Soil, water & weather technologies	(xiii) Farm mechanization
(vii) Farm fresh retail	

The a-IDEA has carried out activities not only for the incubation process but also for ecosystem development, and seed investment in agri-startups. In 2018-19, it has incubated 17 startups and graduated seven startups. It has taken decision to invest in one of the accelerator startup Delmos Research Pvt Ltd from DST seed support fund.

A. Incubatee Startups 2018-19

Sl. No.	Name of Incubatee	Year of induction	Industry Sector	Company Profile
1.	Saro Agri and Dairy Pvt. Ltd.	December 2015	Animal Husbandry	Supply chain of A2 grade cow milk http://saronatural.in/
2.	Bharat Rohan Innovations Pvt. Ltd.	April 2016	Precision/ Smart Agriculture	Hyper spectral aerial (UAV) imagery and advisory services. http://www.bharatrohan.in/
3.	Fresco Greens Pvt. Ltd.	September 2016	Urban/Vertical Farming	Modular rooftop kits plug & play model for organic vegetable & fruit cultivation www.homecrop.in
4.	Stamp IT Business Solutions LLP	February 2017	ICT in Agriculture	IT business process automation for Agri companies (B2B) www.stampit.biz
5.	Vertical Farming Technologies Pvt. Ltd.	November 2017	Urban/Vertical Farming	Enabling city dwellers to grow safe, better and fresh produce by using Innovative technology like Vertical farming & Hydroponics https://citygreens.in/
6.	Zoofresh Foods Pvt. Ltd.	November 2017	Fisheries	Transport & Retailing live fish using high –tech logistics to bring down wastages in fish supply chain.
7.	Transity Digital Solutions, Pvt. Ltd.	January 2018	Supply Chain Technology	Offer a SaaS and technology platform that automates the workflows, vehicle allocation, scheduling and routing. http://transity.co/
8.	Skycraft Aerospace Pvt. Ltd.	January 2018	Precision/Smart Agricultures	Addresses commercial and enterprise drone needs. http://www.skykrafts.com/
9.	Jovaki Agro Foods India Pvt. Ltd.	June 2018	Innovative Food Technology	Working to highlight and understand the importance of setting up a Custard apple value chain for these local farmers and rural families. https://www.jovaki.com/about/
10.	Robic Rufarm India Pvt. Ltd.	July 2018	ICT in Agriculture	Engaged in designing a product that ensures shrimps health like pH and dissolved oxygen level of water to upload in the cloud and provide real-time updates regarding the status of pond using the water quality monitoring system through a portable device. https://robicrufarm.com/
11.	Bloom Foods India	August 2018	Innovative Food Technology	Engaged in production and marketing of the Ready to Eat (RTE) foods with a special focus on traditional Non-veg pickles. https://bloomfoods.co.in/

B. Graduated Startups 2018-19

1.	Millets n Cereals Food Products.	December 2015	Innovative Food Technology	Millet based value added products for social impact http://www.milletbowl.in/
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2.	GenAgritech	August 2016	Sustainable inputs	Naturally derived bio-repellents, nutrient & other plant growth supplements. http://www.genagritech.com/
3.	Greenovate Agrotech Pvt. Ltd.	February 2017	Sustainable inputs	Manufacturing and marketing of crop specific micronutrient technology (technology licensed from ICAR). www.greenovateagro.com
4.	Cocogreens Foods LLP	December 2017	Supply Chain Technology	Linking farmers and primary producers directly across India with retailers and consumers. https://www.cocogreen.co.in/



Events conducted by a-IDEA, NAARM during 2018-19

a-IDEA has conducted various programmes in different parts of the country so as to reach as many agri entrepreneurs as possible. Listed here are the events:

S. No	Name of the Event	Location	Date
1	Startup Samvaad	Hyderabad	July 7, 2018
2	Workshop on Agriculture Data Platforms	Hyderabad	July 28, 2018
3	Startup Samvaad	KL University, Vijaywada	August 3, 2018
4	Product Launch of SkyKrafts	Hyderabad	August 15, 2018
5	The Promise of Agritech	Bangalore	August 30, 2018
6	a-IDEA and WE-HUB co-incubation review meeting of women agri-entrepreneurs	a-IDEA, Hyderabad	September 1, 2018
7	Startup Samvaad	Warangal	September 15, 2018
8	“UPAYA Conclave” organized along with ICAR	New Delhi	October 16-17, 2018
9	STARTUP360 “a peer to peer learning & mentoring session’s for incubatees of a-IDEA & Bionest” in association with Capria	Hyderabad	November 10, 2018
10	One-day Sensitization workshop	GITAM University, Visakhapatnam	November 13, 2018
11	Workshop on “Agri-startup & Incubator Ecosystem	NABARD, Mumbai	November 14, 2018
12	Startup Samvaad	College of Veterinary Science & Animal Husbandry, Khanapara Guwahati	December 1, 2018
13	Startup Samvaad	College of Agriculture, IGKV Raipur	December 28-29, 2018
14	Sensitization workshop	College of Agriculture, Bapla, Andhra Pradesh	January 8, 2019
15	Entrepreneurship Development Program	a-IDEA, NAARM	January 21-25, 2019
16	STARTUP360	a-IDEA, NAARM	January 23, 2019
17	ISBA Managers	Mysore, Karnataka	February 1-3, 2019
18	Pitch Perfect	Hyderabad	February 16, 2019
19	Startup Samvaad	ICAR - Vivekananda Parvatiya Krishi Anusandhan Sansthan (VPKAS), Almora, Uttarakhand	March 25, 2018

Glimpses of the events



Pitch Perfect at NAARM



Startup Samvaad at NAARM



Startup Samvaad at Warangal



Startup Samvaad at Raipur



Startup Samvaad at Guwahati



Sensitization Workshop for Agri Graduates in Bapatla, AP

Workshop on Agriculture Data Platform

a-IDEA conducted a workshop based on insights gained during the project work of Mr. Drew DiPrinzio, US Fulbright-Nehru Student Research Scholar of the Academy. The Fulbright project work was completed at ICAR-NAARM from August 2017 to August 2018. Over the course of 12 months, Drew interviewed stakeholders from the public and private sector and conducted a farmer survey in Tamil Nadu, with the ultimate goal of forming a case study on the emerging agricultural data ecosystem. ICAR-NAARM Director Dr Ch Srinivasa Rao, Former Jt. Director Dr Kalpana

Sastry and Principal Scientists Dr Ranjit Kumar, Dr Dhandapani and Dr K Srinivas were all instrumental in providing insight and connections for this research project. The primary goals of the workshop were to discuss how technology can assist in gathering and using agricultural data, and to outline the roles of various stakeholders in developing agricultural data platforms. About 35 stakeholders across the public and private sector participated, ranging from agricultural research institutions, government bodies, agribusiness corporates, promoters of farmer producer organizations, investors, incubators and startups.



Participants discussing during the workshop



Workshop on Agri Startup & Incubator Ecosystem

a-IDEA in association with NABARD has jointly organized a workshop on “Agri Startup & Incubator Ecosystem”. During the workshop officials interacted

with the Chairman of NABARD Shri Harsh Kumar Bhanwal, Shri Harkesh Kumar Mittal, Secretary NSTEDB, DST & had great opportunity to present success story of a-IDEA, the business incubator of ICAR-NAARM.



Participants during NABARD Workshop



Agri Startup and Entrepreneurship Conclave 2018 organized by ICAR and NAARM

Agri Startup and Entrepreneurship Conclave 2018 for Unleashing Potentials in Agriculture for Young

Agripreneures (UPAYA) was organized at New Delhi during October 16-17, 2018. ICAR-NAARM was an active partner in organizing the programme. Dr K Srinivas, Pr. Scientist and PI of NAARM

business incubator was the organizing secretary. Union Minister for Agriculture and Farmers' Welfare Shri Radha Mohan Singh inaugurated the two-day Agri Startup and Entrepreneurship Conclave organized at Pusa, New Delhi on the occasion of World Food Day, October 16, 2018. In his inaugural address he emphasized the purpose of celebrating the Food Day this year is to achieve the global commitment to eliminate hunger by the year 2030. Minister said that the techniques developed by the Indian Council of Agricultural Research have a great contribution in enhancing agricultural production and ensuring food security in India. He stated that according to the fourth advance assessment in 2017-18 this year, food grains

production is expected to be 284.83 million tonnes. He highlighted that with increased production, many difficulties may be arising in the value chain and the startupentrepreneurs cansolvethesedifficulties,through their innovative solutions. On this occasion, Union Minister of State for Agriculture and Kisan Kalyan, Shri Parshottam Rupala; Dr Trilochan Mohapatra, Secretary, DARE & DG ICAR, Mr. Chabilendra Raul, Special Secretary (DARE) and Secretary (ICAR) and Mr. Bimbadhar Pradhan, Additional Secretary & Finance Advisor (ICAR) were also present. A book on "Agri-startups: Reflection of ICAR technologies in Market (AgRIM)" showcasing 100 successful startups of ICAR was also released on the occasion.



Union Minister for Agriculture and Farmers' Welfare, Shri Radha Mohan Singh and Union Minister of State Sh. G Shehaawat visiting the exhibition stalls and addressing during the workshop



Release of the book "AgRIM"

Prominent activities in which a-IDEA, NAARM participated

Sl. No.	Name of the Event	Location	Date
1	Villgro GES Unconvention Speaker Series	Hyderabad	April 20, 2018
2	National Award for ETBI	Delhi	May 11, 2018
3	German Indian Startup Exchange Program	Bangalore	June 7, 2018
4	Yes Scale Accelerator program launch	Delhi	June 8, 2018
5	Smart Village Program - AIC - SKU	Anantapur	June 12, 2018
6	Smart India Hackathon - Hardware	IIT - Kharagpur	June 23, 2018
7	The Promise of Agritech	Bangalore	August 30, 2018
8	Progress of Bionest (an agriculture Biotech incubator of a-IDEA, with Agri Biotech Foundation) to the BIRAC review team	a-IDEA, Hyderabad	August 31, 2018
9	a-IDEA and WE-HUB co-incubation review meeting of women agri entrepreneurs	a-IDEA, Hyderabad	September 1, 2018
10	Incubator meet	IIT Hyderabad	September 7, 2018
11	Incubation workshop of Venture base camp, Capria	Bangalore	September 25-30, 2018
12	Telangana Startup Yatra for Demo day	Hyderabad	October 4, 2018
13	Prabhav 2018, organized by Impact Investors Council		October 6, 2018
14	EU - India network launch	Bangalore	October 8-9, 2018
15	"Local Startups' Meet" at New Delhi Organized by Entrepreneurship Cell, IIT Kharagpur and Omnivore	New Delhi	October 15, 2018
16	YES Scale Agritech program	Bangalore	November 2, 2018
17	As a panel member at Bio-Economy Conclave 2019 organized by India's leading Biotechnology Industry Association "ABLE"	Bangalore	January 31-February 1, 2019
18	Training organized by NEXUS Startup Hub and conducted by Glenn Robinson, Assistant Director of the IC2 Institute	American Corner, Begumpet, Hyderabad	January 28-29, 2019
19	ISBA Managers Conclave to brainstorm on Unleashing Innovative & Sustainable Incubators	Mysore, Karnataka	February 1-3, 2019
20	"Startup Entrepreneurs 2019"	Hyderabad	January 24, 2019
21	AQUAEX INDIA - 2019 technology expo on fisheries & Aquaculture for South Asia at Hitex Exhibition Center	Hyderabad	January 31-February 2, 2019
22	Discovery team visit at a-IDEA office for a documentary of Urban Kissan under the Planet Healer Program of Discovery Channel	Hyderabad	February 21, 2019
23	Incubators meet at Dlabs	Dlabs, Hyderabad	February 22, 2019
24	Innovative startups to progressive farmers from Tamil Nadu	a-IDEA, Hyderabad	February 27, 2019
25	BIRAC's BioNest Meet at India Habitat Center, New	New Delhi	March 1, 2019
26	Agri startup funding ecosystem "Funding of Agribusiness Startups"	New Delhi	March 7, 2019
27	Incubatee startup "Urban Kissan" being featured in Planet Healer of Discovery Channel.	ISB, Hyderabad	March 8, 2019
28	Agritech panel member organized by AWS in association with Omnivore at T-Hub to discuss on "Will Agritech Create the next Unicorn"	Hyderabad	March 9, 2019
29	International Conference on "Entrepreneurship in Agriculture and Renewable Energy Sector"	Akola, Maharashtra	March 15, 2019
30	7th Foundation Day of BIRAC India on the Theme "Nurturing Innovation: Empowering India"	New Delhi	March 19-20, 2019

Few Snapshots



Villgro Unconventional



National Award for Emerging Technology Business Incubator 2017" to aIDEA NAARM



Smart Village Sarpanch Meet



Agri Vikaas 2018 at Bhubaneswar



Capria Venture Base Camp - STARTUP360



Ideation / Krishi Kalp at BHU, Varanasi

Some Achievements of Incubates of aIDEA, NAARM

S. No.	Forums
1	Skykrafts designed and developed a 20 ltrs drone indigenously and launched at NAARM by Dr Ch. Srinivas Rao, Director, NAARM
2	A case study developed by NASSCOM Community about the products and services of Bharat Rohan Airborne Innovations Pvt. Ltd. http://bit.ly/NCAG24 .
3	Bharat Rohan selected to become a part of the Envision Accelerator Program of DLabs at the Indian School of Business (ISB).
4	AgNext team invited to participate as a panelist at the Global Dialogue on Digital Pathways in Agriculture 2.0. Also, been awarded at Esri India, GIS AgTech Summit.
5	City Greens hosted an inter-ministerial delegation from Narok Country Government, Kenya, consisting of Ministry of Agriculture and Ministry of Trade.
6	Urban Kisaan featured in planet healers of Discovery channel.
7	Ms. Jaya Nabulla, Co-founder, StampIT Business Solutions LLP awarded as the best women entrepreneur by Eenadu Vasundhara.
8	Agricx won MP Krishi Bhushan award for disrupting age old grade assessment method thus empowering farmers and agri supply chain stakeholders to make objective assessment anytime, anywhere.



Best Women Entrepreneur Award by StampIT Business Solutions



Bharat Rohan at Global Future Business Forum



City Greens with delegates from Kenya



MP Krishi Bhushan Award by Agricx

3.2 Centre for Open and Lifelong Learning in Agriculture (COLLAge)

Academy has spearheaded the e-learning movement during last decade to enhance quality and access of educational content in agricultural universities. The initiative has resulted into establishing a state-of-the-art technology enhanced learning facility which has been instrumental in spreading high-end digital content production and management. Considering the rapid developments in the area of technology enhanced learning, the Centre for Open and Lifelong Learning in Agriculture (COLLAge) at the Academy has been set up. The COLLAge will aim to function as a Centre to provide guidance, consultancy and support in initiating advanced digital learning activities across various institutions in NARES. The centre will provide support in latest advances in technology pertaining to learning processes through its continual updating and upgrading in technological advancements.

As a part of activities under COLLAge, existing facilities of TELAge were upgraded and updated. On hardware front,

- Fourth PTZ capturing camera is added with high resolution and optical focus. This will enhance the quality of content besides adding more visual details to the digital content.
- In order to capture the audience (learners) responses during a live session, it is necessary to add audio capture devices without hampering the camera recording process during content production. For this purpose, four surround microphones which are connected to audio mixer are added.
- High end video streaming server is added for developing multimedia enriched content.



PTZ camera



Roof hanging surround microphone



Video streaming server

On software front

- Dragon software for converting audio to text, as needed in videos' production is procured.
- Articulate for content production is added for developing and enriching e-learning content.

Activities were taken up in conjunction with those assigned under TELAge project.

- Capacity building in the area of digital teaching techniques was offered at the Academy during March 14-20, 2019 and as off campus programme

for the participants of UAS, Bangalore during February 26 - March 2, 2019.

- Handholding for the digital content production in consultancy mode for MANAGE and NIPHM. Few more underway for PJTSAU and ICAR-NDRI.



Chapter - 4

Happenings at NAARM

Happenings at NAARM

4.1 Accolades

4.1.1 NAARM's Technology Business Incubator: a-IDEA Bags National Award

a-IDEA, Technology Business Incubator of NAARM won “National Award for Emerging Technology Business Incubator 2017” from Department of Science & Technology, Govt. of India. Dr Ch. Srinivasa Rao, Director, NAARM & President, a-IDEA along with Mr. Vijay Nadiminti, COO, aIDEA received the award from Dr Harsh Vardhan, Hon’ble Union Minister for Science & Technology, Earth Sciences, Environment, Forests and Climate Change, Govt. of India at a function held on Technology Day at Vigyan Bhawan, New Delhi on May 11, 2018. The event was graced by Shri Ram Nath Kovind, Hon’ble President of India.



a-IDEA, NAARM awarded as Emerging Technology Business Incubator

4.1.2 NAARM Bags ‘Best Stall’ Award

The Academy had put its stall at Pragati Maidan, New Delhi during International Agricultural and Horti-Expo-cum Show on Government Achievements and Schemes during July 27 - 29, 2018 organized by NNS Media Group at New Delhi. The stall was visited by different kinds of stakeholders which included farmers, students, teachers, researchers, entrepreneurs, ministers, government employees and general public. NAARM stall was adjudged ‘Best Stall’ during this 2-day International Expo based on the

multidimensional showcase of NAARM’s Education, Extension and Training activities through audio visual as well as verbal presentation.



NAARM's Stall awarded as “Best Stall”

4.2 Special Events

4.2.2 Expert Committee Meeting (Agriculture & Allied Sciences) of DST under Women Scientist Scheme-B (WOS-B)

NAARM hosted the Expert Committee Meeting (Agriculture & Allied Sciences) of DST under Women Scientist Scheme-B (WOS-B) during May 8-9, 2018. Using S&T interventions for the social benefit was the focus area for evaluating the projects. About 40 women participants attended the meeting chaired by Dr Sudha Nair, CEO of Golden Jubilee Bio-Tech Park for Women, Chennai. Dr Vandana Singh from DST was also present during the meeting.



Expert Committee Meeting under Women Scientist Scheme-B (WOS-B)

4.2.3 NAAS Workshop on Ranking of ICAR Institutes

A brainstorming workshop on ‘Development of Proforma for Ranking of ICAR Institutes’ was conducted at the Academy in collaboration with National Academy of Agricultural Sciences (NAAS), New Delhi on July 20, 2018.



Delegates of the workshop

About 35 senior level scientists from several South bound ICAR institutes participated in the workshop. Dr J.C. Katyal, Former DDG (Education) and Convener of Core Group, chaired the inaugural and plenary sessions. Dr A.K. Singh, Secretary, NAAS and Dr Arvind Kumar, Vice Chancellor, Rani Laxmi Bai Central Agricultural University, Jhansi were the Guests of Honour. Dr Ch. Srinivasa Rao, Director, ICAR-NAARM & Convener, NAAS Chapter, Hyderabad also presented various aspects of the ranking criteria and indicators.

4.2.4 National workshop on Artificial Intelligence in Agriculture

A two-days National Workshop on ‘Artificial Intelligence in Agriculture: Status and Prospects’ was jointly organized by ICAR-IASRI, New Delhi with ICAR-NAARM, Hyderabad and ICAR-IIWM, Bhubaneswar at NASC, New Delhi during July 30-31, 2018. The broad objectives of the workshop were to discuss various possible AI based applications in agriculture and development; to explore the possibilities of incorporating AI based systems in potential areas of agriculture, and to provide a platform for agricultural scientists to deliberate on AI and its implementation in agriculture. Sh. Amitabh Kant, CEO, NITI Ayog was the Chief Guest of the inaugural session. Dr Trilochan Mohapatra, Secretary, DARE & DG, ICAR and Sh. Chhabilendra Roul, Special Secretary, DARE & Secretary, ICAR and DDGs of various subject matter divisions of ICAR graced the occasion. Sh. Amitabh Kant delivered the inaugural address and congratulated and appreciated ICAR for taking up the initiative of using AI in agriculture. NAARM Policy Brief No (2), “Artificial Intelligence and Internet of Things: Implications for Human Resources in Indian NARES” was released by the Chief Guest.



Director, NAARM making a presentation



Policy Brief release on AI & IoT

4.2.5 QRT Meeting

The first meeting of 7th Quinquennial Review Team (QRT) of NAARM was conducted on August 6, 2018 under Chairmanship of Dr Mangala Rai, Former Secretary, DARE and Director General, ICAR; Dr B. Venkateswarlu, Former Vice Chancellor, Vasantao Naik Marathwada Krishi Vidyapeeth, Parbhani, Maharashtra; Dr (Mrs.) Ranjini Warriar, Former Advisor, MoEF & Climate Change, New Delhi; Dr Praveen Arora, Advisor & Head, National Science & Technology Management Information System (NSTMIS), DST, New Delhi; Dr Ch. Srinivasa Rao, Director, NAARM; Heads of Divisions; Head, PME Cell; JDA&R and CFAO of NAARM attended

the meeting. The Chairman, QRT, emphasized on research quality and perspective visualization of the future plan of different programmes. He informed that Academy should take lead role in ranking of ICAR institutes and SAUs which should be based on purpose and by discipline. Citing example of automation level of Chitale Dairy, Pune, he stressed that ICAR institutes should develop prototypes for automation and mechanization of small agriculture farms in India. Dr S.K. Soam, Member Secretary presented action taken report and meeting Agenda to the members of QRT. All the Heads of six divisions of the Academy presented the work progress for the period 2012-17. The Chairman emphasized that NAARM should be the think tank for ICAR system.



Dr Mangala Rai interacting with QRT members

4.2.6 Institute Research Council (IRC) Meeting

The 21st IRC of the Academy was held during September 10-12, 2018 at the Committee Room, Faculty Centre, NAARM. The meeting was chaired by the Director, NAARM and all the faculty members participated. Dr P. Krishnan, Member secretary, IRC provided an overview of the research projects operated in the Academy and the collaborations forged by NAARM with various organizations. Progress with respect to 27 research projects were presented by the faculty members. Dr Ch. Srinivasa Rao, Director, NAARM and Chairman, IRC emphasized that the research studies need to have specific focus to strengthen the role of the Academy to serve as think tank of ICAR.

staff members of the Academy attended this programme. Dr Mohapatra said that “the budding scientists should learn from the life of Dr M.V. Rao on how to navigate from dream to destination without compromising on personal and professional front”.



Dr Trilochan Mahapatra, DG, ICAR and Secretary, DARE, delivering the lecture

4.2.7 Dr M.V. Rao Memorial Lecture at NAARM

Dr Trilochana Mohapatra, DG, ICAR and Secretary, DARE, Govt of India delivered Dr M.V. Rao memorial lecture at NAARM on September 22, 2018. The probationer-scientists of 108th FOCARS, trainees of various training programmes of the Academy, scientists,

4.2.8 RAC Meeting

The second meeting of 7th Research Advisory Committee (RAC) of NAARM was conducted

on December 21, 2018 under the Chairmanship of Padma Shri Prof. S.K. Sopory, Former Vice Chancellor, Jawaharlal Nehru University New Delhi. Dr Ch Srinivasa Rao, Director, NAARM welcomed and introduced the Chairman and members. He also presented an overview of the Academy. The RAC reviewed the research and capacity building progress, and suggestions were given for further improvements to attain global visibility. The RAC members viz. Dr R.K. Samanta, Shri M. Dharma Rao, Shri V. Venkateshwarlu and HoDs as co-opted members and Incharges as special invitees from NAARM attended the meeting. Dr S.K. Soam, Member Secretary presented action taken report to members of RAC.



Padma Shri Prof. SK Sopory, interacting with NAARM Faculty

4.2.9 IMC Meeting

The 56th meeting of Institute Management Committee (IMC) was held under the chairmanship of Dr Ch Srinivasa Rao, Director, ICAR-NAARM on December 22, 2018. The meeting was attended by Shri B. Pradhan, Additional Secretary, DARE & Financial Advisor, ICAR and Shri M. Dharma Rao as members of the Committee and Shri K.V. Ramaraju, JDA (Seeds and Crop Insurance), nominee of Special Commissioner of Agriculture, Department of Agriculture, Government of Andhra Pradesh; Smt. G. Narimani, Additional Director of Agriculture and nominee of Director of Agriculture, Department of Agriculture, Government of Telangana.



Group photo of IMC members

4.2.10 Krishi Kala Utsav-II

Krishi Kala Workshop was organized at the Academy during October 27 - November 02, 2018 in association with NAAS-Hyderabad Chapter. In this workshop, a team of 23 artists from Sir JJ School of Arts, Mumbai accompanied by faculty, Prof. Ananth Nikam and Smt. Shilpa Nikam & Shri Deepak Khogre, ACTO (Artist), ICAR-CIFE, Mumbai participated. About fifty two paintings highlighting agriculture, animal husbandry and fisheries activities were developed during this workshop.



Canvas painting by participant

4.2.11 Vigilance Awareness Week

The Vigilance Awareness Week was observed during October 29 - November 3, 2018, with the theme 'Eradicate Corruption - Build a New India'. The activities started with observance of pledge on October 29, 2018. Various events such as debate, lectures, painting competitions, erudition etc. were organized during the week. During closing ceremony, Dr Ch. Srinivasa Rao spoke about importance of vigilance and actions taken to bring transparency in administration.

The Chief Guest, Dr RM Gonela, IAS (Retd.) delivered a lecture on the theme and also distributed certificates to the winners of various competitions.



Observance of pledge during Vigilance Week

4.2.12 Expert Committee Meetings to Review Scientific Cadre Positions of NAARM

The National Committee to review the cadre strength of ICAR chaired by Secretary, DARE & Director General, ICAR, in its recommendations pertaining to the scientific cadre positions for ICAR-NAARM, envisaged that the Academy shall strive to evolve as a Think Tank for ICAR in the field of agriculture, in addition to its role in building the capacities of individuals and institutions under the NARES. The Council constituted an expert committee to review and determine the cadre positions, disciplines and roles, as suitable for the Academy.



Expert Committee discussing Cadre Positions of NAARM

The Committee conducted a detailed review of existing cadre positions of the Academy and interacted with the scientists, senior officials at the council/ASRB. The first meeting was held at New Delhi; following which the Committee met twice at NAARM, Hyderabad on

January 29, 2019 and Feb 18, 2019. The draft report was presented to Dr Trilochan Mohapatra, Secretary, DARE & Director General, ICAR on March 03, 2019. The Committee has finalized its report on the Scientific Cadre Positions for NAARM including a road map to aid in the transformation of the Academy into Think Tank of ICAR.

4.2.13 Monthly Meeting with Pensioners/Family Pensioners

The first monthly meeting with pensioners and family pensioners of ICAR Institutes located in Hyderabad was organized on September 18, 2018 in the Academy under the Chairmanship of Director, ICAR-NAARM. The meeting was organized as a part of good governance and to minimize their grievances related to pension on a pattern similar to the Pension Adalats being organized by Ministry/Department in the Government of India. The objective of the meeting is to grievances promptly within the framework of extant policy guidelines/rules. Heads of Administration & Finance from local ICAR Institutes and many pensioners participated in the meeting. The issues raised by participants in the meeting will be promptly addressed by officers concerned and the same will be reviewed in the next monthly meetings.

4.2.14 Exposure Visit of Skilled Support Staff of ICAR-NAARM to KVKs in Telangana State



Skilled Support Staff exposure to KVK technologies

An exposure visit for 27 Skilled Support Staff (SSS) of the Academy was arranged during March 17-19, 2019 to have an exposure about KVK system and technologies developed like Azola production, Vermi-

composting, Dall Mill processing, Sericulture Unit, Fish culture, plant propagation methods of Mango and Citrus etc. KVKs at Madanapuram, Palem and Gaddipally were visited during these three days.

4.3 Visit of Dignitaries at NAARM

4.3.1 Shri Gajendra Singh Shekhawat, Hon'ble Union Minister of State for Agriculture and Farmers Welfare, GoI Interacts with Farmers, Scientists and other Stakeholders

Shri Gajendra Singh Shekhawat, Union Minister of State for Agriculture & Farmers' Welfare, participated in the interactive session held at the Academy on March 09, 2019. During the session, Shri Shekhawat accentuated that the government's priority is for making the agriculture profitable by reducing the input cost and increasing the remunerative prices for enhancing the farmers' income. The Minister strongly said that by listening to the stakeholders will help in enhancing the government's functioning and decision making process. Shri Shekhawat had an interactive meeting with the farmers, farmer associations, scientists of all the ICAR Institutes located at Hyderabad, officials, students, start-ups and State Agri-Allied Sector Universities, etc. The Minister also released a NAARM Hindi publication titled "AgRIM" during the occasion.



Interaction session with Shri Gajendra Singh Shekhawat, Hon'ble Union Minister of State for Agriculture and Farmers Welfare, GoI

4.3.2 Secretary (DARE) & Director General (ICAR) Addressed the Scientist-Probationers

'Science Communication is important for agricultural scientists' said Dr Trilochan Mohapatra, Secretary

(DARE) & Director General (ICAR) while addressing scientists-probationers undergoing the 109th Foundation Course for Agricultural Research Service (FOCARS) at the Academy on Jan 18, 2019. He called upon the young scientists to come up with innovative ideas to undertake research with proper prioritization for the benefit of the stakeholders. He also shared the importance of new frontier areas like artificial intelligence, block chain technology and IoT in agricultural research in the areas of precision agriculture and post-harvest management. Dr Ch. Srinivasa Rao, Director emphasized the importance of positive thinking and outlined the profile of the FOCARS 109th batch which comprises 32 trainees from 16 disciplines in the Agricultural Research Service (ARS) from various states in the country.



DG addressing 109th FOCARS

4.3.3 Secretary, DBT Visits a-IDEA of ICAR-NAARM

Dr Renu Swarup, Secretary, Department of Biotechnology, Ministry of Science & Technology, GoI and Chairperson, BIRAC visited a-IDEA on February 21, 2019.



Dr Renu Swarup, DBT Secretary visit to a-IDEA, NAARM

During her visit, she also interacted with few start-ups who are being supported by a-IDEA in different areas in agri-value chain. She also visited a-IDEA partner, Agri Biotech Foundation (ABF) and inspected the wet lab facilities and work of Start-ups. The Director briefed that the Academy is a unique institute engaged in capacity building of NARES scientists, Research Policy (think-tank) of ICAR and Agri-business

management education. Dr K Srinivas, and CEO, a-IDEA briefed the delegation about the activities and progress of Incubator and Agri-BioNEST.

4.4 Memorandum of Understanding Signed

4.4.1 The Academy signed Memorandum of Understanding (MoU) with 5 different public and private organizations during this period.



NAARM signing MoUs

4.4.2 A Memorandum of Understanding (MoU) between a-IDEA, NAARM, Hyderabad and Maharashtra Hybrid Seeds Company Private Limited (MAHYCO), Jalna, Maharashtra was signed to promote the startups programmes/events organized by MAHYCO for the larger benefit of Indian agriculture. The MoU was executed on August 6, 2018.

A Memorandum of Understanding (MoU) between a-IDEA, NAARM, Hyderabad and Millet Processing & Incubation Centre, Professor Jayashankar Telangana State Agricultural University (PJ TSAU), Hyderabad was signed to synergize the institutional capabilities and expertise to create complementing entrepreneurship ecosystem for millet based startups. The MoU was executed on August 16, 2018.



MoU signing between aIDEA, NAARM & PJ TSAU

4.5 Celebrations

4.5.1 NAARM Celebrates 43rd Foundation Day

The Academy celebrated its 43rd Foundation Day on September 1, 2018. Dr Peter Carberry, Director General, ICRISAT was the Chief Guest and delivered the Foundation Day address. During his speech, he said that “Indian Agricultural Human Resources are second to none in the world in agricultural research to reaching out to farmers”. He complemented the Academy for its unique service of developing the change agents to meet the challenges faced by the agricultural sector of the entire country. In his presidential address, Dr Ch. Srinivasa Rao, Director, NAARM said that strong collaborations are further needed between ICRISAT and NAARM to lead the small- farmer centric agricultural research and evolving several policies. NAARM Policy Brief No (3), “District-level Food Demand: A Better Estimate for Planning Food

Production and Nutritional Security” was released by the Chief Guest. On this occasion, various awards instituted by the Academy viz. ‘Best Employee of the Year’ and ‘Best Employee of the Decade’, award for various categories of the staff members, including the contractual staff members were given. The staff members were also awarded in recognition of their contribution towards *Swachh Bharath Abhiyaan*, maintenance of infrastructure and energy conservation. Plantation activities were also done on this day.



Dignitaries lighting the lamp



Best Employee Awardees of the Academy

4.5.2 SPICMACAY

The Academy hosted Kathak dance by Sangeetha Natak Academy Award Winner Pandit Rajendra Gangani and Padmasri Dr Nartaki Nataraj in association with the Society for the Promotion of Indian Classical Music and Culture Amongst Youth (SPIC-MACAY) on June 30, 2018 and February 11, 2019. Padmasri Dr Nartaki Nataraj gave a scintillating performance and the programme was the first one rendered by Dr Nartaki after receiving the Padmasri Award from the Govt. of

India on 26th January, 2019. She started the recital with *Sollukettu*, followed up with the piece, *Bo Shambo*

Shiva Shambo, in Rag, Revathi.



SPICMACY dance programme at NAARM

4.5.3 Independence Day Celebration at the Academy



Director, NAARM address on Independence Day

The Academy celebrated the 72nd Independence Day on August 15, 2018. Dr Ch. Srinivasa Rao hoisted the National flag and addressed scientists and staff of the Academy. He mentioned achievements of the Academy during the year. He reiterated to focus on the mandated activities of the Academy to bring newer heights in the coming years. Decorated lightings added beauty to the Academy. Winners of various competitions were given prizes by the Director.

4.5.4 Product launch of SkyKrafts Aerospace

On the eve of 72nd Independence Day of India, a-IDEA at NAARM celebrated with the Product launch of SkyKrafts Aerospace, one of the AgTech Startup's "Spraying Drone". The launch was facilitated by Dr Ch. Srinivas Rao, Director - NAARM, in the presence of staff of SkyKraft, aIDEA, NAARM and PGDMA Students of NAARM.



Launch of drone of SkyKrafts Aerospace

4.5.5 Hindi Pakhwada Celebrated in the Academy

The Academy celebrated Hindi *Pakhwada* during September 14-28, 2018. Various competitions were organized during the two-week long celebrations. Competitions were organized for scientists, office staff, FOCARS trainee scientists as well as for PGDMA students. The valedictory programme was organized on September 28, 2018. Prizes were distributed to the winners of the competitions by the Director on the occasion.



Hindi Pakhwada celebrations at NAARM

4.5.6. Swachhata Hi Seva Campaign

The Academy celebrated *Swachhata Hi Seva* during September 15 - October 02, 2018 which focused on beautification of the campus including pruning of tree branches, hedge cutting, cleaning of the campus. Dr AK Srivastava, Chairman, ASRB participated in the activity on September 19, 2018; and on September 22, 2018, a mass cleaning programme was organized in which Dr T. Mohapatra, Secretary, DARE and DG-ICAR participated. Bottle palm tree plantation was also taken up in the lawns of scientist home. Various competitions like essay writing, elocution and song competitions were conducted during the fortnight and prizes were distributed to the winners.



DG, ICAR participating in Swachhata drive

4.5.7 International Women's Day Celebrations

International Women's Day was celebrated at the Academy on March 8, 2019. The whole day celebration revolved around the theme given by United Nations, 'Think equal, build smart, innovate for change'. Different activities were organized for women at the Academy which included visual quiz and games on women of eminence. Expert lecture by Prof. Pari Plavi who talked on 'Balance for Better'. She talked about how women can balance their dual and triple roles & responsibilities with minimum stress. On this occasion, Dr Ch. Srinivasa Rao, Director released a book on 'Women in Agriculture (The Invisible Partners in Development)' and applauded the womens' role as managers in home and work. Around 75 women from scientific, technical and administrative cadre participated in the programme. As a part of charity work to the underprivileged, members of women cell of the Academy distributed soap, hair oil and clothes to 120 inmates of "Aaramghar: A home for destitute".



International Women's Day Celebrations at NAARM

4.5.8 ICAR-NAARM Organizes National Science Day

The Academy celebrated National Science Day on February 28, 2019 commemorating great contributions of noble laureate Sir CV Raman. The theme was: 'Science for the people and the people for the Science'. More than 100 students from Zilla Parishad High School, Shivarampally and Zilla Parishad High School, Budvel participated in the celebrations. Essay and elocution competitions were also conducted for school students. While addressing the students during closing ceremony, Dr Ch Srinivasa Rao, Director, ICAR-NAARM said "Science has made strides through various inventions for ease of living. The young generation should use technology wisely for the benefit, and should not addict to modern electronic gadgets". Dr SK Soam, Jt. Director explained about the importance of science in day to day life and about scientific development in the field of agriculture.



National Science Day celebrations

4.5.9 NAARM Celebrated Agricultural Education Day

Agricultural Education Day was celebrated at the Academy on December 3, 2018 to commemorate the Birth Day of Bharat Ratna, Dr Rajendra Prasad, first

Agriculture Minister of the country and First President of Independent India. Mr. V. Ram Kaundinya, Director General, Federation of Seed industry of India, was the Chief Guest on the Occasion. He delivered an invited lecture on “New Paradigm in Indian Agriculture”. In his address, he said that the agriculture education has to be promoted right from school education in rural areas as more than 60% of the population are living in rural areas. The final year students from College of Agribusiness Management, Baramati took part in the celebrations.



Mr. V. Ram Kaundinya addressing on Agriculture Education Day

4.5.10 Academy Celebrated Kisan Diwas with School Students

ICAR-NAARM celebrated National Farmers’ Day (Kisan Diwas) on December 23, 2018 to create awareness among young girls and boys on the importance of farmers and farming for a healthy and prosperous nation. The day was aimed to bring an interface between students (future pillars) and farmers, linking past, present and future. On a strong premise of “Seeing is believing” about 40 students consisting of 28 girls and 10 boys from Krishi High School, Rajendranagar, Hyderabad were taken on exposure cum interface visit to innovative farmer’s field. Essay writing and drawing competitions on the themes “Importance of Farming in India” and “Farmer and his Farm”, respectively were organized for the students.

4.6 Sports, Yoga and Meditation

4.6.1 4th International Day of Yoga

NAARM celebrated the 4th edition of International Day of Yoga on June 21, 2018. About 250 participants

including faculty & staffs of NAARM, PGDMA students & trainees of the Academy, employees of other ICAR institutes located at Hyderabad and students from nearby local schools took part in the celebration. The revised Common Yoga Protocol (CYP) as circulated by Govt. of India was performed from 6 - 7am at the lake view lawn of the Academy. Dr Ratnakar Rao, a renowned orthopedic surgeon, Continental Hospital was the Chief Guest on the occasion. He talked about the benefits of yoga, Vitamin D deficiency among school children and ways to manage it. Dr Ch. Srinivasa Rao, Director, NAARM also addressed the participants. Two yoga instructors, Mrs. Haseena and Mrs. Padma demonstrated different asanas.



International Yoga Day organized at NAARM

4.6.2 NAARM participated in the ICAR South Zone Sports Meet-2018

NAARM Contingent participated in the ICAR South Zone Sports Meet-2018 during September 5-9, 2018 at DSA Complex, Kakinada organized by CTRI Rajahmundry. Mrs. Rukmini won the Overall Champion (Women) Award (1st in shotput, 1st in discus, 3rd in javelin, winner in caroms, winner in shuttle badminton, runner in table tennis). The Academy also bagged prizes in men’s shotput (1st-MK Samson), Discus throw (1s - MK Samson), Javelin (2nd - MK Samson), Carrom - men’s singles (Runners-Mr. Sham Bahadur) and Chess - women (Winner - Smt. N. Vijayalaxmi). Among team events, the Academy bagged runners-up in both Table tennis- men’s doubles and basketball.



Sports Contingent of NAARM



Director, NAARM felicitating the winner

4.7 Campus Beautification of the Academy

Landscape Garden and Nursery Management

The existing landscape gardens of Hostels, Faculty block, Academic block, Administrative Block, Directors office, Auditorium and Lake side garden were taken care. Special focus was given on beautification of landscape gardens of Lake side garden, Directors Office, Faculty Block. The lawns and ornamental plants were maintained properly. The Lakeside garden has become an ideal venue for Yoga programmes, official get together, dinners etc.

Orchard crops:

The Agro-technology demonstration block for orchards comprise of 6 hectares which includes fruit crops viz. *mango* (15 varieties), *sapota* (5 varieties), *sweet lime*, *custard apple* (5 varieties), *amla* (3 varieties) & *phalsa*. Some more fruit plant species viz. *guava* (3 varieties) and *karonda* were introduced in the orchard block during the monsoon season of 2018. New drip irrigation system was installed in the fenced sweet lime plot (640 trees).

Rose garden:

The Academy has good collection of rose varieties in the floriculture block. It meets the need of floral bouquets requirement during different programmes & functions of the Academy. 200 rose bushes of 40 different varieties of HT roses, Indian roses, floribundas, polyanthas & miniature roses are grown under open conditions. **The Academy participated in the 37th All India Rose Show** during January 19-21, 2019 at the Institute of Engineers, Hyderabad. The

Academy has **won 14 prizes** under different categories of rose show. 06 First prizes, 05 Second prizes and 03 Third prizes.

Landscape garden:

Lush green lawns, colourful ornamental & flowering plants are maintained in the landscape gardens of the Academy which added beauty and created an aesthetic ambience in the campus. Renovation of lawns of Faculty building, IGH-2, Auditorium building was taken up along with sprinkler (pop-up) irrigation facility during this year. Bougainvillea avenue plantation was taken up along the main roads of the campus. Big size cement pots (*lanco pots*) were arranged with different shades of bougainvilleas near Director Office, auditorium and sports complex

Swachhta activities

The academy is maintaining its campus clean & green and it is declared plastic free. Several cleaning activities are being taken up under Swachhta programmes to keep the campus clean & beautiful. Swachhta pakhwara was observed at the academy twice during this year during 6th September - 2nd October, 2018 (Swachhta hi seva) and 16th-31st December, 2018 (Swachhta pakhwara). Under this program all storm water drains were cleaned, cleaning of surroundings of Type II, III, IV and V quarters was taken up, lime and red oxide painting of avenue tree trunks done, pruning of branches of avenue trees was taken up, cleaning of all the farm plots, farm pond area and lake was taken up, mass cleaning programmes were organised in the campus in which PG students, trainees, faculty, staff, TS & contractual workers are participated. Swachhta

competitions were held during Swachhta pakhwara. The faculty, staff, PG students, FOCARS trainees and school children participated in different categories like essay writing, elocution and song competition. Recycling of solid waste: leaf litter, grass clippings and other farm waste were collected, segregated and

used for making compost at the vermi-compost unit. Door to door garbage collection arrangements were made from residential areas in collaboration with GHMC. Dust bins for dry & wet waste collection were arranged in the residential areas.



Gladiolus in full blooms in the floriculture block of farm



The Faculty building lawn after renovation



Mass cleaning programme in the campus

4.8 Glimpses of Celebrations in the Academy



Cultural night by 108 FOCARS



New Years celebrations at NAARM



National Science Day celebrations



Cultural celebrations by PGDMA students



4.9 Distinguished Guest Faculty from Other Institutes

Name of Guest Faculty	Title of lecture
Dr Anupama Verma Member, NAAS, New Delhi	Is hunger free India, a dream or reality
Dr A.K. Singh Deputy Director General (Agricultural Extension), ICAR, New Delhi	Recent Developments in Agricultural Extension
Dr A Arunachalam Principal Scientific Officer to Secretary (DARE) & DG (ICAR)	Agro forestry policy in India: Status and implementation of India
Dr Ashok Dalwai Chairman, NRAA & Doubling Farmers Income, GoI	Doubling Farmers Income
Dr AK Srivastava Member ASRB, New Delhi	Food and Nutritional Security Challenges and Opportunities in India
Dr Bandla Gangaiah ICAR-Central Island Agricultural Research Institute, Port Blair	Research & Developmental Priorities for Island Ecosystem in India
Dr B Venkateswarlu Former Vice Chancellor Hyderabad	Drought Proofing in India
Dr H Pathak Director ICAR-National Rice Research Institute Cuttack, Odisha	Climate Change and Indian Agriculture
Dr JC Katyal Former DDG, Edu., ICAR	Pursuing Scientific Research as a Career Option
Dr JK Jena DDG, Fisheries, ICAR, New Delhi	Recent Developments in Fisher Sciences
Dr KM Reddy Ex. Principal Scientist NAARM	Creativity and Problem Solving
Dr Narendra Singh Rathore Deputy Director General (Education), ICAR, New Delhi	Research Priorities and Challenges in Agricultural Higher Education

Name of Guest Faculty	Title of lecture
Dr PS Birthal National Professor, NIAP, New Delhi	Transition in India's agri-food policy from food security to farm income security: Challenges and way forward
Prof. Panjab Singh President, NAAS Varanasi	Perspectives in Agricultural research and development
Dr PK Chakrabarty Assistant Director General (Plant Protection, ICAR)	Recent advances in Biotic Stress Management
Dr PK Joshi Director South Asia IFPRI, New Delhi	Indian Agriculture: Sustainable Development Goals
Dr PB Kirti Professor, School of Life Science, University of Hyderabad,	Advances in Biotechnology in Agriculture Systems
Dr RB Gaddagimath Dean, AIS, Bangalore	Scientific Information and Reference Citation
Dr SA Patil Ex. Dir., IARI, New Delhi	Indian Agriculture : Future Prospects
Dr S Senthil Kumar IGNFA, Dehradun	Bureaucracy and science interface and conflict management
Dr S K Choudhary ADG (S&W)	Developments in soil and water management research
Dr Suman P S Khanuja Founder & Chairman Flora Fauna Science Foundation (CIMAP, Lucknow)	Agri-Entrepreneurship towards enhanced farmers' income
Dr Shiv Prasad Kimothi ADG (TC) ICAR	National and International Collaboration of ICAR
Dr TK Adhya Director, South Asia Nitrogen Centre, New Delhi	Mitigation of Green House Gas Emissions in Rice based eco-systems in India
Dr Trilochan Mohapatra Secretary (DARE) & DG (ICAR), New Delhi	Dream to destiny



4.10 Visits of Students/Trainees from Various Agricultural Universities/ Colleges and Farmers

During the year under report, a total of 3440 visitors visited the Academy

S. No.	Details of College and University	Degree	Date of visit scheduled	No. of students (Girls)
1.	Vanavarayar Institute of Agriculture, Manakkadavu, Pollachi	B.Sc (Ag)	May 7, 2018	16 (14)
2.	Veer Kunwar Singh College of Agriculture, Dumraon, Buxar Bihar Agricultural University	B.Sc (Ag)	June 20, 2018	22 (7)
3.	Forest College and Research Institute, Mettupalayam, TNAU	B.Sc (Forestry)	June 27, 2018	44 (15)
4.	Participants of MANAGE Training programme in Agricultural Extension	Trainees	June 27, 2018	12 (3)
5.	Indian Institute of Horticultural Research, Bengaluru	Scientists	July 7, 2018	3
6.	University of Horticultural Sciences, Bagalkot	Ph.D Scholars	August 1, 2018	6 (3)
7.	Imayam Institute of Agriculture and Technology Kannanur, Thuraiyur, TN	B.Sc (Ag)	August 23, 2018	90
8.	RVS Padmavathy College of Horticulture (Affiliated to TNAU, Coimbatore)	IV B.Sc (Horti)	August 24, 2018	36 (20)
9.	Agriculture College and Research Institute, Coimbatore TNAU	B. Tech (Biotechnology) B. Tech (bioinformatics)	August 25, 2018	80 (50)
10.	Agricultural College and Research Institute TNAU, Madurai	B.Sc (Ag)	August 27, 2018	122 (82)
11.	Vanavarayar Institute of Agriculture, Coimbatore (Affiliated to TNAU)	B.Sc (Ag)	August 27, 2018	128 (79)
12.	JKK Munirajah College of Agricultural Science (Affiliated to TNAU)	B.Sc (Ag)	August 28, 2018	68 (34)
13.	Tamil Nadu Agricultural University Agricultural College and Research Institute Killikulam, Vallanadu-628252, Thoothukudi Dt	B.Sc (Ag)	August 28, 2018	107
14.	S. Thangapazham Agril College (Affiliated to TNAU) Vasudevanallur, TN	B.Sc (Ag)	August 28, 2018	49 (33)
15.	PGP College of Agril Sciences (Affiliated to TNAU), Namakkal, TN	B.Sc (Ag)	September 10, 2018	101 (73)
16.	SRS Institute of Agriculture and Technology (Affiliated to TNAU)	B.Sc (Ag)	September 14, 2018	48
17.	Don Basco College of Agriculture	B.Sc. (Ag)	September 15, 2018	74 (43)
18.	College of Agricultural Engineering UAS, GKV, Bengaluru	3 rd B.Sc (Agril Engineering)	Sept. 19, 2018	57 (20)
19.	Adhiparasakthi Agricultural College, Kalavai TNAU	Final B.Sc. (Ag)	September 19, 2018	72 (41)
20.	College of Agricultural Information Technology Anand Agricultural University, Anand, Gujarath	B. Tech. (AIT)	September 24, 2018	25 (13)
21.	Agricultural College and Research Institute TNAU, Thanjavur	IV year (B.Sc (Ag))	September 25, 2018	49(24)
22.	Sethu Bhaskara Agricultural college and Research Foundation Coimbatore (Affiliated to TNAU)	Final B.Sc (Ag)	September 26, 2018	44(31)
23.	Adhiparasakthi Agricultural College, Kalavai, TNAU	Final B.Sc. (Agri)	October 3, 2018	73 (43)
24.	University of Agricultural Sciences, Bengaluru College of Agriculture, GKVK, Bangalore (V batch)	III year (B.Sc (Ag))	October 3, 2018	51 (29)



S. No.	Details of College and University	Degree	Date of visit scheduled	No. of students (Girls)
25	University of Agricultural Sciences, Bengaluru College of Agriculture, GKVK, Bangalore	III year (B.Sc) (Ag)	October 3, 2018	60 (28)
26	Dept of Agril and Rural Management Centre for Agril and Rural Developmental Studies TNAU, Coimbatore-641003	Final B.S. (Agribusiness management)	October 3, 2018	42 (29)
27	College of Agricultural Technology (Affiliated to TNAU), Theni	IV B.Sc (Ag)	October 11, 2018	128 (82)
28	College of Horticulture, University of Horticultural Sciences, Bagalkot	Final B.Sc (Horti)	October 15, 2018	28 (14)
29	Pandit Jawaharlal Nehru College of Agriculture and Research Institute, Karaikal, Puduchery	Final B.Sc (Ag)	October 15, 2018	81 (25)
30	College of Agriculture UAS campus, Raichur, Karnataka	Final B.Sc (Ag)	October 30, 2018	97
31	Agricultural college and Research Institute (TNAU) Kudumiyamalai, Pudukkottai Dt, TN	IV B.Sc (Ag)	October 30, 2018	50
32	Thanthai Roever institute of Agriculture and rural development (Affiliated to TNAU), Perambalur	IV B.Sc (Ag)	October 30, 2018	65 (43)
33	Nammazhvar college of agriculture and technology (Affiliated to TNAU)	Final IV B.Sc (Ag)	November 13, 2018	44 (33)
34	Agril college and Research Institute TNAU, Vazhavachanur, Thiruvannamalai, TN	Final IV B.Sc (Ag)	November 13, 2018	59 (36)
35	College of Sericulture, Chintamani (University of Agril Sciences UAS, Bengaluru)	III B.Sc (Ag)	November 16, 2018	36 (21)
36	College of Sericulture, Chintamani (University of Agril Sciences (UAS, Bengaluru)	III B.Sc (Ag)	November 16, 2018	39 (15)
37	Kumaraguru Institute of Agriculture, (Affiliated to TNAU) Sakthinagar, Erode, TN	Final B.Sc (Ag)	November 20, 2018	64
38	Agril College and Research Institute TNAU, Tiruchirapalli	IV B.Sc (Ag)	November 20, 2018	107 (71)
39	College of agriculture and research institute, Coimbatore	IV B.Sc (Agri)	November 24, 2018	118 (82)
40	Forest College and Research Institute, Mettupalayam (TNAU)	Final B.Sc (Seri)	November 26, 2018	21 (10)
41	Dept of PS&IT Agril Engineering College and Research Institute TNAU, Coimbatore	Final B.Sc (information technology)	November 26, 2018	14 (12)
42	Dr D.Y. Patil College of Agril Eng. and Technology (Affiliated to MPKV, Rahuri)	Final B.Tech (Ag. Engineering)	November 30, 2018	32 (9)
43	Community Science and Research Institute TNAU, Madurai	Final B.Sc	November 30, 2018	44 (26)
44	College of Agribusiness Management, Baramati	3 rd B.Sc (ABM)	December 3, 2018	31
45	School of Agriculture and Animal sciences, Gandhigram Rural Institute, (Deemed to be university) Gandhigram, TN	B.Sc (Ag.)	December 17, 2018	26 (17)
46	College of Horticulture University of Horticultural Sciences, Mysuru	B.Sc (Hort) Final Year Students	December 21, 2018	47 (20)
47	College of Agril engineering and Technology Punjab Agril University, Luthiana	B.Sc (Agril. Eng.)	December 22, 2018	32 (4)



S. No.	Details of College and University	Degree	Date of visit scheduled	No. of students (Girls)
48	College of Fisheries Engineering, Tamil Nadu, J. Jayalalithaa Fisheries University, Panangudi, Nagapattanam	2015-2016 B.E. (Fisheries Engineering)	December 24, 2018	20 (10)
49	College of Horticulture and Forestry Central Agricultural University, Pasighat, Arunachal Pradesh	B.Sc (Forestry)	December 27, 2018	15 (7)
50	College of Horticulture DR YS Parmar University of Horticulture and Forestry Nauni, Solan	B.Sc (Hon) 3 rd Year	December 31, 2018	43 (25)
51	College of Forestry, Dr Yashwant Singh Permar University of Horticulture and Forestry Nauni, Solan, HP	3 rd B.Sc (Hons Forestry)	January 1, 2019	39 (23)
52	College of Horticulture DR YS Parmar University of Horticulture and Forestry Nauni, Solan	B.Sc (Hon) 3 rd Year	January 3, 2019	41
53	BN College of Agriculture, Biswanath Chari, Assam	B.Sc (Agri) 3 rd year	January 9, 2019	36
54	College of Horticulture and Forestry Dr Yashwant Singh Permar University of Horticulture and Forestry Neri, Hamirpur, HP	B.Sc. Hons (Forestry) & (Biotechnology)	January 16, 2019	55 (33)
55	ATMA, Odisha	--	January 18, 2019	33 farmers
56	Agril College, PJTSAU Aswaraopet, Khammam, Telangana	B.Sc. (Hons) 3 rd year	January 28, 2019	64 (30)
57	KRC College of Horticulture (University of Horticultural Sciences, Bagalkot) Arabhavi, Karnataka	Final B.Sc (Hort.)	January 30, 2019	55 (26)
58	Department of Agricultural Extension College of Agriculture, GKVK, University of Agricultural Sciences, Bengaluru	PG + Ph.D	February 7, 2019	19
59	Sri Siddeheswar FPO company Ltd, Belagur, Karnataka	--	February 11, 2019	55 Farmers
60	College of Horticulture University of Horticultural Sciences Hosalingapur, Munirabad, Koppal, Karnataka	Final B.Sc. (Horti)	Feb. 16, 2019	51
61	College of Horticulture, Bagalkot University of Horticultural Sciences, Bagalkot	Final B.Sc. (Horti)	February 16, 2019	51 (19)
62	MGM College of Agricultural Bio-Technology, Gandheli, Aurangabad.	Final year B.Sc. Ag. Biotech.	February 21, 2019	46 (3)
63	District Watershed Development Agency Salem, Dindigul	--	February 27, 2019	85 Farmers
64	College of Home Science ANGRAU, Guntur, AP	B.Sc (Home science)	March 11	(90)



Chapter - 5
Infrastructure &
Facilities

Infrastructure & Facilities

The Academy is established in a green serene and sprawling campus of about 50 hectares. It has various facilities like Faculty Block, Academic Block, Administrative Block, Library, Medical and health Centre, Sports Complex, Guest houses and Trainees living facilities, Housing and Residential Quarters.

5.1 Library

The NAARM Library has about 31,000 books, 71 International journals, 70 Indian Journals, 15 International online journals, 22 free access online journals, and 13 online databases / software, i.e. EBSCO Business Source Premier, Indiatat.com,

Commodities, Economic Outlook, Districts of India, iThenticate, Lexis Nexis.com, Total Patent, Patent Optimizer, Orbit.com, SciVal, ABI-Inform, Web of Science etc. The Library has many of its publications in digital formats like CD, VCD and its own digital repository (EPrints@NAARM). The Academy is also a member of CeRA, and has access to journals and databases through CSIRO, Springer and Open J-Gate etc. Library management has been completely digitized through the open source software “Koha”. The Academy also has a rich repository of online patent search providers, GIS software and other statistical software.



Library Facilities at NAARM

5.2 Health Centre

ICAR-NAARM runs a one health centre headed by Dr (Maj.) A Debnath and have facilities for daily OPD, pharmacy facilities, physiotherapy, referral service and day are service for emergencies. A cardio-diabetic health camp was organized at NAARM Health Center on August 4, 2018. Dr Chaitanya, Cardiologist from Maxcure Hospital came for consultation and FBS/RBS/

PPBS, blood glucose, ECG, 2D Echo Biothesiometry/ neuropathy, ECG tests were done. Special lectures were also organized on September 18, 2018. Dr Alluri Vasu, Cardiologist spoke on stress related cardio vascular diseases and Dr Ratnakar Rao, Orthopedic surgeon addressed the audience on musculo-skeletal diseases.



Health camp at NAARM

5.3 Sports Complex

The sports complex at NAARM has several facilities which includes a stadium for outdoor events like cricket, football, lawn tennis and various athletic

events. Indoor complex comprises of badminton court, table tennis, billiard/snooker and gymnasium.



Sports facilities at NAARM



Chapter - 6
Recognition,
Publication and
Participation

Recognition, Publication and Participation

6.1 Recognition

6.1.1 Visits Abroad

Dr Ch. Srinivasa Rao, Director, NAARM attended the 23rd Inter-Governmental Meeting of the Asia Pacific Network (APN) for Global Change Research at **Bangkok, Thailand** during July 9-12, 2018. He participated and co-chaired Capacity Development Committee Meeting, APN in which 17 multi-year projects of various countries were reviewed. Discussions were held on finalization and modalities of capacity building projects. He also co-chaired Steering Committee Meeting as a nominee of Indian focal point. He participated in South Asia Sub Regional Sessions where discussions were held on research priorities in continuation with the last workshop which was held at ICAR-NAARM, Hyderabad. During the event he evaluated Thailand youth poster session on Global Change Research and Development.



*Dr Ch. Srinivasa Rao, Director, NAARM
co-chairing the meeting*

Dr Ch. Srinivasa Rao, Director and Dr S.K. Soam visited **Singapore, Australia** and **New Zealand** during May 19-29, 2018 as a part of 'Senior Executive Development Programme on Developing Effective Organizational Leadership for Senior Officers of ICAR' conducted by Administrative Staff College of India (ASCI), Hyderabad. During the visit, they were

exposed to various applications/technologies being adopted in Australian Centre for Field Robotics at University of Sydney, Hawkesbury Institute for the Environment, Western Sydney University, Australian Meat Industry Council, Department of Agriculture and Water Resources, Grain Research and Development Corporation and Australian Centre for International Agricultural Research. In Wellington, New Zealand, they visited Ministry of Primary Industries (MPI) Office, The Global Research Alliance (GRA), T&G Fresh Produce Market, Horticulture, New Zealand and attended a talk by Mr. Mike Chapman, CEO, Cooperative Bank House.



*Dr Ch. Srinivasa Rao, Director and Dr S.K. Soam
with other delegates*

Drs. Bharat S. Sontakki, D. Thammi Raju, S. Senthil Vinayagam and P. Venkatesan visited **Kandy, Sri Lanka** to attend International Conference on 'Transforming Agricultural Extension Systems: Towards Achieving the Relevant Sustainable Development Goals (SDGs) for Global Impact' during May 10-12, 2018. Drs. D. Thammi Raju and S. Senthil Vinayagam acted as panelists and presented lead papers in the conference. The conference was attended by more than 200 researchers from ten Asian and other countries.



NAARM faculty at Kandy, Sri Lanka

Dr K. Srinivas visited Krishi Gobeshona Foundation, **Dhaka, Bangladesh** as a resource person for training programme on 'IP and Technology Management' for personnel of National Agricultural Research System

(NARS), Bangladesh at BRAC, Savor, Dhaka during June 24-29, 2018. He dealt sessions on technology valuations, deal making, entrepreneurship and business incubation. There were about 43 participants from different organizations viz. Bangladesh Agricultural University, line departments and private industries.



Dr K Srinivas delivering lecture at Bangladesh

6.1.2 Awards Received by the Faculty Members of NAARM

Name of Staff	Name of award	Date	Purpose	Awarding body
N Srinivasa Rao	Distinguished Scientist	11-08-2018	In the field of Computer Science	Centre for Advanced Research and Design of Venus International Foundation, Chennai
Manju Gerard	Best Paper Award	07-10-2018	National Conference 'Empirical Research on Alternative Medicine: Impingement on Health Care System'	Society of Pharmacognosy (Formerly Indian Society of Pharmacognosy)
K Srinivas	Dr NA Majumdar Prize award	03-11-2018	78 th Annual Conference of ISAE	ISAE, Mumbai
M Krishnan	Dr Anamitra Saha prize for best paper award	03-11-2018	78 th Annual Conference of the Indian Society of Agricultural Economics	ISAE, Mumbai
P Venkatesan	Best Paper Presentation Award-2018	17-11-2018	9 th National Extension Education Congress-2018	SEE Agra at CAEPHT, Ranipool, Sikkim (CAU, Imphal)
Surya Rathore	Excellence in Research Award	24-02-2019	International Conference on Advances in Agriculture and Allied Science Research	Rama University, Kanpur

6.1.3 Honours and Awards Received by the PGDMA Students

Name of Student	Occasion	Date	Position	Awarding Body
Anushree Jha	Screening of minds-SPARDHA	February 20, 2019	First	Vaikunth Mehta National Institute of Co-operative Management (VAMNICOM)
	IGNOVITA'19			
	Extempore-SPARDHA	February 20, 2019	First	Vaikunth Mehta National Institute of Co-operative Management (VAMNICOM)
	IGNOVITA'19 Annual Business Fest			
K Sai Prajeeth Reddy & Siddarth P	Dhwani'19 - The Marketing Event	January 18, 2019	First	Shri Dharmasthala Manjunatheshwara Institute for Management Development - Mysore

6.1.4 NAARM in Media

1. Special news article about PGDMA course, ICAR-NAARM in Eenadu telugu daily newspaper on February 5, 2019.
2. News item about the offering of distance education programme of One Year Post Graduate Diploma in Educational Technology and Management (PGD-ETM) by ICAR-NAARM in collaboration with University of Hyderabad, 2019. Andhra Jyothi and Mana Telangana telugu daily newspaper and The Hindu English daily on November 1, 2018.
3. News item about the Graduation Day ceremony of PGDMA of ICAR-NAARM held at ICAR-NAARM on October 6, 2018, Andhra Jyothi, Mana Telangana, Andhra Prabha, Vaartha, Sakshi, Eenadu telugu daily newspapers and Hindi Milap daily newspaper on October 6, 2018.
4. Curtain-raiser news item about the Graduation Day ceremony of PGDMA at ICAR-NAARM on October 6, 2018 in Vaartha, Hindi Milap Hindi daily Newspaper; Navatelangana, Mana Telangana, Andhra Bhumi, Eenadu, Andhra Jyothi telugu daily newspaper on October 6, 2018.
5. News item about the PGDMA B-Fest held at ICAR-NAARM on October 3 – 4, 2018, Daily Hindi Milap on October 6, 2018.
6. News item about the 43rd ICAR-NAARM Foundation Day Celebrations at ICAR-NAARM on September 1, 2018, Eenadu, Andhra Prabha, Andhra Jyothi Telugu daily newspaper, indi Vaartha - Hindi daily newspaper and The Hindu, English daily newspaper on September 1, 2018.
7. News item about the document prepared on Farmers' feedback on Doubling Farmers' Income by 2022. Proceedings and recommendations of 'Let's Listen to Farmers' Workshop on Farmers' Feedback on Doubling Farmers' Income by 2022', ICAR-NAARM, Hyderabad. Eenadu Telugu daily newspaper on July 18, 2018.
8. News item about the signing of Memorandum of Understanding between ICAR-NAARM and Coromandel International Limited on capability development on May 16, 2018. Hindi Milap Hindi daily newspaper and Andhra Jyothi telugu daily newspaper on May 17, 2018.
9. News item about the Training programme for Farmers on Natural Farming held at Swarna Bharat Trust (Hyderabad Chapter), at Muchhintal village, Ranga Reddy district of Telangana, on May 6, 2018. Namasthe Telangana telugu daily newspaper on May 7, 2018.
10. News coverage about the Regional Consultation on 'Climate Smart Agricultural Policies, Strategies and Agricultural Development Programmes towards Climate Change Adaptation jointly organized Regional Consultation on Climate Resilient Agricultural Policies Strategies and Programmes , jointly Organized by SAARC Agriculture Centre (SAC), Dhaka, Bangladesh; Asia-Pacific Network for Global Change Research (APN), Kobe, Japan; Ministry of Environment,



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NEW BEGINNINGS



Sahendra Kumar Joshi, Chief Secretary of Telangana Inaugured regional consultation on "Climate Smart Agriculture Policies, Strategies and Agricultural Extension Programmes towards Climate Change Adaptation" yesterday at ICAR National Academy of Agricultural Research Hyderabad (NAAR), Hyderabad. The regional consultation was jointly organized by SAARC Agriculture Centre (SAC), Bangladesh, Asia-Pacific Network for Global Change Research (APN), Japan, Ministry of Environment, Forest and Climate Change (MoEF), India and ICAR-NAAR. The workshop was attended by senior officers and agricultural experts from Afghanistan, Bangladesh, Malawi, India, Maldives, Nepal, Pakistan and Sri Lanka.

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Forest and Climate Change (MoEFCC), New Delhi, India and ICAR-NAARM during April 17-19, 2018, at ICAR-NAARM, Hyderabad. Andhra Jyothi, Sakshi, Eenadu, Namasthe Telangana, Nava Telangana, The Hans India, Indian Express, Hindi Milap on April 18, 2018.

11. New coverage of the 4th Convocation of the Tata Institute of Social Sciences, Hyderabad campus, held at ICAR-NAARM on April 23, 2018. Eenadu,

Sakshi, Andhra Jyothi daily telugu newspapers, on April 24, 2018.

12. Video coverage of the news item about the SAARC Regional Consultation on 'Climate Resilient Agricultural Policies, Strategies and Programmes Smart agriculture news coverage on April 18, 2018 in hmtv channel, nelatalli programme. <https://youtu.be/RJ6n3TI8Zkw>.

6.1.5 TV/Radio Talk delivered

Sl. No.	Name	Topic of the talk	Date	Name of the Channel and the Programme
1	Dr. Ch. Srinivasa Rao	About the SAARC Regional Consultation on 'Climate Resilient Agricultural Policies, Strategies and Programmes' held at NAARM during April 17-20, 2018	18 th Aug. 18	HMTV News channel Nelatalli Programme HMTV agri https://youtu.be/RJ6n3TI8Zkw https://youtu.be/7UF_wtfZyBA ETV News channel Jaikisan https://youtu.be/cv1X8FsOhKk
2	Dr. Surya Rathore	Misal - e-Pantnagar	20 th Aug.18	Pantnagar Janwani

6.1.6 Invited Lectures Delivered by the Faculty & Staff

Name of Staff	Lecture	Programme/Organized by
A. Dhandapani	Official Statistics: A closer look	National Conference on "Challenges and Opportunities in Statistics and Informatics for Futuristic Humanosphere especially in Agriculture (COSIFHA)" held during 29-31 January, 2019, at S.V. Agricultural College, Tirupati, ANGRAU, A.P.
B. Ganesh Kumar	Marketing strategies for livestock products	Training programme on 'New Dimensions in Extension Management' for officials of Department of Animal Husbandary by MANAGE, Hyderabad on 09 Aug, 2018
	Marketing scenario and alternate marketing systems in fisheries	Refresher training programme on 'New Dimensions in Extension Management' for middle-level extension functionaries of Fisheries Departments by MANAGE, Hyderabad on 29 Aug, 2018
	Livestock based livelihood in Rural Areas	ITEC training programme on "Natural Resource Management for Sustainable Rural Livelihoods" for international participants by NIRD&PR, Hyderabad on 04 Sep, 2018
	Supply chain management in fish and meat	Training programme on 'Extension skills for supply chain management in Agri and allied sectors' by MANAGE, Hyderabad on 05 Sep, 2018
	Linking livestock farmers to the market: Issues and challenges	Induction training programme on 'Extension management approaches for livestock development' for LDOs of Govt. of Maharashtra by MANAGE, Hyderabad on 27 Oct, 2018

Name of Staff	Lecture	Programme/Organized by
B. Ganesh Kumar	Supply chain management of fish	Training programme on 'Extension skills for supply chain management in Agri and allied sectors' by MANAGE, Hyderabad on 11 Dec, 2018
	Fish and meat value chains	Training programme on 'Value chain management for agricultural produce by MANAGE, Hyderabad on 08 Jan, 2018
	Linking farmers to the market for sustainable livelihood through Animal Husbandry	Training programme on 'Extension management approaches for sustainable livelihood through Animal Husbandry' by MANAGE, Hyderabad on 22 Jan, 2019
	Marketing Strategies for livestock commodities	Training programme on 'Recent advances in livestock sector' for the extension functionaries of State Department of Animal Husbandry, Tamil Nadu by MANAGE, Hyderabad on 01 Feb, 2019
B. S. Sontakki	Capacity Development Levels in Extension and Advisory Services, The New Extensionist: Core Competencies for Individuals, Training for capacity development	Training Programme on Enhancing New Extensionist for Better Livelihood Transformation by Extension Education Institute, Hyderabad on 03 Jan, 2018
	Student Centred Learning, Basics of Scientific Writing	Training Programme for Newly Recruited Assistant Professors in SKLTSHU, Hyderabad on 18 Mar, 2019
Ch. Srinivasa Rao	Research towards climate-smart agriculture in South Asia	Regional Consultation on Climate Resilient Agricultural Policies Strategies and Programmes (April 17-20, 2018) by Jointly Organized by SAARC Agriculture Centre (SAC), Dhaka, Bangladesh; Asia-Pacific Network for Global Change Research (APN), Kobe, Japan; Ministry of Environment, Forest and Climate Change (MoEFCC), New Delhi, India and ICAR-National Academy of Agricultural Research Management (NAARM), Hyderabad on 18 Apr, 2018
Ch. Srinivasa Rao	Climate Change Mitigation and Adaptation Tools in Agriculture Sector	Training Programme on Climate Change and Management of Water Resources by Engineering Staff College of India, Hyderabad on 8 Jun, 2018
	Innovation Platforms for Entrepreneurship	Regional Conference on Motivating and Attracting Youth in Agriculture (MAYA) by Jointly organized by TAAS in collaboration with ICAR and Ministry of Skill Development and Entrepreneurship and MSSRF, APAARI, YPARD on NASC Complex, New Delhi on 30 Aug, 2018
	Climate Change and Natural Resource Management – National Perspective	Mid-Career Training Programme for IFS Officers by Indira Gandhi National Forest Academy (IGNFA), Dehradun on 1 Oct, 2018
	Integrated Nutrient Management – Soil Carbon Sequestration and Climate Resilient Agriculture	Training Programme on Integrated Nutrient Management and Nutrient Budgeting through Advanced Models to improve crop productivity by ICAR-IISWC Research, Udhagamandalam (Ooty) on 26 Oct, 2018



Name of Staff	Lecture	Programme/Organized by
Ch. Srinivasa Rao	Multiple facets of improved nutrient use efficiency in agriculture and food systems	IPNI Cooperators' Meet 2018 and International Symposium on Advancements in Soil, Water and Plant Nutrition Research' by IPNI South Asia Programme, VANAMATI, Nagpur on 2 Nov, 2018
	Agri-incubation through public funding	Workshop on Ag-startup and Incubator Ecosystem, NABARD, Mumbai on 14 Nov, 2018
	Use of Fly ash in agriculture	8 th International Conference of Sustainable Waste Management by Swachh Andhra Corporation (SAC), Govt of A.P. and ISWMAW, Acharya Nagarjuna University, Guntur, A.P. on 23 Nov, 2018
	Climate Resilient Agriculture	National Workshop on 'Sustainability of Indian Agriculture: Biodiversity, Environmental and Climate Change Perspectives' by ICAR-NIAP, New Delhi on 26 Nov, 2018
	45 th R.V. Tamhane Memorial Lecture 'Soil Carbon Sequestration: Research, Technology and Policy Implementation Needs in India'	83 rd Annual Convention of Indian Society of Soil Science, New Delhi by IISS, New Delhi, AAU, Anand on 28 Nov, 2018
	Synergy of research technology-policy implementation for resilient rainfed-dryland production systems	13 th International Conference on Development of ylands - Converting Dryland Areas from Grey into Green by International Dryland Development Commission (IDDC) and Arid Zone Research Association of India (AZRAI) and hosted by the ICAR-Central Arid Zone Research Institute (CAZRI), at Jodhpur, India on 11-14 Feb, 2019
D. Thammi Raju	Extension Through application of ICT tools and techniques	Training programme on New Dimensions in Extension Management by MANAGE, Hyderabad on 7 Aug, 2018
	Innovative ICT initiatives in Agriculture and Allied Sectors	Training programme on ICTs for effective knowledge and extension delivery organized by Extension Education Institute, Hyderabad on 8 Aug, 2018
	Teaching Methods	21 Days winter school on Advances of ICT in libraries of NARES to support Education and Research by PJTSAU, Hyderabad on 17 Sep, 2018
	Extension through application of ICT tools and techniques	Extension Management Approaches for Livestock Development by MANAGE, Hyderabad on 28 Nov, 2018
	Application of ICT and Social Media for effective advisory	Recent Advances in Livestock sector by MANAGE, Hyderabad on 22 Jan, 2019
	Application of ICT for effective livestock extension services	Recent Advances in Livestock sector by MANAGE, Hyderabad on 1 Feb, 2019

Name of Staff	Lecture	Programme/Organized by
D. Thammi Raju	Basics of Teaching and Learning and Innovative Teaching Methods	Training on Digital Teaching Techniques by University of Agricultural Sciences, Bengaluru on 26 Feb, 2019
	Curriculum Design and Lesson Planning	
	Micro Teaching	
	Application of ICT for effective livestock extension services	Recent Advances in Livestock sector by MANAGE, Hyderabad on 1 March, 2019
	Innovative Teaching Methods	Training Programme for newly recruited Assistant Professors by College of Horticulture, Sri Konda Lakshman Telangana State, Hyderabad on 15 March, 2019
G.R.K. Murthy	Curriculum Design and Lesson Planning	Training Programme for newly recruited Assistant Professors by College of Horticulture, Sri Konda Lakshman Telangana State, Hyderabad on 19 Mar, 2019
	Digital learning and teaching practices	Orientation programme on digital learning by NMIT, Bengaluru on 25 May, 2018
	Technology Enhanced Learning	Agricultural Extension Management by MANAGE, Hyderabad on 27 Jun, 2018, 15 Nov, 2018
	E-learning	Orientation programmes to Teaching faculty by MANUU, Hyderabad on 12 Jul, 2018, 5 Oct, 2018, 29 Oct, 2018, 5 Mar, 2019
	Digital Teaching methods	Exploring OER, developing e-content and creating online courses using MOODLE by W College, Gudur, AP on 7 Feb, 2019
	World of MOOCs	Workshop on MOOCs, E-Content Development and Open Educational Resources by OU, Hyderabad on 13 Feb, 2019
	Digital Agriculture	Training programme for Agronomists by Coramandel Fertilizers, Hyderabad on 21 Feb, 2019
	Digital Teaching Methods	Orientation programme for the teaching faculty by MGNRCE, Hyderabad on 6 Nov, 2018, 19 Jan, 2019
	Introduction to E-learning	Induction Programme of Horticulture faculty by Sri Konda Lakshman, Telangana State, Hyderabad on 13 Mar, 2019
K Srinivas	Agri Based Startups: Cases and Impact	CAFT training of ICAR-IARI Division of Agriculture Extension by Division of Ag Extension, IARI, New Delhi on 16 Jan, 2019
	IP Valuation Theory	Training Programme on “Intellectual Property Rights and Technology Commercialization in Agriculture” by Krishi Gobesona Foundation, Dhaka on 24-27 Jun, 2018
K.H. Rao	Role Play – A training methodology	International Training Programme on Training Methodology for Development Professionals by NIRD and PR, Hyderabad on 31 Jan, 2019
M. Krishnan	Essentials of Monitoring & Evaluation Indicators and Outcome monitoring”	Training Programme on “Monitoring & Evaluation of fisheries Projects” by MANAGE, Hyderabad on 21 May, 2018
	Importance of Monitoring & Evaluation of Fisheries Projects	Training Programme on “Monitoring & Evaluation of fisheries Projects” by MANAGE, Hyderabad on 21 May, 2018



Name of Staff	Lecture	Programme/Organized by
M. Krishnan	ICT Innovations in Agricultural Educational Technology	Training program on Agril development, planning and implementation for inclusive growth by MANAGE, Hyderabad on 08 May, 2018
	Impact of fisheries schemes for increasing farmers welfare	Training program on Enhancing the impact of agril schemes in increasing farmers welfare by MANAGE, Hyderabad on 06 Jul, 2018
	Emerging technologies for processing and packaging of fish and fishery products	21 day CAFT at PG & Research centre, by PJTSAU, Rajendranagar, Hyderabad on 25 Jul, 2018
Manju Gerard	Scope & Challenges: Emerging Agri Biotech in Industrial Era	SRIX Incubator, SR Engineering College/aIDEA, Warangal on 15 Sep, 2018
	Harnessing the potential of Agribusiness Industry in North East India	Startup Samvaad at Vaterinary College, Guwahati, Assam on 01 Dec, 2018
	Key note talk on “Future of Chemical Sciences: A perspective”	International Conference on “Chemical Sciences” National & Global Prospective” by Lucknow. Christian College on 29 Oct, 2018
	Building Conducive ecosystem for scaling Indian Agri Startups	International conference & Awards for Innovation by Agri Startups on “Agri Startup & Incubator Ecosystem by FICCI, New Delhi on 16 Nov, 2018
	Key Speaker- Innovations and IPR	National Seminar on IPR at St. Teresa’s Ernakulum. Kerala on 22 Feb, 2019
N. Sivaramane	Modelling human decisions in adoption of agricultural technologies – LOGIT, PROBIT and TOBIT” and “Conjoint Analysis”	National Workshop on Advances in Social and Behavioural Science Research by CTCRI, Thiruvananthapuram on 15 Nov, 2018
	Communication Skills	Off-campus Training Programme on Motivation, Positive Thinking and Communication Skills for Technical Officers (T-5 and above) of ICAR Institutes by IISWC, Dehradun on 14 Mar, 2019
	Managing Procurement for Agricultural Products	MDP on Agribusiness Supply Chain for Corporate Executives by MANAGE, Hyderabad on 06 Feb, 2019
	Market segmentation, targeting and positioning for formulating winning marketing strategy	MDP on Agricultural Input Marketing for Corporate Executives by MANAGE, Hyderabad on 18 Mar, 2019
P D Sreekanth	Geospatial Technologies in Agriculture	Special Lecture by PJTSAU, Hyderabad on 24 Oct, 2018
	Introduction to GIS, RS and geospatial technologies in agriculture	Application of Remote Sensing and Geospatial Information System for Agricultural Development by Extension Education Institute (EEI), Hyderabad on 30 Oct, 03
	Use of GIS in Agricultural Extension	ICAR sponsored winter school on ICT interventions for Agricultural Development by ICAR-NAARM, Hyderabad on 28 Nov - 18 Dec, 2018



Name of Staff	Lecture	Programme/Organized by
P D Sreekanth	Geospatial Analysis for Agriculture	Google Earth Engine Advanced User meet by Google, Bangalore on 15-17 Mar, 2019
P. Krishnan	Key requirements of fisheries professionals in the current scenario	TNJFU Alumni Network Meet '18, by TNJFU Alumni in Fisheries and Aquaculture, Nagapattinam on 15 Dec, 2018
	Revenue Generation through Research: Issues and Strategies	Developing Winning Research Proposals (DWRP) by Tamilnadu J. Jayalalitha Fisheries University (TNJFU), Nagapattinam on 02-04 Aug, 2018
	Revenue Generation through Research: Issues and Strategies	Developing Winning Research Proposals (DWRP) by Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya (RVSKVV), Gwalior on 25-27 Sep, 2018
	Revenue Generation through Research: Issues and Strategies	Developing Winning Research Proposals (DWRP) by University of Agricultural Sciences, Bengaluru during 26-28 Dec 2018
P. Ramesh	Psychological Principles of Teaching and Learning	UGC-HRDC-Orientation programme for the lecturers/Assistant Professors of different degree colleges and Universities by Maulana Azaad National Urdu University (MANUU), Hyderabad on 22 Jan, 2019
P. Venkatesan	Tools of facilitation skills for community mobilization	Community forestry & JFM for the in-service SFS officers of State Forest Department by Central Academy of State Forest Services (CASFOS), GOI, Coimbatore on 26 Jun, 2018
	Effective Presentation	Community forestry & JFM for the in-service SFS officers of State Forest Department, Central Academy of State Forest Services (CASFOS), GOI, Coimbatore on 22 Dec, 2018
	Farmer Producer Organizations (FPOs) and their Success	Market Led Extension for Agriculture and Allied Sectors by EEI, Hyderabad on 05 Feb, 2019
	New Agricultural reforms SREP and ATMA convergence with emphasis to Market led Extension in Andhra Pradesh	Induction training programme for Agriculture officers of Andhra Pradesh by Human Resource Development Institute, Bapatla (Andhra Pradesh) on 08 Nov, 2018
	Entrepreneurship convergence with emphasis to Market led Extension	Skill training course by Bharatiya Skill Development University, Jaipur on 30 Nov, 2018
	Private sector extension service providers	Skill Development Training Programme on Agriculture Extension Service Provide by Extension Education Institute, Hyderabad on 26 Feb, 2019
	Story boarding and script writing	"Utilization of ICT tools for effective transfer of technology" by PJTSAU, Hyderabad on 03 Jul, 2018
	Traditional media and communication technologies -Burra Katha style	Dissemination of BCI technologies by PRDIS, Hyderabad on 06 June, 2018
	Participatory methods and approaches for effective extension services	Participatory methods and approaches for effective extension services by SAMETI, Hyderabad on 14 Feb, 2019



Name of Staff	Lecture	Programme/Organized by
P. Venkatesan	Process documentation & presentation skills	Process documentation & presentation skills by SAMETI, Hyderabad on 11 Jan, 2019
	Market led Extension	PGDAEM by MANAGE, Hyderabad on 29 October, 2018
P.C. Meena	Agricultural Policies	Training Programme for National Seed Corporation by MANAGE, Hyderabad on 02 Jun, 2018
Ranjit Kumar	Understanding farmers' buying behaviour for agricultural inputs	MDP on Agricultural Input Marketing for Corporate Executives by MANAGE, Hyderabad on 17 Mar, 2019 & 26 Dec, 2018
	Food Consumption Pattern in Telangana State	Induction training of newly recruited Horticulture officer in Telangana state by Telangana Horticulture Training Institute, Hyderabad on 09 Nov, 2018
S. Senthil Vinayagam	Project Appraisal for Planning	DPR Preparation and Project Appraisal related to Cooperate Sector by MANAGE, Hyderabad on 11 Apr, 2018
	Appraisal of Project Proposals - Essential and Experiences	DPR Preparation and Project Appraisal related to Cooperate Sector by MANAGE, Hyderabad on 30 May, 2018
	Encouraging Sustainable Agriculture to ensure better consumption of people	ToT on Integration of Sustainable Development goals (SDGs) in Preparation of Gram Panchayat Development Plan (GPDP) by NIRD&PR, Hyderabad on 08 Aug, 2018
	Competency enhancement for Technology based learning	21 days Winter School on "Advances in Application of ICT in Libraries of NARES to Support Education and Research" by PJT-SAU, Hyderabad on 18 Aug, 2018
	Inter-personnel and Communication Skills for Library Professionals	21 days Winter School on "Advances in Application of ICT in Libraries of NARES to Support Education and Research" by PJT-SAU, Hyderabad on 18 Aug, 2018
	Appraisal techniques in Project planning	International Training Programme on "Planning and Management of Rural Development Programs" by NIRD, Hyderabad on 04 Oct, 2018
	Competency Enhancement for Technology Transfer in Oilseeds with particular reference to Soybean	ICAR short course on "Recent advances in adaptation and management strategies for sustainable oilseeds production under climate change scenario" by IIOR, Hyderabad on 08 Oct, 2018
	IT for Agriculture and Rural Development	Training programme on Good Governance for Management of Rural Development Programmes during by NIRDPR, Hyderabad on 02 Nov, 2018
	Interpersonal Communication for Effective Teaching	Professional Development Programme / Short-Term Course for the Lecturers/Assistant Professors of different degree colleges and universities by MANUU, Hyderabad on 28 Nov, 2018
	Handled 15 sessions on Agricultural Extension Management	Course on Agricultural Extension Management (2+0 credits) by NIAM, Jaipur on 21-25 Dec, 2018
	New Dimension in Agricultural Extension Management: Innovations	First International Training Programme on "Computer Applications in Agriculture" under Indian Technical and Economic Cooperation (ITEC) by MANAGE, Hyderabad on 20 Feb, 2018



Name of Staff	Lecture	Programme/Organized by
S. Senthil Vinayagam	Communication skills	Off-campus Training Workshop on Digital Teaching Techniques) (For the Faculty of UAS, Bengaluru) by UAS, Bengaluru on 26 Feb, 2018
	Competency Enhancement Digital Teaching	Off-campus Training Workshop on Digital Teaching Techniques) (For the Faculty of UAS, Bengaluru) by UAS, Bengaluru on 26 Feb, 2018
	Sustainability Issues in Agriculture. & Food Security	International Training Programme on “Participatory Planning for Poverty Reduction and Sustainable Development” by NIRD, Hyderabad on 14 Mar, 2018
	Open Educational Resources	Workshop on MOOCs, e-Content Development and Open Educational Resources by MANNU, Hyderabad on 15 Mar, 2018
S.K. Soam	Application of Artificial Intelligence in Agriculture	ICAR National Workshop on Artificial Intelligence in Agriculture: Status and Prospects by IASRI, New Delhi, NASC, New Delhi on 30 Jul, 2018
	Writing research proposals using software	Orientation course on communication and writing skills by NCIPM, New Delhi on 26 Jul, 2018
	Project logical framework approach	Project management skills for officers of agriculture and allied sector by MANAGE, Hyderabad on 14 Jun, 2018
	Identification of Parameters and Indicators for Monitoring and Evaluation - Logframe	Monitoring and Evaluation of Fisheries Projects by MANAGE, Hyderabad on 11 Dec, 2018
	Intellectual Property Rights in Communication	Orientation lecture series for PG students , UoH, on 20 Aug., 2018
Sanjiv Kumar	Project Management	Diploma programme in Cooperative Management by Institute of Co-operative Management, Hyderabad on 30 Jul, 2018
	Advanced Data Analysis using MS Excel	Training Programme on Monitoring and Evaluation of Fisheries Project for NFDB officials by MANAGE, Hyderabad on 12 Dec, 2018
	Data Analysis	Workshop on Research Methodology and Data Processing in Social Sciences by MANUU, Hyderabad on 30 Jan, 2019
	Data Analysis using Excel	Workshop on MOOCs, e-content development and open educational resources by MANUU, Hyderabad on 14 Mar, 2019
Surya Rathore	Doubling Farmers' Income	<i>Kisan Kalyan Karyashala</i> by State Department of Agriculture, Village Mukunoor, Ibrahimpatnam Mandal, Telangana on 02 May, 2018
	IOT based Smart Irrigation System and Nutrient detection with disease analysis	Training Programme on Smart Irrigation using IOT by Andhra Pradesh Human Resource Development Institute, Bapatla (Andhra Pradesh) on 7-9 May, 2018
	Artificial Intelligence in Indian Agriculture	Training Programme on Machine Learning and Artificial Intelligence for Agriculture sector from by Andhra Pradesh Human Resource Development Institute, Bapatla (Andhra Pradesh) on 19-21 Jul, 2018
	Prospects and avenues of Agricultural education in India and abroad	Career guidance for Agri-graduates by College of Agriculture, Gwalior, RVSKVV, Gwalior (M. P.) on 25 Sep, 2018
	Use of IOT Applications in Agriculture, Documentation of Success Stories	ICAR Winter School on Extension and Communication approaches for Sustainable Agriculture during Sept. 19- Oct. 9, 2018 by Dept. of Extension Education, College of Agriculture, RVSKVV, Gwalior (M.P.) on 26 Sep, 2018

Name of Staff	Lecture	Programme/Organized by
Surya Rathore	Startup Success by Agri-graduates: dreams to destination	India's 6 th International Conference on Agri-business through Innovation Eco-system by Kenes Exhibitions, Hitex, Hyderabad on 04-05 Oct, 2018
	Impact Assessment of Agricultural Extension Programmes, Documentation of Success Stories of Farmers	Induction Training Programme for Agricultural Officers by Andhra Pradesh Human Resource Development Institute, Bapatla (Andhra Pradesh) on 09 Nov, 2018
V.V. Sumanth Kumar	Scope of Artificial Intelligence & Machine Learning in Agriculture Sector	Three Day Residential Training Programme on "Machine Learning & Artificial Intelligence for Agriculture Sector" by AP HRDI, Bapatla on 19 Jul, 2018

6.1.7 Invited Lectures Delivered by Technical Staff

Mrs. G. Aneja	How to maintain balance between work and home- A critical task for working women	Departmental training programme for Women employees of Income Tax Department, Hyderabad on October 25, 2018
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6.2 Publication

NAARM faculty contributed 151 publications in the year 2018-19 in various forms. Research papers formed more than 37 % of the total publications. Ten book chapters were authored by the NAARM faculty and thirteen books were written or edited by them. The faculty contributed in developing nine working papers/project reports and brought out four policy papers.

Publication	Total
Research Papers	57
Books Authored	13
Book Chapters	10
Popular Article/newspaper article	07
Training Manuals /Technical bulletins	18
Papers in proceedings of Conferences / Seminars/ Workshops	33
Research Projects/Working Papers	09
Policy Papers	04

6.2.1 Research Papers in Refereed Journals

Abhilash, KR., Sankar, R., Purvaja, R., Deepak, SV., Sreeraj, CR., Krishnan, P., Sekar, V., Biswas, AK., Kumarapandiyam, G. & Ramesh, R. Impact of long-term seaweed farming on water quality: A case study from Palk Bay, India, *Journal of Coastal Conservation*. <https://doi.org/10.1007/s11852-018-00678-4>.

Ahire LM, Srinivasarao Ch, Venkattakumar R and Vijender Reddy P (2018). An innovative concept

to earn an extra income from agri -tourism – The case of an agri-tourism centre in Maharashtra, India. *Multilogic in Science*, 8, Special Issue (E).

Anu, SS., Azhar, Abbas, Subhas, P., Kaechele, H., Ranjit Kumar & Muller, K. (2018). Linking food security with household's adaptive capacity and drought risk. *Implications for sustainable development. Springer- Social Indicators Research*, Published online, 15 May 2018. DOI 10.1007/s11205-018-1925-0.

Anupam B and Balakrishnan M (2018) Smart phone applications: Role in agri-information dissemination. *Agricultural Reviews*, 39(1), 82-85.

Atheeq Sultan Ghori, Md. & Balakrishnan, M. (2018). Statistical analysis of Turmeric crop in Telangana State. *Jour of Adv. Research in Dynamical & Control Systems*, Vol. 10(4): 208-216.

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- Haqq, S.M., Pandey. H., Gerard, M. & Chattree, A. (2018, October). *Synthesis of silver nano engineered herbal formulation of chrysanthemum spp. For the treatment of bacterial infections in swiss albino mice*. Paper presented at Empirical Research on Alternative Medicine, Allahabad.
- Kalayani, B., Venkatesan, P., Sontakki, B.S. & Sivaramane, N. (2018, November). *ICT Based Concept Building of the Perception of the Ultimate Stakeholders*. Paper presented in the 9th National Extension Education Congress-2018, on Climate Smart Agricultural Technologies: Innovations and Interventions.
- Krishnan, M. & Srinivasa Rao, Ch. (2018). *SME Aqua-Start ups for smart farming*, p.16 in Ayyappan S., Kataktalware, M.A. & Lethi Devi, G. (Eds) E-compendium on discussion notes, In National Consultation on sustainable and remunerative small farming, organized by NABARD Chair Professor, ICAR-NIANP, Bengaluru, October 11-12, 2018, pp. 1-101.
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- Krishnan, M. (2019). *Transformation of socio-economic conditions among seaweed farmers in India, Impacts of seaweed farming on socio-economic development of coastal communities*, invited lead talk presented in India International Seaweed Expo & Summit 2019, World Trade Centre, Mumbai, January 22-24, 2019.
- Krishnan, M., Senthil Vinayagam, S., Murthy, GRK., Thammi Raju, D. & Sumanth Kumar, V.V. (2018, May). *Open and Distance Learning Systems- MOOCS and Digital learning Education in Fisheries for Reaching the Unreached*, *Proceedings*, 3rd International Symposium on Aquaculture and Fisheries Education (ISAFE 3), ICAR- Central Institute of Fisheries Education, Mumbai, May 16-18, 2018, 115-123.
- Mohapatra, T., Vyas, A.K., Vinyagam, S.S & Jain, N.K. (2018, August). *Initiatives of Youth Empowerment in Agriculture in India*. Paper presented in conference organized by Trust for Advancement of Agricultural Sciences (TAAS) in collaboration with ICAR.
- Munawar, W., Vinayagam, S.S. & Reddy, D.R. (2018, May). *Participatory Management process in NRM by women groups*. Poster presented at International Conference on Transforming Agricultural Extension Systems at Kandy, Sri Lanka.
- Naorem Dinesh Singh & Krishnan, M. (2019, March). *Market share competitiveness & comparative advantage of Indian Seafood exports*, in Souvenir of the International Fisheries Conference 2019, College of Fisheries (DBSKKV) Ratnagiri & ISASaT, Dapoli, Maharashtra, January 17-20., 2019.
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- Qureshi Neha, W. & Krishnan, M. (2019). *Fish Marketing in Kashmir – A case study of Srinagar*, p.6, in Ashalatha K.V., Kiresur, V.R., Ramesh Bhatt, Suma Hasalkar & Dolli, S.S. (Eds.) Compendium of Abstracts, National Conf on Adv Res Methodologies in Soc. Sciences, 144p. UAS, Dharwad, February 26-28, 2019.
- Rathore, S. & Ravichandran, S. (2019, March). *What is and What ought to be the Status of ICT in*

- our SAUs to be World Class. Paper presented at National Conference on Strategic Approaches for developing World Class Agricultural Universities of Acharya N. G. Ranga Agricultural University, Bapatla.
- Rathore, S. (2019, February). *Monitoring, Evaluation and Impact Assessment of Extension Programmes*. Paper presented at International Conference on Advances in Agriculture & Allied Science Research of Rama University, Kanpur & Samagra Vikas Welfare Society, Lucknow.
- Rathore, S., Rao, VKJ. & Sontakki, B. S. (2019, February). *Entrepreneurial process followed by Entrepreneurs: A Case of Indian Agri-graduates*. Paper presented at International Conference on Advances in Agriculture & Allied Science Research of Rama University, Kanpur & Samagra Vikas Welfare Society, Lucknow.
- Ravichandran, S. (2018). *VARMA Statistical Modelling and Forecasting India's Cereal Production*. Lead Paper presented in the 72nd Annual Conference of the Indian Society of Agricultural Statistics, ICAR-CIAE, Bhopal.
- Ravichandran, S. (2019). *Big Data Analytics for Crop Yield Forecasting*. Lead paper presented in the 21st annual of the "Society of Statistics, Computer and Applications (SSCA)" held at SV College of Agriculture (ANGRAU), Tirupati during 29-31 January 2019.
- Sanjiv Kumar, Kalamkar, SS., Parihar, TB. & Meena, PC. (2018). *Fodder Production and Marketing Practices in Gujarat*. International Conference on Emerging Global Economic Situation, Impact on Trade and Agribusiness in India, BJVM, VV Nagar, Anand, Gujarat, September 28-29, 2018.
- Sontakki, Bharat, S., Rao, I. S. & Bhuvana, N. (2018, May). *Extension Approaches for Sustainability of Agriculture and Food Eco System*. Lead Paper presented in the International Conference on Transforming Agricultural Extension Systems: Towards Achieving the Relevant Sustainable Development Goals (SDGs) for Global Impact (May 10-12, 2018) at Kandy, Sri Lanka.
- Sontakki, Bharat. S. & Praveen Kumar, R. (2018, October). *Capacity development and skill development needs of Agri Startups*. Invited presentation as a panelist in the National Consultation on Agri Startups for Smart Farming at ICAR-NIANP, Bengaluru.
- Srinivasa Reddy, M., Thammi Raju, D. & Kanthisri, B.S. (2018, May). *Livelihood opportunities among Dairy farmers for their economic empowerment*. Paper presented International Conference on Transforming Agricultural Extension Systems towards achieving relevant SDGs for Global Impact May 10-12, 2018 at Kandy, Sri Lanka.
- Srinivasarao, Ch., Vishal Tripathi, Ramakant Dubey & Sudha Rani, Y. (2018, November). *Fly Ash and its Utilisation in Agriculture: Constraints and Opportunities*. Paper presented at the 8th International Conference on Sustainable Waste Management at Vijayawada.
- Thammi Raju, D. (2018). *Curriculum Development in Veterinary and Animal Husbandry Extension Master's and Doctoral Programmes* during Stakeholders Meeting in the discipline of Extension Education held at EEI on December 7, 2018.
- Thammi Raju, D. (2018). *Curriculum Development in Veterinary and Animal Husbandry Extension* during Workshop on Issues and challenges in Strengthening Extension Services in Animal Husbandry Sector at MANAGE, Hyderabad on August 20, 2018.
- Thammi Raju, D. (2018, May). *Revamping Extension for quality education and learning of farmers*. Paper presented International Conference on Transforming Agricultural Extension Systems towards achieving relevant SDGs for Global Impact May 10-12, 2018 at Kandy, Sri Lanka.
- Thammi Raju, D., Vinayagam S.S & Srinivasa Rao, Ch. (2018, May). *Curriculum Development in Higher Education: Creating Quality Human Resource for Tomorrow*. Paper presented at the 3rd International Symposium on Aquaculture and Fisheries Education at CIFE, Mumbai.



- Venkatesan, P. & Sontakki, B.S. (2018, May). *Farmer Innovations for Sustainable Agricultural Development*. Paper presented in International Conference on Transforming Agricultural Extension Systems towards achieving relevant SDGs for Global Impact at Kandy, Sri Lanka.
- Venkatesan, P. & Sontakki, B.S. (2019, February). *Farmers Innovation for Transformation in Agriculture through Farmer FIRST Programme*. Paper presented in the XIV Agricultural Science Congress Innovations for Agricultural Transformation.
- Vinayagam, S.S. (2018, May). *Strengthening of Extension System for Poverty Alleviation and Promotion of Rural Livelihood*. Paper presented at International Conference on Transforming Agricultural Extension Systems at Kandy, Sri Lanka.
- 6.2.7 Research Project report, Working Papers, etc.**
- Murthy GRK, Technology Enhanced Learning in Agricultural Education – A perspective from ICAR-NAARM(2018) (Niche Project report on Technology Enhanced Learning in Agricultural Education).
- Gerard, M and Srinivas, K, (2018) Rising Startups of Agrigraduates: A success story, EMR- ICAR Project Report
- Kareemulla, K. & Ravichandran, S. (2018). Agricultural Research and Development Infrastructure in AP and Telangana. DST-NSTMIS Project. ICAR-NAARM, Hyderabad. pp.72.
- Kareemulla, K. & Ravichandran, S. (2018). Annual Report- Agricultural Research and Development Infrastructure in Andhra Pradesh and Karnataka. pp.68.
- Kareemulla, K., Ravichandran, S., Ganesh Kumar, B., Krishnan, P., Yashavanth, B.S. & Sekar, I. (2019). Environmental Impact Assessment of Tobacco Curing in AP and Karnataka. Tobacco Board Sponsored Project. ICAR-NAARM, Hyderabad. pp.121.
- Krishnan, M. (2018). Doubling Fishers and Fish Farmers' Income In Dr. Ashok Dalwai (Ed.) Report of the Committee on Doubling Farmers' Income, Department of Agriculture and Cooperation, Govt. of India, New Delhi.
- Rathore, S., Rao, VKJ. & Sontakki, B. (2018). Project Report of Documentation of Success Stories of Agri-graduates turned Entrepreneurs
- Venkatesh G, Gopinath, K.A., Sammi Reddy, K., Sanjeeva Reddy, B., Prasad, J.V.N.S., Rajeshwar Rao, G. Pratibha, G., Srinivasarao, Ch., Ravindra Chary, G., Prabhakar, M., Visha Kumari, V., Shankar, A.K. and Venkateswarlu, B. (2018). Biochar production and its use in rainfed agriculture: Experiences from CRIDA. CRIDA-NICRA Research Bulletin 02/2018, ICAR-CRIDA, Hyderabad, India. pp50.
- 6.2.8 Policy Papers**
- Ranjit Kumar, Sivaramane, N., Sanjiv Kumar, Meena, PC., Dhandapani, A. & Srinivasarao, Ch. (2018). *District-level Food Demand: A Better Estimate for Planning Food Production and Nutritional Security*. Policy Brief, No. 3, August 2018. ICAR-NAARM, Hyderabad. DOI: 10.13140/RG.2.2.20685.69604
- Ranjit Kumar, Sanjiv Kumar & Meena, PC. (2018). *Smart micro-mandi, for better price realization and reduced market risks for the smallholders*. Policy Brief, April 2018, No. 1. ICAR-NAARM, Hyderabad. DOI 10.13140/RG.2.2.32106.00969.
- Soam, SK., Balakrishnan, M., Sumanthkumar, VV. & Srinivasarao, Ch. (2018). *Artificial Intelligence and Internet of Things: Implications for Human Resources in Indian NARES*. NAARM Policy Brief, July 2018 No (2). ICAR- National Academy of Agricultural Research Management, Hyderabad.
- Srinivas I., Srinivasarao, Ch., Sammi Reddy K., Sanjeeva Reddy B., Adake R.V., Dhimate A.S., Osman M., Prasad J.V.N.S and Rama Rao C.A. (2017) *Up-Scaling Farm Machinery Custom Hiring Centres in India: A Policy paper*. ICAR-CRIDA, Hyderabad, pp26.

6.2.9 Copyrights

Title	Acknowledgement Number	Authors
Agro-biodiversity Knowledge Management Portal	7525/2019-CO/SW	DR. M Balakrishnan, Dr. Sk Soam and Dr. PD Sreekanth
AHP Analyser	7525/2019-CO/SW	Dr. SK Soam, Ch. Sridhar Babu, T. Raj Kumar and B Raghupathi
Project Logframe Writer	7535/2019-CO/SW	Dr. SK Soam, B Raghupathi and T. Raj Kumar
Research Concept Writer	7533/2019-CO/SW	Dr. SK Soam, B Raghupathi and T. Raj Kumar
Decision Support System for Technology Valuation	7522/2019-CO/SW	Dr. Manoj P Samuel, Dr. Kalpana Sastry R and Satheesan Periyattuveetil

6.3 Participations

6.3.1 Participation in Workshop/Conferences/Symposia/ Meetings

Month/Year	Name	Programme Name	Venue	Period
February' 18	N . Sivaramane & P.C.Meena	One day workshop on “Development of Market linkages for FPOs”	SFAC in collaboration with FICCI and APEDA, Hyderabad	February 08, 2018
March' 18	K Srinivas	Sensitization workshop for students in BHU	Institute of Agricultural Sciences, BHU	March 13, 2018
April '18	S. Senthil Vinayagam	To prepare the action plan in connection with International Conference (Srilanka - Kandy)	EEI, Hyderabad	April 10, 2018
	Ch. Srinivasa Rao	SAARC - Regional Consultation on ‘Climate smart agricultural policies, strategies and agricultural development programmes towards climate change adaptation and mitigation’	NAARM, Hyderabad Jointly organized by SAARC Agriculture Centre (SAC)&NAARM	April 17-19, 2018
	V.K. Jayaraghavendra Rao	Workshop on dimensions of Agrarian Distress	MANAGE, Hyderabad	April 20-21, 2018
	Ch. Srinivasa Rao	38 th Meeting of Central Joint Staff Council (CJSC)	ICAR, New Delhi	April 27, 2018
	M. Krishnan	National Conference on Marine Debris	ICAR-CMFRI, Kochi	April 11-12, 2018
May'18	Ch. Srinivasa Rao	Workshop on EIA on Tobacco curing	Tobacco Board, Guntur, Andhra Pradesh	May 02, 2018
	P. Krishnan	Environment Impact Assessment of Tobacco Cultivation in Andhra Pradesh and Karnataka	Headquarters of Tobacco Board, Guntur	May 02, 2018
	P. Krishnan	Drafting Committee of National Mariculture Policy	National Fisheries Development Board	May 14, 2018
	D Thammi Raju	3 rd International Symposium on Aquaculture and Fisheries Education (ISAFE3)	Central Institute of Fisheries Education, Mumbai	May 16-18, 2018



Month/ Year	Name	Programme Name	Venue	Period
June'18	P D Sreekanth	Workshop on National Dialogue on AI and IoT Applications in Agriculture	ICAR-NAARM, Hyderabad	June 01-02, 2018
	K Srinivas	National Seminar on Student Startup	CCH HAU, Hisar	June 01-02, 2018
	P. Krishnan	Inception Meeting by Department of Science and Technology (DST), Ministry of Science and Technology, New Delhi	The Energy and Resources Institute (TERI), New Delhi	June 06, 2018
	D Thammi Raju	Workshop on Training Need Assessment	TSIRD, Rajendranagar, Hyd	June 08, 2018
	Ranjit Kumar	First Round Table on The Role of Public Private Partnerships- Agricultural Research, Technology & Innovation	ICRISAT, Patancheru	June 11, 2018
	Ch. Srinivasa Rao	Brainstorming Session on Holistic Strategy for assessment of Soil Health	NASC Complex, New Delhi;	June 26, 2018
	Ranjit Kumar, B. Ganesh Kumar, P.C. Meena Surya Rathore	Agri-entrepreneurship conclave AgriVikas 2018	Bhubaneswar, Odisha	June 29-30, 2018
July'18	Surya Rathore	Consultation meeting for formulating the proposals built upon the idea of Skill Vigyan	Telangana State Council of Science and Technology	July 04, 2018
	S. Senthil Vinayagam Manju Gerard D Thammi Raju K Srinivas	Startup Samvaad	a-IDEA, ICAR-NAARM	July 07, 2018
	D Thammi Raju	Stakeholders Meeting in the discipline of Extension Education	EEI, Hyderabad	July 12, 2018
	D Thammi Raju	BSMA Committee meeting on Basic Veterinary Sciences	SVVU, Tirupati	July 16, 2018
	Ch. Srinivasa Rao	National Conference on Reforming Agrarian Economy : Role of Insurance	India International Centre, New Delhi;	July 21, 2018
	Manju Gerard S. Senthil Vinayagam	5 th JAC on PGDTMA/PGDETM	ICAR-NAARM	July 25, 2018
	Dr. P D Sreekanth	Workshop on Networking and Collaborative Research and Educational Activities of Scientific Institutes in Hyderabad	Forest College of Research Institute, Mulugu, Hyderabad	July 25, 2018
	Surya Rathore	International Agricultural and Horti-Expo cum show on Govt. Achievements and Schemes	Pragati Maidan, New Delhi by NNS Media group	July 27-29, 2018

Month/ Year	Name	Programme Name	Venue	Period
July'18	D Thammi Raju Manju Gerard	Roundtable Workshop "AgData Update: How can open data platforms support entrepreneurship and collaboration between the public and private sector in agriculture?"	ICAR – NAARM, Hyderabad	July 28, 2018
	Ch. Srinivasa Rao GRK Murthy	National Workshop on Artificial Intelligence (AI) in Agriculture : Status and Prospects	IASRI/NASC Complex, New Delhi	July 30-31, 2018
	Manju Gerard	Starup Samvaad	KLEF University, Guntur	August 03, 18
	D Thammi Raju	Workshop on Designing curricula for Master's and Doctoral degree programme in the discipline of Veterinary and Animal Husbandry Extension	Madras Veterinary College, TANUVAS, Chennai	August 10, 2018
August'18	Manju Gerard	IP Strategy & Monetization: Derive the Maximum Value from your Intangible Assets	CII, Hyderabad	August 10, 18
	Surya Rathore	Workshop on Agricultural Communication: Yesterday, Today and Tomorrow	Dept. of Agricultural Communication, G. B. Pant University of Agriculture & Technology, Pantnagar (UK)	August 19, 2018
	D Thammi Raju	Workshop on Issues and challenges in strengthening Extension Services in Animal Husbandry Sector	MANAGE, Hyderabad	August 20, 2018
	D Thammi Raju	Group discussion on Curriculum Revision in Agricultural Extension	TISS, Hyderabad	August 30, 2018
	S. Senthil Vinayagam	RISC-2018 - As one of the jury members for identifying/selection of the best Innovator/Startups	NIRD&PR	August 30-31, 2018
	Ch. Srinivasa Rao	Regional Conference on Motivating and attracting Youth in Agriculture (MAYA)	NASC Complex, New Delhi by Trust for Advancement in Agricultural Sciences (TAAS)	August 30-31, 2018
	Manju Gerard	Impact of New Breeding Technologies in Agriculture	Key Gene Technologies, Hyderabad	September 03, 2018
	Ch. Srinivasa Rao	Village Seminar by FET Team of 108 th FOCARS	Pondicherry Organized by KVK, Pondicherry	September 04, 2018
September' 18	P D Sreekanth	ESRI GIS User National Conference	ESRI, Hyderabad	September 06, 2018
	Ranjit Kumar	Broad Subject Matter Area Committees (BSMA) meeting for Social Science (Agri-Business Management and Agricultural Economics)	Rajasthan Agricultural University, Bikaner	September 17-18, 2018



Month/ Year	Name	Programme Name	Venue	Period
September' 18	B. S. Sontakki	Follow-on workshop on Agricultural Extension and Sustainable Development Goals (SDGs)	MS Swaminathan Library Building, IARI, New Delhi Indian Agricultural Research Institute (IARI), New Delhi & Participatory Rural Development Initiatives Society (PRDIS), Hyderabad	September 17, 2018
	M. Krishnan	National Consultation on Sustainable and Remunerative Small Farming	ICAR-NIANP, Bengaluru	September 18-19, 2018
	D Thammi Raju	State level workshop on Problems and prospects of Fishery Extension Services in Andhra Pradesh- Way Forward	SVVU, Tirupati	September 28, 2018
	P. Krishnan	Drafting Committee for National Mariculture Policy	National Fisheries Development Board (NFDB), Hyderabad	September 28, 2018
October' 18	D Thammi Raju	Workshop on Restructuring of Masters and Ph D Curriculum and syllabi for the discipline of Veterinary Extension Education	IVRI, Bareilly	October 08, 2018
	B. S. Sontakki	National Consultation on Agri. Startups for Smart Farming	ICAR-NIANP, Bengaluru NABARD, NDRI, NIANP & ICAR	October 11-12, 2018
	Manju Gerard K Srinivas Surya Rathore Ch. Srinivasa Rao	Agri-startup and Entrepreneurship Conclave (Unleashing potential in Agriculture for Young Agripreneurs)	NASC Complex/ ICAR- New Delhi	October 16-17, 2018
	Manju Gerard	International Conference on Chemical Sciences: National and Global Prospective	Lucknow Christian College, Lucknow	October 29, 2018
	Ch. Srinivasa Rao	Meeting on Framework for Ranking & Ratings of R&D Labs	New Delhi Organized by Niti Aayog	October 30, 2018
	Manju Gerard	International Conference on “ Empowering & Enabling Women in Science UTKARSH	NAS Complex, New Delhi, organized by CSIR-NEERI, Nagpur	October 30-31, 2018



Month/ Year	Name	Programme Name	Venue	Period
November '18	V V Sumanth Kumar	Attended NAHEP Project formulation meeting.	IASRI	November 01-02, 2018.
	Ch. Srinivasa Rao	International Symposium on Advancements in Soil, Water and Plant Nutrition Research”	Vasant Rao Naik State Agricultural Extension Management Training Institute (VANAMATI), Nagpur	November 02, 2018
	V V Sumanth Kumar D Thammi Raju	Brainstorming Session on NAHAEP-Component 2A	NASC Complex, New Delhi	November 03, 2018
	P. Venkatesan	Seed and Farmers Day	ICAR-IIRR, Hyderabad	November 03, 2018
	Manju Gerard K Srinivas	Agri Startup & Incubator Ecosystem	NABARD	November 14, 2018
	B. S. Sontakki P. Venkatesan	9 th National Extension Education Congress on Climate Smart Agricultural Technologies: Innovations and Interventions	College of Agricultural Engineering and Post-Harvest Technology (CAEPHT-CAU), Ranipool, Gangtok, Sikkim	November 15-17, 2018
	K Srinivas	26 th Annual Conference of AERA	NDRI Karnal	November 15-17, 2018
	N. Sivaramane	One day workshop on “Doubling the farmers’ income by 2022”	Institute of Co-operative Management, Hyderabad	November 15, 2018
	Manju Gerard	International Conference & Business Excellence Awards for Innovations by Agri Startups	FICCI Federation House, New Delhi	November 16., 2018
	P. Krishnan	National Survey on Resources Devoted to Scientific and Technological (S&T) Activities 2018-19	TERI, New Delhi	November 20, 2018
	K Kareemulla	Annual Conference of Indian Society of Agricultural Marketing	Institute for Socio Economic Change, Bengaluru	November 20-24, 2018
	B. Ganesh Kumar	National Conference on Revisiting Agricultural Research and Monitoring System for Developing Innovations: To Meet the Newer Challenges	ICAR-CIWA, Bhubaneswar	November 24-25, 2018
	P. Krishnan	Role of technology in community level disaster mitigation	Lal Bahadur Sastri National Academy for Administration (LBSNAA), Mussourie	November 26-30, 2018.



Month/ Year	Name	Programme Name	Venue	Period
December'19	Manju Gerard K Srinivas	Startup Samvaad	College of Veterinary Sciences, Khanapara, Guwahati	December 01, 2018
	K Srinivas	Parliamentary standing committee meeting on startup ecosystem	Parliament Annxe, Parliament, New Delhi	December 20, 2018
	D Thammi Raju	Workshop on Restructuring of Masters and Ph.D Curriculum and syllabi for the discipline for the disciples under Agricultural Sciences- Basic Veterinary Sciences	LUVAS, Hisar	December 28, 2018
	K Srinivas	Startup Samvaad and WISE	IGKV, Raipur	December 28-29, 2018
January'19	P. Krishnan	Brainstorming Meeting on "Fisheries Informatics"	National Fisheries Development Board (NFDB), Hyderabad	January 11, 2019
	Ch. Srinivasa Rao	Meeting with Consultative Group on International Agricultural Research Centres (CGIAR) to discuss their ongoing activities with India focus as well as their future plans in 2019	NASC Complex, New Delhi	January 24-25, 2019
	Surya Rathore	State level workshop on Skill development in Agriculture and Allied sectors	SAMETI, Telangana	January 29, 2019
	Ch. Srinivasa Rao	ICAR Directors' Conference	NASC Complex, New Delhi	January 31-February 01, 2019
February'19	K Srinivas	National Agri-Business Entrepreneurship conclave	ICAR complex for NEH, Shillong	February 08-09, 2019
	Ch. Srinivasa Rao	13 th International Conference on Development of Drylands - Converting Dryland Areas from Grey into Green	International Dryland Development Commission (IDDC) and Arid Zone Research Association of India (AZRAI)	February 11-14, 2019.
	K Srinivas	Committee on Enhancing Visibility of the ICAR/ Institutes' Own Brand	ICAR HQ	February 13, 2019
	V V Sumanth Kumar	13th International Conference on Development of Drylands	CAZRI, Jodhpur, India	February 11-14, 2019
	Ch. Srinivasa Rao	Executive Board Meeting of International Dryland Development Commission (IDDC)	IDDC, Egypt at Jodhpur	February 11, 2019
	Ch. Srinivasa Rao	Dryland Agrobiodiversity for Adaptation to Climate Change	ISPGR, Bioversity International and Asia- Pacific Association of Agricultural Research Institutes (APAARI)	February 13, 2019

Month/ Year	Name	Programme Name	Venue	Period
February' 19	Surya Rathore	One-day Workshop on Doubling the Farmers' Income by 2022	Institute of Cooperative Management, Hyderabad	February 15, 2019
	D Thammi Raju	Workshop on Dual Vocational Education and Training	The Plaza, Hyderabad organized by Steinbeis Centre for Technology Transfer India and Telangana Academy for Skill and Knowledge	February 16, 2019
	Ch. Srinivasa Rao	Executive Council Meeting of NAAS	NAAS, New Delhi	February 19, 2019
	P. Venkatesan	XIV Agricultural Science Congress "Innovations for Agricultural Transformation"	ICAR-IARI & NAAS New Delhi	February 20-23, 2019
	P. Venkatesan	Kian Samman Nidhi Program	IIRR, Hyderabad	February 24, 2019
March'19	S. Senthil Vinayagam Manju Gerard K Srinivas	Member, ABI Screening Committee Meeting	NAARM, Hyderabad	March 04, 2019
	P Krishnan	Expert Member in the Committee to select scientists for National Centre for Sustainable Coastal Management (NCSCM)	Chennai	March 04-05, 2019
	P D Sreekanth	Interactive meeting cum workshop of Scientists in IT/Computer Applications	IASRI, New Delhi	March 06, 2019
	B. Ganesh Kumar	Module Development Framework Writeshop for SLACC (Sponsored by World Bank and MoRD) for Community Resource Persons and Mission Staff	NIRD&PR, Hyderabad	March 07-08, 2019
	P Krishnan	Project Monitoring Committee (PMC) to review the implementation of the projects commissioned by Dept. of Biotechnology (DBT) and Department of Space (DoS), on biodiversity characterization at landscape level using RS-GIS	New Delhi	March 07, 2019
	P D Sreekanth	Google Earth Engine Advanced User meet	Google, Bangalore	March 15-17, 2019



6.3.2 Training Programmes Attended by the Staff of the Academy

Employee Name	Programme Name	Venue	Period
Ch. Srinivasa Rao	Senior Executive Development Programme on Developing Effective Organizational Leadership for Senior Officers of ICAR (In-country component)	Hyderabad Administrative Staff college of India (ASCI), Hyderabad	April 09-13, 2018
K.H.Rao	National Facilitators Development Program in Agricultural Marketing	CCS-NIAM, Jaipur	November 26-05 December, 2018
V.K. Jayaraghavendra Rao			
P D Sreekanth	GIS Approach in soil, water and plant health management	National Institute of Plant Health Management (NIPHM), Hyderabad	May 14-18, 2018
V.K. Jayaraghavendra Rao	Teaching with Simulations by HBS	SPJIMR College Mumbai	March 15-16, 2019
Manju Gerard	Gene Editing in Agriculture: Science, Policy, Story	Cornell -Sathguru Foundation for Development, Hyderabad	February 25-28, 2019
	MDP on Management of Creativity and Innovation	Indian Institute of Management, Calcutta	March 04-08, 2019
P. Ramesh	Managerial Effectiveness	Indian Institute of Management, Ahmedabad	February 18-23, 2019
P.C. Meena	Case Writing and Teaching Workshop	Indian School of Business (ISB), Hyderabad	September 21-22, 2018
Sanjiv Kumar			
V.V. Sumanth Kumar	Artificial Intelligence & Machine Learning	IIIT-H and Talent Sprint at Hyderabad	June 16-September 16, 2018
G. Aneeja	Executive Educational Programme on taking People Along: Managing by Persuasion	IIM, Ahmedabad	August 6-10, 2018
	New Media for Agricultural Extension	MANAGE, Hyderabad	July 09-13

6.3.3 Other Responsibilities and Assignments

Scientist	Responsibilities/Assignments
B. Ganesh Kumar	Member, Institute Management Committee for ICAR-CTRI, Rajahmundry. To participate in the IMC meetings of the institute. ICAR on 2017-2020 Expert Member, Training Quality Improvement Measures Committee (TQIMC) of NIRD & PR, Hyderabad. To guide the training/Workshops/Seminars organized by the Centre for natural Resource Management (CNRM) and Centre for Climate Change and Disaster Mitigation (CCC&DM) of NIRD &PR, Hyderabad on Sep, 2016 to till date Expert Member, Screening & Promotion Committee of ANGRAU, Guntur for faculty in Agricultural Economics under Career Advancement Scheme on 19 Jun, 2018 Member, CAS Promotion Committee of ICAR-NRC on Meat, Chengicherla, Telangana for a scientist in the discipline of Veterinary Extension on 19 Jun, 2018



Ch. Srinivasa Rao	Chairman, ICAR Committee for screening of proposals for new ABI centres in ICAR Institutes under Component-II (Incubation Fund) Member, Executive Council, National Academy of Agricultural Sciences (NAAS), as ICAR/DARE nominee – 2019-21 Member, National Steering Committee for organizing National Agri-Innovation and Entrepreneurship Conclave at New Delhi during October 16-17, 2019. Member, Steering Committee Meeting to review and monitor the performance of 'National Agriculture Innovation Fund' Scheme, operated by IP&TM Unit of ICAR Member, NAAS Sectional Committee for NRM Member, ICAR Training Board Member, Executive Board of the International Dryland Development Commission (IDDC), Cairo, Egypt. (2019-20) Member, International Technical Program Committee for 13th International Conference on Development of Drylands, held at Jodhpur during February 11-14, 2019. President, Association for Management of Agricultural Research and Agripreneurship (AMARA), Hyderabad. Member, Executive Council, National Council for Cooperative Training (NCCT), Department of Agriculture, Cooperation & Farmers Welfare, Ministry of Agriculture & Farmers' Welfare, Govt of India. Member, Management Committee of Extension Education Institute (Southern Region), Department of Agriculture, Cooperation & Farmers Welfare, Ministry of Agriculture & Farmers' Welfare, Govt. of India. Member, Advisory Board, The Journal of Research – ANGRAU, Andhra Pradesh 45th Dr RV Tamhane Memorial Lecture on 'Soil Carbon Sequestration: Research, Technology and Policy Implementation Needs in India' at 83rd Annual Convention of the Indian Society of Soil Science (ISSS) at AAU, Anand on 28 November 2018. Meeting of Consultative Committee of the Ministry of Agriculture and Farmers Welfare to discuss on Agri-entrepreneurship and Startups for enhancing farmers income, held at Parliament House Annex, New Delhi; Organized by DARE, Ministry of Agriculture & Farmers Welfare, GoI, on December 20, 2018.
D Thammi Raju	PGDMA Coordinator for 2018-20 batch Member Secretary of Awards Committee Member of Broad Subject Matter Area Committee on Basic Veterinary Sciences, constituted by ICAR Invited member for 5th Joint Advisory Committee Meeting (JAC) on PGDETM Reviewer for International Journal of Livestock Research Reviewer for Journal of Agricultural Extension Management Journal of Animal Health Production and Technology Expert Panel Member of SVVU for thesis evaluation
K Srinivas	Member of QRT of Indian Institute of Maize Research, Ludhiana 2017 to 2018 Co-Organized "Agri Startup & Incubator Ecosystem" at NABARD Headquarters on November 14, 2018
Manju Gerard	PGDTMA Coordinator 2018 batch Member Secretary, Joint Advisory Committee, NAARM & UoH, PGDTMA Member, ICAR-ABI Screening Committee Meeting, 2nd March, 2019 Invited Member of Institutions Innovation Council, PJTSAU



M. Krishnan	External Subject Expert, Assessment committee for promotion of scientists under revised CAS for Agril Economics at ICAR-CMFRI, Kochi Served as external subject expert for recruitment of Assistant Professor (Fisheries Economics) Cochin Univ of Science and Technology (CUSAT) on 25 Jul, 2018 Invited expert for meeting on Strategic investments for blue revolution, under Chairmanship of Prof. Ramesh Chand, Member (Agril) NITIAYOG on 7 Aug, 2018 Invited panelist in Nat Consultation on sustainable and remunerative small farming : Finance-Market Linkages ICAR-NIANP on 19 Sep, 2018 Invited lead talk, International Fisheries Conference 2019, College of Fisheries (DBSKKV) Ratnagiri & ISASaT, Dapoli, Maharashtra on 17-20 Jan. ,2019 Invited lead talk and Chairman of session on Impacts of seaweed farming on socio-economic development of coastal communities, India International Seaweed Expo & Summit 2019 World Trade Centre, Mumbai on 22-24 Jan, 2019 Invited expert for syllabus revision committee for Fisheries Economics and Statistics at ICAR-CIFE, Mumbai on 18-19 Feb, 2019 Invited Chair Analysis of household social system : challenges & opportunities in the National Conf on Adv Res Methodologies in Soc. Sciences, UAS, Dharwad on 26-28 Feb, 2019 Invited Chair, Adv analytical methods for processing qualitative data & participatory methods in Social Sciences, in the National Conf on Adv Res Methodologies in Soc. Sciences, UAS, Dharwad on 26-28 Feb, 2019
M. Krishnan, S. Senthil Vinayagam,	HiTex Exhibition Centre, ICAR-NAARM Stall @AquaEx India 2018 on 15-17 March, 2018
M. Krishnan and S. Senthil Vinayagam	Hi Tex Exhibition Centre, Hyderabad, ICAR-NAARM Stall @AquaEx India on 31 Jan - 2 Feb., 2019
Ranjit Kumar	Member, Institute Management Committee for ICAR-CTRI, Rajahmundry. To participate in the IMC meetings of the institute. ICAR on 2017-2020
Sanjiv Kumar	Centre Superintendent for centre at T.K.R.College of Engineering & Technology, Medbowli, Meerpet, Hyderabad for AIEEA UG (All India Entrance Examination for Admission for under graduation held on Aug 19, 2018.



Chapter - 7

Financials

Financials

7.1 Budget Allocation and Expenditure during 2018-19 (Rs. in lakh)

Head of Account	Final Revised Estimate	Expenditure
A. Grants in Aid - Capital	750.30	750.13
B. Grants in Aid - Salaries (REVENUE)		
Establishment Expenses		
i. Establishment Charges	2462.10	2462.10
ii. Wages	201.70	201.66
iii. OTA	0.20	0.20
Total Grants in Aid-Salaries	2664.00	2663.96
C. Grants in Aid-General (REVENUE)		
Pension & Other Retirement Benefits	3200.00	3200.00
Traveling Allowance	125.00	125.00
Research & Operational Expenses	217.55	217.54
Administrative Expenses	1770.42	1770.39
H.R.D.	36.80	36.80
Miscellaneous Expenses	150.23	150.22
Total Grants in Aid-General	5500.00	5499.95
Grand Total (Capital+ Salaries+ General)	8914.30	8914.04

7.2 Resource Generation

7.2.1 Sponsored Research Projects Budget (Rs. In Lakhs)

	Name of the Project	Budget outlay	Fund available/ received including Opening Balance for 2018-19	Expenditure during 2018-19
1	RKVY - Telangana State Govt. Developing SAIDP Project	24.72	4.66	4.66
2	Developing Guidelines & Methodologies for IP Policies for PJTSAU	5.00	3.26	0.81
3	PJTSAU Food Consumption Project	10.17	4.64	0
4	PJTSAU Food Consumption Project	0.25	0.25	0.25
5	Environmental Impact Assessment of Tobacco Guring	20.87	13.16	10.42
6	DBT-NAARM Project - Socio Economic Analysis of Finger Millet in India (Ragi)	15.99	1.65	1.65
7	R&D Strategies and Policies under SERB, DST	12.42	6.70	4.84
8	Agril. Res. & Dev. Infra. in Select Indian States under NSTMIS Scheme of DST	30.00	19.32	9.01
9	RGMVP, Raibareli	2.12	2.12	2.12
10	IFPRI Sponsored Project - ASTI in Indian NARES- ASTI Project	28.35	11.35	10.54

11	Consultancy Project on Adoption of Insect Resistance Mgmt. (Mhyco, Monsanto)	26.00	20.35	10.92
12	Climate Change & Indian Agril: Impact & Adoption Strategies GOI-GEF-UNDP NATCOM	8.50	5.10	5.10
13	ICAR-NAARM-NIPHM Project “Plant Bio-security through MOOC Platform”	7.20	7.20	3.75
14	ICAR-NAARM-MANAGE Project “Agri-Warehousing Management”	9.90	8.37	0
15	ICAR-NAARM-PJTSAU for MOOC	4.50	4.50	0
16	ICAR-NAARM-ITC “Economic Analysis of Eucalyptus and Subabul based Agro-forestry Ecosystems in Telangana & A.P	11.28	6.96	1.2
17	Profiling the Resources Devoted in Agril. R&D Dev. of National framework for Evidence-based S&T Governance in India Scheme of DST	24.88	21.00	3.52
18	Skill Development in Bio-informatics Infrastructure Facility on Agril. Bio Technology under DBT	19.42	19.42	0
19	Adoption of Natural Farming and its effect on Crop Yield and Farmers Livelihood in India NITI Ayog Project	35.00	12.60	4.86
20	Expert committee Meeting of DST under Woman Scientists under DST	9.00	9.00	5.43
Total		305.57	181.61	79.08

7.2.2. Off - Campus and Sponsored Programmes

(Rs. In Lakhs)

Name of the Programme	Amount(Gross)
Climate Smart Agril. Policies Strategies & Agril. Dev. Progr. From 17-19 th April, 2018 (SAARC Programme)	5.88
Induction course for newly recruited Asst. Professors of PJTSAU (2 Batches)	10.99
Coramandel International Ltd. CIL-NAARM Collaborative training programmes-5	10.59
Trg. On “Scientific Communication & DWRP” to the teaching staff of TNJFU - 2 programmes	2.68
Orientation Trg. To PJTSAU faculty on Geo Informatics & nanotechnology” from 27-29 th June,2018	1.04
ICTs for Agril. Development Programme from 21-8-2018 to 19-9-2018	6.79
DWRP in Agril. Res. for Faculty members of RVSKVV, Gwalior from 25-29 th Sept. 2018	2.60
PRA Commn. & Strategies for IFS Probationers IGFNA, Dehradun from 6-7 th Octo.2018	4.45
Foundation Course to the newly recruited Asst. Professors PJTSAU from 5-14 th Nov.2018	3.90
DWRP for faculty members of UAS, Bangalore from 26-28 th December, 2018	2.18
Emotional Intelligence for personal & Work Excellence for faculty of CIFE, Mumbai from 11-13 th Feb. 2019	1.63
FDP at UAHS, Shivamogga, Karnataka from 4-8 th Feb. 2019	2.23
Digital Teaching Techniques for UAS, Bangalore from 19-23 rd Feb. 2019	2.94
DWRP in Agril. Res. For faculty members of RVSKVV, Gwalior from 7-9 th March, 2019	2.60
Entrepreneurial Skill Dev. Among Agri-Graduates of ANGRAU - 2 programmes	22.50
DWRP in Food Tech. & Allied Disciplines for Faculty members of NIFTEM, Sonapat 13-15 th Oct. 2018	1.33
ICT & its appln. in Livestock Production and Health under NAHEP for students of SVVU, Tirupathi	4.28
Total	88.61



7.2.3 Resource Generation from Educational Programmes

(Rs. In lakhs)

Name of the Programme	Amount (Gross)
PGDMA 2017-19	102.53
PGDMA 2018-20	90.05
PGDETM-2018	7.85
PGDTMA 2018	6.84
Total	207.27

7.2.4 Resource Generation from Other Activities

(Rs. In Lakhs)

Activities	Amount(Gross)
Receipts from Farm produce	1.79
Receipts from Guest House and Quarters	49.05
Receipts from Institute Training Programmes	10.00
Receipts for sale of application fee	6.87
Receipts from services rendered	3.89
Miscellaneous Receipts	57.27
Total	128.87

7.2.5 Net receipts from Sponsored Research Projects

(Rs. In Lakhs)

Name of Project	Amount (Gross)
RKVY (TS) Project SAIP	4.31
PJTSAU Medicinal Plants	0.25
RGMVP, Raibareli	0.69
Revenue from contract research	0.92
Net receipts from Sponsored programmes	50.88
Net receipts from Educational Programme PGDTMA 2016-18	100.00
Net receipts from Educational Programme PGDMA - III	7.96
Total	165.01

7.2.6 Plan Schemes

(Rs. In Lakhs)

Name of the Scheme	Budget outlay (Rupees in Lakhs)	Fund available/received including Opening Balance for 2018-19	Expenditure during 2018-19
NAIF - ZTMC	15.00	9.75	7.99
NAIF - ABI	32.00	18	14.85
NICHE - TELAgE Project	9.19	15.72	7.44
KRISHI Project	26.55	19.05	18.53
FFP Project	8.90	8.9	8.88
Extra Mural Res. Project (ESM)	5.50	5.5	1.23
Summer/Winter Short Courses	6.51	5.25	5.23
Total	103.65	65.91	64.15



Chapter - 8

Personnel

Personnel

8.1 Human Resources at the Academy (as on 31.03.2019)

Category	Sanctioned Strength	In position	Vacant
Scientific - RMP	02	01	01
Scientific - Faculty	60	32	28
Technical	**35	32	03
Administrative	49	33	16
Skilled Support Staff	39	32	07
Total	185	130	55

**Excludes posts which came under dying cadre

Research Management Positions

1. Dr Ch. Srinivasa Rao, Director
2. Dr S K Soam (Incharge Joint Director)

Scientific Staff

Agribusiness Management Division

1. Dr. Ranjit Kumar, Head
2. Dr. B. Ganesh Kumar, Principal Scientist
3. Dr. N. Sivaramane, Senior Scientist
4. Dr. Prem Chand Meena, Senior Scientist
5. Dr. Sanjiv Kumar, Scientist

Education Systems Management Division

1. Dr. M Krishnan, Head
2. Dr. G.R. Ramakrishna Murthy, Principal Scientist
3. Dr. D. Thammi Raju, Principal Scientist
4. Dr. S. Senthil Vinayagam, Principal Scientist
5. Dr. V.V. Sumanth Kumar, Scientist

Extension Systems Management Division

1. Dr. Bharat S. Sontakki, Head
2. Dr. (Smt.) N. Sandhya Shenoy, Principal Scientist
3. Dr. V.K. Jayaraghavendra Rao, Principal Scientist
4. Dr. Surya Rathore, Principal Scientist
5. Dr. P. Venkatesan, Senior Scientist

Human Resources Management Division

1. Dr. R.V. S Rao, I/c Head, HRM Division
2. Dr. P. Ramesh, Principal Scientist
3. Dr. K.H. Rao, Principal Scientist
4. Dr. Alok Kumar, Principal Scientist

Information and Communication Management Division

1. Dr. SK Soam, Head
2. Dr. A. Dhandapani, Principal Scientist
3. Dr. S. Ravichandran, Principal Scientist
4. Dr. M. Balakrishnan, Principal Scientist
5. Dr. N. Srinivasa Rao, Principal Scientist
6. Dr. P.D. Sreekanth, Senior Scientist
7. Dr. B.S. Yashavanth, Scientist

Research Systems Management Division

1. Dr. I. Sekar, Head
2. Dr. K. Srinivas, Principal Scientist
3. Dr. K. Kareemulla, Principal Scientist



4. Dr. M.B. Dastagiri, Principal Scientist
5. Dr. P. Krishnan, Principal Scientist
6. Dr. Manju Gerard, Principal Scientist

Administrative Staff

Joint Director (Admn.) & Registrar

Sh. Ashish Roy (upto February 28, 2019)

Chief Finance & Accounts Officer

Sh. S. George

Senior Administrative Officer

Shri J.N.L. Das

Administrative Officer

Sh. W. Sreenivasa Bhat

Asst. Admn. Officer

1. Sh. P.G. Kohad
2. Sh. K.R. Ghanshyam
3. Sh. M. Dinesh

Private Secretary

1. Sh. N. Raghunath
2. Mrs. A. Mercy
3. Mrs. T. Vanisri

Junior Accounts Officer

1. Mrs. N. Vijayalakshmi

Assistants

1. Sh. T. Srinivas
2. Sh. G. Raj Reddy
3. Sh. C. Phani Raj
4. Mrs. B. Padma Saroja
5. Sh. P. Srinivasu
6. Mrs. G. Jessie Ecclicia
7. Sh. R. Chandra Babu
8. Sh. C. Julius Samuel
9. Sh. M. Sridhar

Personal Assistants

1. Sh. T. V. Ramadas
2. Mrs. Y. Anuradha
3. Mrs. S. Sesha Sai
4. Mrs. K.K. Rukmani Ammal
5. Mrs. V. Shailaja
6. Mrs. S. Shanthi

Upper Division Clerks

1. Mrs. Y. Gayathri
2. Sh. M.K. Samson
3. Mrs. Rajashri Bokde
4. Sh. K. Suryanarayana

Lower Division Clerk

1. Sh. P. Swamy
2. Sh. M. Narsing Rao
3. Sh. M. Ashok
4. Sh. C. Bikshapathi

Technical Staff

Grade T-9 (Category-III)

1. Dr. V. Murali, Chief Technical Officer (Garden Superintendent)
2. Dr. Ch. Janardhan Rao, Chief Technical Officer (Video cameraman) (Relieved on 10.04.2017 (AN) with lien for two years to join the post of Reader at UoH.)
3. Dr. M.A. Basith, Chief Technical Officer (Jr. Farm Superintendent)
4. Sh. K.V. Kumar, Chief Technical Officer (Statistical Assistant)
5. Sh. P. Vijender Reddy, Assistant Chief Technical Officer (Research Assistant)

Grade T(7-8) (Category-III)

1. Sh. P. Namdev, Assistant Chief Technical Officer (Graphic-cum-Visualizer)
2. Sh. Sohail Ahmad Khan,

Assistant Chief Technical Officer (Junior Engineer-Civil)

3. Mrs. G. Aneeta, Assistant Chief Technical Officer (Assistant Editor)

4. Sh. M. Shekhar Reddy, Assistant Chief Tech. Officer (Dark Room Assistant)

5. Dr. Ahire Laxman Mahar, Assistant Chief Technical Officer (Horticultural Technical Assistant)

6. Sh. M. Ravi, Sr. Technical Officer (Photographer-cum-Artist)

7. Sh. Sham Bahadur, Asst. Chief Technical Officer (Catering in-charge)

Grade T-6 (Category-III)

6. Sh. P. Mohan Singh, Sr. Technical Officer (Computer Assistant)

7. Sh. Ravi Viswanathan, Sr. Technical Officer (Editor-cum-Information Officer)

8. Ms. Savithri Murali, Sr. Technical Officer (Catering in-charge)

Grade T-5 (Category-II)

1. Sh. N. Naresh Kumar, Technical Officer (VCR-cum-Tele cine Operator)

2. Sh. B.S.N. Murthy, Technical Officer (Technician Electrical)

3. Sh. K. Obulapathi, Technical Officer (Technician Electrical)

4. Sh. M.K. Shamshuddin, Technical Officer (Technician Electrical)

5. Sh. M.K. Sonkusare, Technical Officer (Technician Electrical)

6. Sh. K. Shivaiah, Technical Officer (Technician)



7. Sh. D.Rajagopal Rao, Technical Officer (Media Operator)

8. Sh. S.Rajukumar, Technical Officer (Electronic Computer Operator)

9. Sh. S.Sunder Raj, Technical Officer (Media Operator)

10. Sh. T.Laxman, Technical Officer (Driver)

Grade T-4 (Category-II)

1. Sh. A.C.P.Rama Nageswara Rao, Sr.Technical Assistant (Lineman)

2. Sh. Pitla Srinivas, Sr.Technical Assistant (Proof Reader)

3. Sh. G.Muthyalu, Sr.Technical Assistant (Driver)

4. Sh. N.Ashok, Sr.Technical Assistant (Driver)

5. Sh. N.Prabhakar Rao, Sr.Technical Assistant (Plumber)

6. Sh. M.Srinivasa Rao, Sr.Technical Assistant (Pump Driver)

7. Sh. K.Veera Narasaiah, Sr. Technical Assistant (Carpenter-cum-Painter)

Grade T-3 (Category-II)

1. Sh. B.K.Venkatram, Technical Assistant (Pump Driver)

2. Sh. R.Siva Prasad, Technical Assistant (Driver)

Skilled Support Staff

1. Sh. S. Swamy, Asst .Gestt. Operator

2. Sh. Sirigiri Venkatesham, Xerox Machine Operator

3. Sh. P.Balraj

4. Mrs. B.Santhamma

5. Mrs. S.Shakuntala

6. Mrs. B.Bharathamma

7. Sh. S.Jangaiah

8. Sh. Kumba Satyanarayana

9. Sh. K.Pentaiah

10. Sh. P.Yadaiah

11. Sh. G.Pentaiah

12. Sh. J.Chandraiah

13. Sh. M.Krishnaiah

14. Sh. R.Sattaiah

15. Ms. Indirapati Bharatamma

16. Sh. C.Chandramouli

17. Sh. G.Dasaratha

18. Ms. Nazia Parveen

19. Sh. L.Satyanarayana

20. Ms. N.Sukunamma

21. Ms. M.Yadamma

22. Ms. C.Narsubai

23. Ms. C.Kausalya

24. Sh. M.Ganesh Kumar

25. Sh. Chilumula Venkatesham

26. Sh. Sirigiri Narsimha

27. Sh. Kyasam Satyanarayana

28. Sh. B.Premdas

29. Mrs. A.Saroja

30. Sh. S.Nayab Rasool

31. Sh. B.Ashok

32. Sh. K.Daniel

Appointments/

New Joinings

1. Sh. Danam Murahari appointed as T-3 Technical Assistant (Audio Visual Operator)w.e.f. 21.06.2018

2. Smt. K.Mariyamma, CLTS–regularized her services against the post of Skilled Support Staff w.e.f. 18.05.2018

3. S. Jaiprakash Bisen, Scientist (Agricultural Economics, 107th FOCARS – joined on 16.04.2018 for 3-months Professional Attachment Training

4. Sh. M.Satyanarayana, CLTS has been regularised against the vacant post of SSS w.e.f. 20.10.2018 (FN)

5. Sh. D.Srisailam, CLTS has been regularised against the vacant post of SSS w.e.f. 20.10.2018 (FN)

6. Sh. B.Ratnaiah, Casual Labourer with temporary status (CLTS) has been regularised against the vacant post of SSS w.e.f. 22.02.2019 (FN)

7. Sh. D.Babaiah, Casual Labourer with temporary status (CLTS) has been regularised against the vacant post of SSS w.e.f. 22.02.2019 (FN)

Promotions

1. Dr.N.Sivaramane, Sr. Scientist (Agribusiness & Marketing Management) – promoted under Revised CAS to the Grade of RGP 9000 w.e.f. 28.03.2015 for which orders received in April 2018

2. Dr. V.V.Sumanth Kumar, Scientist (Computer Application in Agriculture) – promoted under Revised CAS to the Grade of RGP 7000 w.e.f. 16.04.2009 for which orders received in April 2018

3. Smt. T.Vanisri, PA – promoted to the post of Private Secretary w.e.f. 14.06.2018

4. Dr. P.C.Meena, Sr. Scientist–promoted as Principal Scientist w.e.f. 24.04.2017 under revised CAS.

5. Dr. P.Venkatesan, Sr. Scientist promoted as Principal Scientist w.e.f. 05.05.2017 under revised CAS.



6. Dr. N.Sivaramane, Sr.Scientist promoted as Principal Scientist w.e.f. 28.03.2018 under revised CAS.

7. Smt. V.Shailaja, Stenographer Grade-III promoted as Personal Assistant w.e.f. 28.02.2019 (FN)

8. Smt. S.Shanti, Stenographer Grade-III promoted as Personal Assistant w.e.f. 28.02.2019 (FN)

Financial Upgradations (under MACP Scheme for Admn. and Skilled Support Staff)

1. Mrs. A. Mercy, Private Secretary, 3rd : GP4800 (Level # 8) from 01.09.2018

2. Mrs. T. Vanisri, Private Secretary, 3rd : GP4800 (Level # 8) from 01.09.2018

3. Mrs. B. PadmaSaroja, Assistant, 3rd : GP4600 (Level # 7) from 26.08.2017

4. Sh. P. Srinivasu, Assistant, 3rd : GP4600 (Level # 7) from 11.01.2018

5. Sh. T.V.Ramdas, Personal Assistant, 3rd : GP4800 (Level # 8) from 01.09.2018

6. Mrs. Y.Anuradha, Personal Assistant, 3rd : GP4800 (Level # 8) from 01.09.2018

7. Mrs. Y. Gayathri, UDC, 3rd : GP4200 (Level # 6) from 08.01.2019

8. Sh. M.K.Samson, UDC, 3rd : GP4200 (Level # 6) from 25.01.2019

9. Sh. P.Swamy, LDC, 3rd : GP4200 (Level # 6) from 10.08.2015

10. Sh. M.Narsing Rao, LDC, 3rd : GP2400 (Level # 4) from 26.08.2017

11. Sh. C.Bikshapathi, LDC, 3rd : GP2400 (Level # 4) from 26.08.2017

12. Sh. M.Ashok, LDC, 3rd : GP2400 (Level # 4) from 26.08.2017

13. Mrs. N. Sukunamma, SSS, 3rd : GP2400 (Level # 4) from 26.08.2017

14. Mrs. M. Yadamma, SSS, 3rd : GP2400 (Level # 4) from 29.08.2017

15. Mrs. B.Bharatamma, SSS, 3rd : GP2400 (Level # 4) from 26.08.2017

16. Sh. S.Jangaiah, SSS, 3rd : GP2400 (Level # 4) from 26.08.2017

17. Sh. G.Pentaiah, SSS, 3rd : GP2400 (Level # 4) from 26.08.2017

18. Sh. G.Dasaratha, SSS, 3rd : GP2400 (Level # 4) from 26.08.2017

19. Sh. J.Chandraiah, SSS, 3rd : GP2400 (Level # 4) from 27.08.2017

20. Sh. C.Chandramouli, SSS, 3rd : GP2400 (Level # 4) from 26.08.2017

21. Sh. M.Krishnaiah, Ex-SSS, 3rd : GP2400 (Level # 4) from 26.08.2017

22. Sh. R.Sattaiah, SSS, 3rd : GP2400 (Level # 4) from 26.08.2017

23. Sh. L.Satyanarayana, SSS, 3rd : GP2400 (Level # 4) from 25.06.2018

24. Mrs. I.Bharatamma, SSS, 3rd : GP2400 (Level # 4) from 06.03.2019

25. Sh. K.Pentaiah, SSS, 2nd

: GP2000 (Level # 3) from 27.09.2015

26. Sh. P.Yadaiah, SSS, 2nd : GP2000 (Level # 3) from 27.09.2015

27. Mrs. C.Narsubai, SSS, 2nd : GP2000 (Level # 3) from 30.01.2016

28. Mrs. C.Kausalya, SSS, 2nd : GP2000 (Level # 3) from 04.07.2018

Transfers

1. Sh. Jaiprakash Bisen, Scientist (Agricultural Economics, 107th FOCARS) posted at ICAR-NRRI, Cuttack after undergoing professional training at ICAR-NAARM w.e.f. 04.01.2018 to 16.07.2018.

Retirements

1. Sh. M.Venkatesh, Personal Assistant Retired Voluntarily w.e.f. 05.05.2018

2. Sh. Zameer Ahmed, T-9 Chief Technical Officer [Manager (Hostel Services)]-Retired on superannuation w.e.f. May 31.05.2018

3. Sh. S.Sunder Raj, T-5 Technical Officer (Media Operator) – Retired on superannuation w.e.f. 31.05.2018

4. Sh. P.Bal Raj, Skilled Support Staff – Retired Voluntarily w.e.f. 31.05.2018

5. Sh. S.Swamy, Skilled Supporting Staff (Asst. Gestetner Operator) – Retired on superannuation w.e.f. 31.08.2018

6. Dr. A.Debnath, T-9 Chief Technical Officer (Medical Officer) – Retired on superannuation w.e.f. 30.09.2018

7. Sh. M.K.Shamshuddin,



Technical Officer (Technician Electrical) – Retired on superannuation w.e.f. 30.09.2018

8. Smt. M.Yadamma, Skilled Support Staff – Retired on superannuation w.e.f. 30.09.2018

9. Sh. M.Krishnaiah, SSS Retired Voluntarily w.e.f. 20.10.2018 (FN)

10. Shri M.K.Sonkusare, T-5 TO (Technician Electrical) Retired on superannuation w.e.f. 30.11.2018 (AN)

11. Smt. N.Sukunamma, Skilled Support Staff – Retired on superannuation w.e.f. 31.12.2018

12. Sh. Phool Kumar, Skilled Support Staff – Retired on superannuation w.e.f. 31.01.2019 (AN)

13. Sh. Ashish Roy, Joint Director (Admin.) & Registrar - Retired on superannuation w.e.f. 28.02.2019 (AN)

14. Sh. B.Rajaiah, Casual Labourer Temporary Status (CLTS) – Retired on superannuation w.e.f. 30.06.2018 (AN)

15. Sh. P.Venkat Rao, Casual Labourer Temporary Status (CLTS) – Retired on retired voluntarily w.e.f. 05.11.2018 (FN)

Clearance of Probationary Period (PP) and Confirmation

1. Dr. Alok Kumar, Principal Scientist – PP cleared w.e.f. 01.09.2015 and confirmed w.e.f. 02.09.2015 in the grade of Senior Scientist

2. Dr. Prem Chand Meena, Senior Scientist – PP cleared w.e.f. 22.06.2011 and confirmed w.e.f. 23.06.2011 In the grade of Scientist

3. Dr. Sanjiv Kumar, Scientist – PP cleared w.e.f... 30.06.2017 and confirmed w.e.f. 01.07.2017 In the grade of Senior Scientist

4. Dr. B.S. Yashavanth, Scientist (Agricultural Statistics) – Probationary Period cleared w.e.f. 04.07.2018 and confirmed w.e.f. 05.07.2018

Others

1. Dr. S.Senthil Vinayagam, Principal Scientist has been appointed by ICAR as Vigilance Officer of the Academy w.e.f. 19.03.2019

8.2 List of Committees (as on 31.03.2019)

Higher Education Committee

1. Dr Ch Srinivasa Rao, Director - Chairman
2. Dr SK Soam, Head, ICM Division & Joint Director I/C - Member
3. Dr Bharat Sontakki, Head, XSM Division - Member
4. Dr M Krishnan, Head, ESM Division - Member
5. Dr I Sekar, Head, RSM Division - Member
6. Dr Ranjit Kumar, Head, ABM Division - Member
7. Dr RVS Rao, Head, HRM Division Member
8. Dr P Krishnan, Pr. Scientist & Head, PME Cell - Member
9. Dr P Ramesh, Principal Scientist & Controller of Examinations (CoE) - Member
10. Shri Ashish Roy, Joint Director (A) & Registrar - Member
11. Dr P Vijender Reddy, CTO - Member Secretary
12. Coordinators of PGDMA, PGD-TMA, PGD-ETM - Coopted Members

Board of Studies for PGDMA, PGDTMA and PGDETM

1. Dr Ch Srinivasa Rao, Director - Chairman
2. Dr SK Soam, Head, ICM Division & Joint Director I/C - Member
3. Dr Bharat Sontakki, Head, XSM Division - Member
4. Dr Ranjit Kumar, Head, ABM Division - Member
5. Dr KH Rao, Pr. Scientist & In-charge, Placement Cell - Member
6. Dr A Dhandapani, Pr. Scientist, ICM Division - Member
7. Dr D Thammi Raju, Principal Scientist, ESM Division - Member
8. Shri Ashish Roy, Joint Director (A)& Registrar - Member
9. Shri S George, CFAO - Member
10. Shri. KR Ghanshyam, AAO - Member
11. Dr PC Meena, Senior Scientist, ABM - Member

12. Coordinators of PGDMA, PGD - TMA, PGD - ETM - Coopted Members

Anti-Ragging Committee

1. Joint Director / Dean - Chairman
2. Dr I Sekar, Head, RSM Division - Member
3. Joint Director (Admn.) & Registrar - Member
4. Academic Coordinators, PGDM (A) Programme - Members
5. Office Incharge, Training & P.G. Studies - Member-Secretary

Anti-Ragging Squad

1. Dr (Mrs.) Surya Rathore, Pr. Scientist - Chairman
2. Dr B Ganesh Kumar, Pr. Scientist - Member
3. Chief Finance & Accounts Officer - Member
4. Officer Incharge, Hostel - Member Secretary

Ombudsman/ Grievance Committee of the Academy

1. Director, Chairman
2. Joint Director /Dean - Member
3. Joint Director (Admn.) & Registrar - Member
4. Academic Coordinators of PGDM(A) Programme, Members
5. Senior Admn. Officer - Member
6. Shri S Rajukumar, Technical Officer - Member
7. Dr LM Ahire, ACTO & Incharge, Training Unit, Member Secretary

Committee for Regulatory Affairs

1. Dr N Sivaramane, Sr. Scientist - Chairman
2. Senior Admn. Officer - Member
3. Shri S. Rajukumar, Technical Officer - Member
4. Dr LM Ahire, Asst. Chief Technical Officer - Member
5. Shri KR Ghanshyam, AAO - Member Secretary



Swachh Bharat Abhiyan Implementation Committee

1. Dr RVS Rao, Chairman, Farm Advisory Committee - Chairman
2. Dr MA Basith, CTO & OIC, Farm - Member
3. Shri JNL Das, Senior Admn. Officer - Member
4. Dr LM Ahire, ACTO, Academic Cell - Member
5. Shri M Shekhar Reddy, ACTO - Member
6. Dr P Vijender Reddy, CTO - Member Secretary

Farm Advisory Committee / Landscape and Beautification Committee

1. Dr RVS Rao, I/C Head - HRM Division - Chairman
2. Dr P Ramesh, Pr. Scientist, HRM Division - Member
3. Dr GRK Murthy, Pr. Scientist, ESM Division - Member
4. Dr Surya Rathore, Pr. Scientist, XSM Division - Member
5. Dr BS Yashavanth, Scientist, ICM Division - Member
6. Dr V Murali, CTO - Member
7. Dr Ahire Laxman Mahar, ACTO - Member
8. Shri P Mohan Singh, STO - Member
9. Shri M Shekhar Reddy, ACTO - Member
10. Shri T Laxman, - Member
11. Dr MA Basith, AC - Member Secretary

Library Advisory Committee

1. Dr SK Soam, Head -ICM Division and JD I/C - Chairman
2. All HoDs - Member
3. Dr P Krishnan, Pr. Scientist, RSM Division - Member
4. Dr VV Sumanth Kumar, Scientist, ESM Division - Member
5. Joint Director (A) & Registrar - Member
6. Chief Finance & Accounts Office - Member
7. Dr V Murali, CTO Member - Secretary
8. Dr N Srinivasa Rao, Pr. Scientist - Member Secretary

Newsletter Committee

1. Dr Ranjit Kumar, Head-ABM Division - Chairman
2. Dr Sanjiv Kumar, Scientist, ABM Division - Member

3. Dr BS Yashavanth, Scientist, ICM Division - Member
4. Smt. G Aneja, Asst. Chief Tech. Officer - Member Secretary

Annual Report Committee

1. Dr M Krishnan, Head, ESM Division - Chairman
2. Dr I Sekar, Head, RSM Division - Member
3. Dr Bharat Sontakki, Head, XSM Division - Member
4. Dr Ranjit Kumar, Head, ABM Division - Member
5. Dr RVS Rao, Head, HRM Division - Member
6. Dr D Thammi Raju, Pr. Scientist, ESM Division - Member
7. Dr PD Sreekanth, Pr. Scientist, ICM Division - Member
8. Smt. G Aneja, Asst. Chief Tech. Officer - Member
9. Dr Manju Gerard, Pr. Scientist, RSM Division - Member Secretary

Campus Residents Welfare Committee

1. Dr SK Soam, Joint Director I/c - Chairman
2. Dr M Balakrishnan, Pr. Scientist - Member
3. Dr Manju Gerard, Pr. Scientist - Member
4. Dr BS Yashavant, Scientist - Member Secretary
5. Shri Shitantshu Kumar, SAO, ICAR-IIOR - Outside Member
6. Shri Ch. Suresh Kumar, Assistant, ICAR-IIMR - Outside Member

Examination Unit

1. Dr P Ramesh, Pr. Scientist, HRM Division - CoE
2. Dr Manju Gerard, Principal Scientist, RSM, In - charge, CoE
3. Shri JNL Das, Senior Admn. Officer
4. Shri Pitla Srinivas, Senior Technical Assistant
5. Mrs. B. Padma Saroja - Member
6. Shri B Ashok, SSS, Training Unit

Hostel Management Committee

1. Dr SK Soam, Joint Director I/c.- Chairman
2. Dr Surya Rathore, Pr. Scientist - Member

3. Dr M Balakrishnan, Pr. Scientist
4. Joint Director (A) & Registrar - Member
5. Chief Finance & Account Officer - Member
6. Dr Yashavanth B.S., Scientist - Member
7. Smt. Savithri Murali, Sr. Tech. Officer-Member
8. Sh. Sham Bahadur, Asst. Chief Tech. Officer - Member Secretary

Institute Technology Management Committee

1. Dr Ch Srinivasa Rao, Director, NAARM - Chairman
2. Dr SK Soam, Jt. Director, I/c, Member
3. Dr GRK Murthy, Pr. Scientist, ESM Division - Member
4. Dr K Srinivas, Pr. Scientist, RSM - Member
5. Joint Director (Admin) Registrar - Member
6. Sh. S George, CFAO, Member
7. Dr RM Sundaram, Pr. Scientist, IIRR. Hyd.- Member
8. Dr Manju Gerard, Pr. Scientist, RSM Division, Member Secretary

Sports Complex Maintenance Committee

1. Dr P Vijender Reddy, ACTO - Chairman
2. Dr MA Basith, CTO - Member
3. Sh. Sohail Ahmd Khan, ACTO - Member
4. Dr Laxman M Ahire, ACTO - Member
5. Sh. M Shekhar Reddy, ACTO - Member
6. Sh. P Mohan Singh, STO - Member
7. Sh. Sham Bahadur, ASTO - Member
8. Sh. MK Samson, UDC - Member
9. Smt. KK Rukmani Ammal, PA - Member Secretary

Campus Canteen Committee

1. Dr Manju Gerard, Pr. Scientist, RSM Division, Chairperson
2. Dr PC Meena, Senior Scientist, ABM - Member
3. Sh. PG Kohad, AAO, Member
4. Smt. Y Gayarthri, UDC, Member
5. Representative from FOCARS, Member
6. Two representatives each from PGDMA I & II, Members
7. Secretary, IJSC Member
8. Sh. K Veeranarsaiah, Tech. Assistant, Member Secretary

Internal Complaints Committee

1. Dr Surya Rathore, Principal Scientist, XSM - Chairperson
2. Mrs. Vijayalaxmi - Jr. Account Officer Member
3. Mrs. G Aneeraja, Asst Chief Technical Officer - Member
4. Mrs. P. Padmavathi, Katurba Gandhi National Memorial Trust - Member
5. Mrs. K.K. Rukmani Ammal, PA - Member Secretary

Bio-metric & Performance Monitoring Committee

1. Dr Ranjit Kumar, Head, ABM - Chairman
2. Dr RVS Rao, I/c Head, HRM
3. Sh. Rajukumar, Technical Officer - Member

